

PROJECT CASCADE,
COMPUTATIONAL ANALYSIS OF SPACEFLIGHT
CONDITIONS, ASTRONAUT DATA, AND EPIDEMIOLOGICAL
EVIDENCE
MAPPING THE EVIDENCE BEHIND THE SCORE: EXAMINING THE RISK
ARCHITECTURE OF HUMAN SPACEFLIGHT THROUGH NATURAL LANGUAGE
PROCESSING AND NETWORK ANALYSIS OF NASA’S EVIDENCE BASE AND LEVELS
OF EVIDENCE SCORING ACROSS 29 HUMAN SYSTEM RISKS, (2020-2025)

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Dedication

NOTE TO SELF - NEED TO ADD HERE DON'T FORGET KM

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by

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Epidemiology

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Disclosure

This research was conducted under a Space Act Agreement between the NASA Johnson Space Center and The University of Texas Health Science Center at Houston School of Public Health, agreement number SA-SA-22-36464. The views expressed in this dissertation are those of the author and do not constitute official policy or position of the National Aeronautics and Space Administration or the United States Government. No astronaut-identifiable health data were accessed in the conduct of this research.

Preface

This dissertation began with a simple question: when decisions about human safety rest on expert judgment, how well does the evidence on record match the confidence those decisions assign to it? Project CASCADE pursues that question for NASA’s human system risk portfolio, but the underlying concern belongs to epidemiology and to any field that must act on mapping risk evidence.

A word on what this work is and is not. It examines how evidence is documented and graded across an entire portfolio, not whether any individual safety decision was correct. It is examination of evidence architecture, offered as a transparent, reproducible preliminary method and a dated baseline rather than a verdict on the agency or its experts.

Methodologically, it draws together tools that rarely meet in one place: systematic evidence extraction, natural language processing, network science, and the reasoning of epidemiology. The appendices hold the full machinery, the lexicons, the formulas, and the data, so that any reader can retrace every step and judge the work on its own terms. This work also lives beyond the page, in three tangible deliverables, an interactive dashboard and Jupyter Notebook, the Evidence Architecture Explorer application, and a portfolio website, each built to let others see, navigate, and interrogate the evidence for themselves, and together they are among the most durable and reusable contributions of this study.

The lasting contribution is the framework itself, designed to outlive its pilot findings and to prove useful wherever evidence must be appraised, graded, and turned into decisions under uncertainty.

Acknowledgements

I extend special acknowledgment to NASA's Human System Risk Board for its guidance and partnership in developing this dissertation framework, and to the wider NASA community whose science and evidence are woven throughout this work.

I gratefully acknowledge the Southwest Center for Occupational and Environmental Health (SWCOEH) at UTHealth Houston School of Public Health, whose Research Accelerator Just-in-Time Award made Project CASCADE possible. This work extends the principles of occupational and environmental medicine to one of the most extreme occupational cohorts of all, astronauts in human spaceflight, whose in-mission health rests on the same rigorous evaluation of evidence that protects workers terrestrially.

I am deeply grateful to the members of my dissertation committee. My chair, Dr. Troisi, opened the door to this journey when she interviewed me for admission to the doctoral program, and across community health, leadership, and the epidemiological sciences her guidance has been steady and her belief in me unwavering.

Dr. Fujimoto laid the foundation for my understanding of network analysis and sharpened my use of statistical tools. When I brought her an idea well outside a traditional epidemiology dissertation, she met it with curiosity and helped me frame it with rigor, showing me how to bring perspectives from different fields into epidemiology and public health.

Dr. Xiao, through her experimental methods course and an early informational interview spanning her experience with wet lab work, NASA earth sciences research, and epidemiological perspectives, shaped my thinking from the start and showed me how skills across these domains could be bridged into a single line of inquiry.

My most heartfelt gratitude belongs to Dr. Van Baalen, whose impact on this work and on me has been profound. As a role model, supervisor, and leader in the space sciences, she moves fluidly across disciplines and brought that same range to our conversations, ones that reshaped how I understood the gaps in our knowledge and the processes by which we learn and move forward. She was integral to this work at every stage. Through her leadership at NASA and her expertise across all of the risk domains, she deepened my thinking and opened my eyes to questions I had not known to ask, and she stood by this project with steadiness even through unprecedented disruption, and governmental uncertainties. Her curiosity, and her insistence on pushing the boundaries of science toward the questions that matter most, modeled a kind of scholarship I hope to carry throughout my career. Above all, she taught me to test the limits of an idea, to sit with hard problems rather than around them, and to bring new strategies to bear when the familiar ones fall short, making me a sharper and more refined *critical thinker* than I was when I began. That growth does not end here; I will keep building on what she has given me and keep pressing at the boundaries she taught me to question. For her mentorship, her example, and her belief in this work, I am more grateful than these words can hold.

To my family and friends, and friends who have become family, all around the world, thank you for your unwavering support and for your grace with the time this work kept me away, and for the way you magnify every moment we spend together. I am deeply grateful for each of you. I look forward to summiting mountains with you again, and to reaching for the moon together, here and beyond.

Finally, to my dogs, whose namesake this project carries: thank you for keeping me steady and earthbound while I chased ideas far beyond it. Bruce and Cascade, you are the best boys.

Abstract

Background. NASA’s Human System Risk Board (HSRB) grades each active human system risk with a Level of Evidence (LoE) across mission profiles from low Earth orbit to Mars, informing crew-safety decisions. Whether the evidence in the packages tracks the LoE assigned had never been examined systematically.

Objective. Project CASCADE is the first systematic examination of whether documented evidence matches the assigned LoE, and a map of its risk-evidence network.

Methods. The study used a systematic, mixed-methods, cross-sectional computational design: a complete census of 93 approved risk summary packages (29 agency-prioritized risks, 32 likelihood-by-consequence posture sub-tables, 2020 through 2025). A dual-path pipeline paired structured table reading with natural language processing of text and OCR-recovered images. A reproducible, fixed-rule classifier sorted evidence into the eight pre-defined types, validated against a human-annotated gold standard at almost-perfect agreement (Cohen’s kappa 0.923). A pre-specified two-tier design tested alignment at the portfolio level and by mission across Design Reference Missions (DRMs); weak relationships were probed with equivalence tests, Bayes factors, a comparison against randomly rewired networks, and deductive thematic coding.

Results. At the portfolio level, evidence volume, the quality-weighted composite, and diversity did not predict a risk’s LoE: every test was non-significant and a Bayes factor near 0.19 gave moderate support for no link, stable across alternative weightings. By mission, the LoE weakened from low Earth orbit toward Mars as evidence breadth thinned, leaving the thinnest evidence where the operational margin is smallest. The network held meaningful communities of similar risks, yet cross-risk dependency appeared in fewer than half the packages: interconnected in structure but documented in isolation.

Conclusions. NASA’s LoE scoring reflects mission applicability and expert judgment more than the amount or diversity of evidence documented, and the evidence base is thinnest for exploration-class missions, where directly applicable human evidence does not yet exist. Beyond its pilot scope, the study delivers three durable products: a reproducible extraction and analysis pipeline, a dated evidence baseline, and a framework for flagging alignment-tension cases. The approach carries to any field that grades expert-reviewed evidence under limited data, making the evidence behind high-stakes risk governance transparent, comparable, and reproducible.

Table of Contents

Dedication	ii
Disclosure	iv
Preface.....	v
Acknowledgements.....	vi
Abstract	viii
Table of Contents	x
List of Tables	xv
List of Figures	xvi
List of Appendices	xvii
Background	1
Literature Review: The Hazards of Human Spaceflight and the Case for Risk Governance	1
The Human System Risk Board and NASA’s Risk Management Structure	2
Risk Custodians, Risk Teams, and the Value of Expert Consensus	4
The Risk Summary Package: Structure, Standardization, and Lifecycle	5
Risk Posture: The Likelihood by Consequence (LxC) Matrix and Risk Disposition	8
Design Reference Missions and the Mission Range	10
The LoE and the Eight-Type Evidence Hierarchy	10
Development and Assignment of the LoE	12
From Scores to Decisions: Risk Disposition, Mitigation, and Human Risk Buydown	14
The Unexamined Relationship Between Documented Evidence and LoE.....	16
Directed Acyclic Graphs, the DAG Narrative, and the Scope of Causal Analysis	18
The Value of Network Science and the Case for Computational Evidence Examination	19
Network Science as an Analytical Framework for Risk Architecture	21
Foundational Network Topologies and Their Metrics	22
Scale-Free Networks and the Identification of Hubs	22
Small-World Networks and the Propagation of Influence.....	23
Community Structure and Modularity	24
Triad Census and Network Motifs	25
Validating Network Structure	26
Thematic Analysis in the Epidemiological and Health Sciences.....	27
The Systematic Review Framework: PRISMA and STROBE	27
Natural Language Processing and How It Processes Text.....	28
Corpora, Lexicons, and Dictionary-Based Classification.....	31

Evaluation Against a Human Annotated Gold Standard	33
The Selection of the 2020 to 2025 Study Window	35
The Project CASCADE Contribution	36
Public Health Significance.....	39
The Importance of Spaceflight Health Risk Assessment.....	39
Project CASCADE as a Public Health Contribution	40
Transferability and Broader Public Health Impact	41
Research Question, Specific Aims, and Hypotheses	43
Overarching Research Question	43
Aim 1: Systematic Evidence Extraction and LoE Evaluation	44
Aim 2: Multilayer Network Topology Modeling	46
Aim 3: Thematic Synthesis and Recommendations	48
Methods.....	51
Study Design.....	51
Study Subjects and Units of Analysis	53
Sample Size and Statistical Power	55
Data Sources and Provenance	56
Corpus Construction and the PRISMA Selection Process.....	58
Structured Ingestion and LoE Validation	60
The Natural Language Processing Engine	62
The Evidence Type Lexicon	63
Mission and Analog Catalogs	64
Three Track Matching and Contextual Constraints	65
Deduplication, Image Handling, and Subject Role Disambiguation	69
Three Evidence Streams and Triangulation.....	71
The Master Dataset and the Quality Weighting Scheme	71
The Extraction Examination	73
Human-Annotated Gold Standard.....	76
The Gold-Standard Principle: Reliability Versus Validity	76
The Annotation Protocol and Its Basis	77
Implementation in the Analytical Notebook.....	79
How Classification Accuracy Propagates to the Findings	80
Reliability Benchmarks and Interpretation	81
Reproducibility and Computational Environment	87

Statistical Analysis Plan.....	88
The Two-Tier Design and the Primary Outcome	88
The Linking Framework and Variable Definitions.....	91
Statistical Tests	93
Mission-Stratified Primary Analyses	99
Network Analysis Procedures.....	105
Thematic Analysis Procedures.....	107
Mission-Impact Consequence Categories	108
Robustness and Validation.....	108
Bias and Confounding.....	109
Human Subjects Considerations	110
Operational Definitions of the Eight Evidence Types	111
Posture table Standardization and Data Schema.....	112
Results.....	113
Systematic Evidence Extraction and LoE Evaluation	113
The Documented Evidence Landscape	113
Portfolio-Level Alignment of Evidence and LoE	117
Mission-Stratified Behavior of Evidence and LoE	123
Longitudinal Trajectories of the Evidence Base	128
Multilayer Network Topology Modeling.....	130
Network Structure of the Risk-Evidence Portfolio	130
Network Centrality and Triad Structure	139
Complexity, Outliers, and Notable Risks	141
Documentation Completeness, Missingness, and Bias.....	148
Thematic Synthesis and Recommendations.....	149
Thematic Structure of the Documentation	149
Validation and Robustness Across the Aims	151
Linking the Results to the Research Question and Aims.....	153
Discussion	155
Limitations	161
Conclusion	167
References.....	169
Works Consulted.....	182
Appendix A. Risk Roster, Mission Taxonomy, Evidence Hierarchy, and Abbreviations.....	187

A1. The 29 Active Human System Risks	187
A2. Design Reference Mission Continuum	187
A3. The Eight-Type Evidence Hierarchy	188
A4. Abbreviations	188
Appendix B. Analytic Workflow and Standardized Risk Summary Package Template Sections	191
B1. Analytic Workflow Across the Three Aims	191
B2. Standardized Risk Summary Package Template Sections	193
Appendix C. Statistical and Network Formulas	196
Appendix D. Lexicon, Catalogs, and Extraction Provenance	198
D5. Thematic codebook (Aim 3). The deductive thematic analysis coded each package narrative against a fixed list of themes set in advance rather than allowed to emerge from the text, so that every package was read against the same defined categories. The themes were drawn from the structural features of the corpus, and the coding was applied programmatically using fixed, keyword-anchored rules, so identical language was credited the same way in every package. Table D5 states, in plain language, what each theme captures, how it was identified, and why it was included, together with the share of the 93 packages in which it appeared.....	201
The purpose of the coding is to connect how evidence is described and reasoned about in the narratives to the quantitative evidence and network findings, and in particular to test whether mechanistically related risks are documented in isolation or with explicit reference to one another. Two of the themes both concern evidence gaps but at different levels of specificity: the general evidence-gap theme captures any acknowledgment that the evidence is incomplete or uncertain, whereas the explicit (specific) evidence-gap theme captures cases in which the narrative names a particular missing study, dataset, or measurement. The first measures how often uncertainty is admitted at all; the second measures how often that uncertainty is tied to a concrete, addressable need.....	201
Table D5. Deductive thematic codebook: what each narrative theme captures, how it was identified, and its prevalence across the 93 packages.	202
Appendix E. Supplementary Figure Series	203
Appendix F. Reproducibility, Data, and Code Availability.....	218
F1. Interactive Dashboard and Notebook	218
F2. Interactive Application	220
F3. Data and Code Availability Statement	222
Appendix G. Mission-Stratified and Risk-Level Data Summary	223
G2. Risk-Level Evidence and LoE Summary	224
Appendix H. Study Characteristics (Epidemiological Summary)	226
Appendix I. Core Datasets and Data Manifest.....	228
I.1 Dataset manifest	228

I.2 Data dictionary: df_master_n93_longitudinal.csv (93×91).....	228
I.3 Data dictionary: cascade_master_consolidated.csv (513×32).....	229
I.4 Reproduced summary tables.....	229

List of Tables

Table	Title
Table 1	Documented evidence landscape across the corpus of 93 approved submissions.
Table 2	Prevalence of the eight evidence types across the 29 risks.
Table 3	Primary confirmatory tests of portfolio-level alignment (Tier 1; N = 27 scored risks; Bonferroni $\alpha = 0.0167$).
Table 4	Mean LoE across the Design Reference Mission continuum (operations outcome; lower is stronger).
Table 5	Prevalence of deductive themes across the 93 package narratives.
Table A1	The 29 active human system risks (32 posture sub-tables).
Table A2	The eight Design Reference Missions, ordered along the mission continuum.
Table A3	The eight evidence types, ordered from strongest to weakest, with the analytic weight applied.
Table A4	Abbreviations and acronyms used throughout the dissertation.
Table D1	Worked examples of the eight JSC-66705 evidence types: detection cues, an illustrative package sentence, and the classification decision for each.
Table D2	Composition of the classification lexicon and reference catalogs.
Table D3	Provenance of the 9,098 accepted evidence matches across the document structure.
Table D4	Structured LoE entry validation across the corpus.
Table D5	Deductive thematic codebook: what each narrative theme captures, how it was identified, and its prevalence across the 93 packages.
Table G1	Mission-stratified summary across the Design Reference Mission continuum (operations outcome).
Table G2	Risk-level evidence and LoE summary across the 29 active human system risks.
Table H1	Study population, variables, and analytic design (STROBE-aligned).
Table I1	Core analytic datasets, dimensions, and contents.
Table I2	Evidence-support rating by DRM (posture counts).
Table I3	Per-risk evidence type presence (1 = present). Columns I–VIII are the eight JSC-66705 evidence types.
Table I4	Per-risk count and list of evidence types present.
Table I5	Weighted evidence-diversity (w2) ranking by DRM $\times$ outcome (H1d Part B).
Table I6	Exploratory Bradford–Hill cue detection per risk.
Table I7	Results of pre-specified hypotheses to analytic approach, result, and preliminary finding

List of Figures

Figure	Title
Figure 1a	The Project CASCADE analytical pipeline and the architecture of the assembled corpus.
Figure 1b	The Project CASCADE natural language evidence-classification pipeline.
Figure 2	Distribution of documented evidence across the eight evidence types and the 29 risks.
Figure 3	Coverage and validity of the structured LoE entries across risks and Design Reference Missions.
Figure 4	Portfolio-level statistical summary of the primary confirmatory tests, effect sizes, confidence intervals, and the equivalence-testing result.
Figure 5	LoE and evidence breadth across the Design Reference Mission continuum.
Figure 6a	Evidence-profile similarity network of the portfolio, resolved to 32 sub-risk nodes, with four recovered communities.
Figure 6b	Small-world test of the evidence-similarity network against a random reference.
Figure 6c	Hub spotlights for the three most-connected risks in the similarity network.
Figure 7	Evidence type co-occurrence and conditional-dependency structure across the 93 packages.
Figure 8	Composite complexity ordering of the 29 risks.
Figure 9a	Multi-dimensional comparison of the most complex risk, Immune, and the simplest risk, Pharmacy.
Figure 9b	Evidence type profile comparison of Immune and Pharmacy.
Figure 10	Prevalence and frequency of the deductive themes across the corpus.
Figure B1	Conceptual analytic workflow linking the research question, the evidence corpus, the three specific aims, and the integrated outputs.
Figure D1	Decision logic of the evidence type classifier: how a cited evidence statement is routed to one of the eight JSC-66705 evidence types.
Figure E1	Data-flow hierarchy from the two ingestion paths to the consolidated master files.
Figure E2	Epidemiological study-characteristics table for the risk cohort.
Figure E3	Master-dataset structure and completeness examination.
Figure E4	Evidence type quality-weighting scheme.
Figure E5	Longitudinal evidence trajectory across successive approved submissions.
Figure E6	Disposition-evidence alignment across the portfolio.
Figure E7	Spearman correlation of documented evidence against LoE.
Figure E8	Bonferroni correction and multiple-test summary for the primary family.
Figure E9	Mission-stratified Tier-2 primary analysis across Design Reference Missions.
Figure E10	Triad census and topology metrics for the risk-evidence network.
Figure E11	Multilayer network visualization of risks and evidence types.
Figure E12	Network topology tests: small-world, scale-free, and community structure.
Figure E13	Portfolio heatmap of risks by evidence type.
Figure E14	Focused network analysis of quantified risk-evidence relationships.
Figure E15	Weight-scheme robustness of the evidence-to-LoE association across alternative weightings.

List of Appendices

Appendix A. Risk Roster, Mission Taxonomy, Evidence Hierarchy, and Abbreviations.

Appendix B. Standardized Risk Summary Package Template Sections.

Appendix C. Statistical and Network Formulas.

Appendix D. Lexicon, Catalogs, and Extraction Provenance.

Appendix E. Supplementary Figure Series.

Appendix F. Reproducibility, Data, and Code Availability.

Appendix G. Mission-Stratified and Risk-Level Data Summary.

Appendix H. Study Characteristics (Epidemiological Summary).

Appendix I. Core Datasets and Data Manifest.

Background

Literature Review: The Hazards of Human Spaceflight and the Case for Risk

Governance

The next era of human space exploration, including the Artemis lunar campaign, the Gateway orbital outpost, and eventual multi-year expeditions to Mars, will expose crews to five recognized hazards of spaceflight: space radiation, altered gravity fields, hostile and closed environments, distance from Earth, and isolation and confinement. Each hazard places distinct stresses on the human body, and together they are linked to more than 20 documented human health and performance risks (1). These hazards produce consequences along two related but separate dimensions that the National Aeronautics and Space Administration (NASA) tracks independently: acute effects that threaten crew health and performance during the mission itself, and long-term health effects that may not manifest until months or years after return to Earth (1, 2). Space radiation, for example, contributes both to in-mission central nervous system and behavioral effects and to long-term cancer and cardiovascular risk, while altered gravity contributes to in-mission sensorimotor impairment and to long-term bone loss (1, 2). The dual character of these consequences is fundamental to Project CASCADE, because the agency evaluates the supporting evidence and the resulting risk posture for in-mission operations and for long-term health as distinct assessments.

The consequences of evidentiary uncertainty are not constant across mission types. In low Earth orbit, the proximity of the International Space Station (ISS) to Earth permits resupply, evacuation, and continuous ground support, which narrows the operational margin lost when evidence is incomplete (3). Beyond low Earth orbit, that margin contracts sharply, because a crew

weeks or months from definitive medical care cannot rely on return as a contingency (2). As exploration campaigns extend to greater distances and longer durations, the demand on the evidence base that underwrites crew health decisions therefore intensifies, which establishes the central importance of how that evidence is gathered, documented, graded, and translated into formal risk decisions. The systematic evaluation of those documentation and grading practices is the subject of this dissertation.

The premise that a risk decision must rest on characterizable, scrutinized evidence has a long regulatory lineage. In the 1980 Benzene Case (Industrial Union Department, AFL-CIO v. American Petroleum Institute), the U.S. Supreme Court struck down a federal occupational limit on benzene and held that, before acting, a regulating agency must first show that a significant risk exists and, where feasible, quantify it (4). The decision established that, before regulating, an agency must first demonstrate that a significant risk exists and, where feasible, quantify it, thus, anchoring the principle that risk must be estimated and defended with evidence rather than asserted (4). The human system risk enterprise at NASA inherits both principles, that a posture must be anchored in characterizable evidence, and that the quality and quantity of that evidence are themselves objects of scrutiny, and Project CASCADE operationalizes this by testing, systematically, whether the documented evidence base is commensurate with the LoE values assigned across the human system risk portfolio.

The Human System Risk Board and NASA's Risk Management Structure

NASA occupies an uncommon position among federal entities in that it integrates regulatory authority, primary clinical care, and research oversight for its astronaut corps within a single operational model. This authority is codified under Title 14 of the Code of Federal Regulations (CFR), Part 1241, the regulation To Research, Evaluate, Assess, and Treat (TREAT) Astronauts,

which permits NASA to collect, analyze, and apply astronaut health data for biomedical research and medical monitoring before, during, and after missions (5). Under this regulation, NASA functions simultaneously as health care provider and research steward, an arrangement that enables longitudinal surveillance and risk characterization across the astronaut life span and that is operationalized through the agency's Health and Medical Technical Authority (HMTA) (6). The corresponding NASA Policy Directive (NPD) formalizes the integration of research with operational health standards by mandating continuous ethical oversight and systems-level risk mitigation planning (6). Protected astronaut health data collected through the Lifetime Surveillance of Astronaut Health program serve as a recurring input to these evaluations, which underscores the agency's distinctive obligation to treat, monitor, and study the same small population under conditions of extreme environmental exposure (5).

Within this structure, the Human System Risk Board (HSRB), referred to throughout this dissertation as the Board, is the body responsible for identifying, evaluating, and communicating the health and performance risks of human spaceflight (7, 8). NASA partitions spaceflight risk into three categories, namely human system risks, programmatic risks, and center or institutional risks, of which human system risks are those that arise directly from the five spaceflight hazards and on crew health and performance; these are the exclusive focus of this dissertation (7, 9). The Board does not operate on a fixed review schedule. Reassessment of a given risk is instead triggered by the arrival of new evidence, by changes in mission architecture, or by shifts in programmatic priority (7). This responsive model keeps the Board current with evolving mission demands, but it also introduces variability in how frequently individual risks are revisited and in how their supporting evidence is interpreted and recorded over time. That variability, and the documentation

practices that accompany it, are precisely what a systematic and reproducible examination of the evidence corpus is positioned to examine.

Risk Custodians, Risk Teams, and the Value of Expert Consensus

Responsibility for the scientific content of each human system risk rests with a risk custodian, a designated subject matter expert who owns the risk and maintains its documentation over time, supported by a risk team (3). The risk team is a multidisciplinary group that may draw together clinicians, physiologists, behavioral scientists, radiation specialists, epidemiologists, and operational flight surgeons, depending on the domain of the risk. This team is responsible for locating and appraising the available evidence, constructing the causal representation of the risk, proposing the risk posture, and authoring the formal record that the Board reviews (3). The composition of the team reflects the interdisciplinary nature of spaceflight risk, in which a single outcome may depend on environmental exposure, physiological mechanism, and operational context simultaneously, and in which no single discipline holds the complete picture (10).

The expert consensus generated by these custodian teams is not incidental to the process but is itself a recognized and consequential input to risk characterization. Where direct human spaceflight data are sparse, which is common for the conditions of greatest concern on exploration-class missions, the structured judgment of the custodian team supplies the interpretive bridge between heterogeneous and often indirect evidence and a defensible posture. Expert consensus is formally represented within the evidence hierarchy applied in scoring, and it shapes both the assignment of the LoE and the framing of the supporting rationale (7). This reliance on expert judgment confers a genuine strength, in that it allows domain expertise to integrate evidence that no mechanical rule could weigh, and it simultaneously introduces a source of variability and subjectivity that the agency has worked to constrain through standardization. The balance between

the value and the limitations of expert consensus is a recurring theme in the evaluation of the evidence base and is examined directly in the analyses that follow. More broadly, the risk posture is the agency's principal instrument for translating evidence and expert judgment into action: it expresses, for each mission profile, how likely and how consequential a risk is judged to be, and it is the posture, rather than the underlying evidence alone, that informs resource allocation, countermeasure prioritization, and flight-readiness decisions (7). The quality and consistency of the evidence standing behind that posture are therefore of direct consequence to the agency, which is the central concern of this study.

The Risk Summary Package: Structure, Standardization, and Lifecycle

The formal record that a custodian team produces for each risk is referred to in this study as a Risk Summary Package. The Board currently maintains a portfolio of 29 active human system risks (detailed in Appendix A), each represented by such a package that is submitted as a collective of data and insights into a PDF. Two of these risks used within Project CASCADE are decomposed by NASA into independently scored sub-risks with their own LoE scoring, with Non-Ionizing Radiation separated into laser and solar exposure and Toxic Exposure separated into high, medium, and low concentration tiers, so that the 29 risks are represented by 32 distinct risk-posture tables for the purpose of this dissertation. A Risk Summary Package is more than a literature summary; it is a structured decision document that records the causal logic of the risk, the evidence assembled in its support, the formal posture assigned across mission types, and the strategic disposition the agency has adopted (3). Read together across the combined risk summary packages as a portfolio, these packages constitute the documented evidence base for crew health risk governance and are the primary corpus analyzed in Project CASCADE.

To ensure that risks are documented consistently across custodian teams and across time, the Board requires that every Risk Summary Package follow a standardized template, a presentation-based format that specifies the sections each package must contain that was formally introduced between 2019-2020. The most recent effort to standardize the structure of the submitted material is part of the basis for the analysis window chosen here, since the standardized format took full effect and was deployed across the 2020 to 2025 period. The standardized Risk Summary Package template defines the expected structure of the record, including the directed acyclic graph that represents the causal architecture of the risk, the Likelihood by Consequence (LxC) matrix, the per-mission risk posture tables, the Overall Assessment of the Evidence, the State of Knowledge, the supporting metrics, the standards and requirements flow, and the high-value risk mitigation targets. The complete set of standardized template sections is provided in Appendix B. This standardization is the operational feature that makes a systematic computational review of the corpus feasible, because each package should be organized around the same defined sections, evidence can be located, extracted, and compared across risks in a reproducible manner rather than read characteristically from free-form documents.

Standardization, however, has been a process rather than a fixed state, and the degree to which packages conform to the template has itself evolved. Earlier records were more heterogeneous in their section structure and in the consistency of their rationale documentation, whereas more recent packages adhere more closely to the standardized format. As a consequence, the presence and completeness of individual template sections differ across the corpus, and the assessment of that variation is one of the study's analytic objectives. The structure of this dissertation follows the development of a corpus and natural language processing framework built to capture the information provided in the standardized structure; elements recorded outside those standardized

features may be under scrutinized, which is acknowledged as a limitation and a basis for future recommendations on data capture elements.

Each package also carries a lifecycle. A risk custodian team initiates a revision through a structured change process whenever new evidence emerges or a mission context changes, producing an updated record that supersedes the prior version while preserving the history of the risk through sequential revision labels such as baseline, Revision A, and Revision B. Because revisions accumulate in this way, a single risk is often represented by more than one approved package within the analysis window. All 29 risks carried at least one Board approved package between 2020 and 2025, and most carried several, with individual risks documented as many as four times; this is what yields the 93 packages analyzed here and supports the longitudinal view of evidence accrual reported in the Results. Every package is subjected to formal Board deliberation before it is approved, which is significant for interpretation because an approved package reflects the expert assessment of the Board at the time of approval, not only the documented evidence (7, 11). The longitudinal accumulation of these approved packages across successive submissions is what permits this study to examine both the cross-sectional state of the evidence base and its change over time.

The human spaceflight evidence that ultimately informs these judgments is curated and made accessible through the NASA Life Sciences Portal (NLSP), the Human Research Program resource that consolidates the Life Sciences Data Archive (LSDA) and the Lifetime Surveillance of Astronaut Health (LSAH) program and provides de-identified spaceflight and ground-analog research data reaching back to 1961 and the packages are managed and archived by the HSRB internal to the agency (3, 12). Access to NLSP or LSAH data is governed by a formal data request

process, reflecting the same integration of clinical care, research, and surveillance that characterizes the agency's risk governance more broadly.

Risk Posture: The Likelihood by Consequence (LxC) Matrix and Risk Disposition

Each human system risk is assigned a risk posture derived from the combination of its likelihood and its consequence, evaluated using a standardized 5x5 Likelihood by Consequence (LxC) matrix (7). Likelihood is scored on an ordinal scale from 1-5, in which a score of one denotes an event that is exceptionally unlikely to occur, a score of three denotes an event with moderate probability under specific conditions, and a score of five denotes an event that is almost certain to occur during typical mission activities (7). Consequence is scored on a parallel ordinal scale from 1-5, in which a score of one denotes negligible impact, a score of three denotes noticeable but temporary health effects, and a score of five denotes a catastrophic outcome that could result in permanent disability, loss of life, or mission failure (7). The matrix is applied separately across mission profiles and across the two impact categories introduced above, In Mission Operations (ops) and Long Term Health (lth), so that a single risk carries multiple posture assessments reflecting the distinct ways it manifests across mission types and time horizons.

Likelihood ↓ / Consequence →	C1	C2	C3	C4	C5
L5					
L4					
L3					
L2					
L1					

Likelihood by Consequence matrix. Likelihood (L1 lowest to L5 highest) is paired with consequence (C1 lowest to C5 highest); the shading shows the escalating acceptability bands the agency summarizes as green, yellow, and red. Each risk receives one such posture per mission profile and impact category.

Each posture is accompanied by a risk disposition, which expresses the agency's current strategic response and ranges from dispositions that require active mitigation to dispositions that accept the risk with continued monitoring (7, 9). Risks that lack sufficient data to support a full

matrix classification are designated as Concerns, a status that signals the need for further investigation without meeting the threshold for formal risk categorization (7). In internal tracking, dispositions are also summarized through a color classification of red, yellow, and green that communicates the perceived acceptability of a risk for spaceflight at a glance (7). The matrix position and the disposition describe how serious a risk is judged to be and how the agency intends to respond to it, but neither expresses how much confidence the underlying evidence warrants. That dimension is supplied by the LoE, described below, and it is the alignment between the strength of the evidence and these scoring outputs that this study evaluates.

The convention governing this product is consequential for interpretation. The risk custodian scores the most likely level at which a risk would manifest for a given Design Reference Mission (the standardized mission profile defined in the next section) and pairs it with the highest credible consequence should the risk be realized (7). This deliberately asymmetric rule typical likelihood, worst credible consequence is a conservative standardization that prevents low-probability, high-severity outcomes from being scored to negligible postures. Given that nearly every human system risk is internally heterogeneous, spanning multiple outcomes, mechanisms, and time courses, the Board assesses the risk as a single integrated package rather than scoring each sub-outcome separately. Spaceflight induced bone loss illustrates the trade-off: trabecular and cortical loss, site-specific decline, observed in risk packages for review, and differing recovery and fracture trajectories could each be scored differently in isolation, yet the risk receives one posture taken from a whole-risk vantage. This exchanges within-risk specificity for cross-risk comparability, which is precisely what renders a multi-risk portfolio mutually rankable under a single standard; a posture cell should therefore be read as a standardized summary judgment of the risk as a whole rather than a statement about any single sub-outcome.

Design Reference Missions and the Mission Range

Risk postures are not assigned in the abstract but are evaluated against Design Reference Missions (DRMs), the standardized mission profiles that NASA uses to represent the range of operational contexts for which crew health risk must be assessed (7). The Design Reference Missions span a continuum defined principally by mission duration and distance from Earth, ranging from short-duration low Earth orbit operations through lunar orbital and surface missions to long-duration Mars transit and planetary scenarios. Given that a single risk may behave very differently depending on how long a crew is exposed and how far they are from definitive care, the posture for one risk is not a single value but a profile across this mission continuum, evaluated separately for in-mission operations and for long-term health.

This continuum is consequential for the evidence base itself. The great majority of direct human spaceflight evidence derives from low Earth orbit operations aboard the ISS, whereas the longer and more distant missions, for which the operational margin is smallest, are precisely those for which directly applicable human evidence is sparsest, as for long-duration Mars transit and planetary operations, where no directly comparable human mission has yet been flown and the evidence rests largely on analog, animal, and modeling studies (3, 9). The applicability of a given body of evidence therefore varies systematically across the mission continuum, with the strongest evidentiary support concentrated at the low Earth orbit end and the thinnest support at the Mars end. How the documented evidence base behaves across this low-Earth-orbit to Mars gradient is a central element of the analysis and informs the study's principal structural finding.

The LoE and the Eight-Type Evidence Hierarchy

The LoE (LoE) is the graded judgment of the certainty associated with the causal relationship between a spaceflight hazard and an adverse human outcome provided by the risk custodian and

team assigned to the risk, recorded on a four-tier ordinal scale in which 1 denotes Strong evidence, 2 denotes Moderate evidence, 3 denotes Weak evidence, and 4 denotes Speculative evidence (7, 11). A lower numerical score therefore represents greater confidence in the underlying evidence. As with the posture and disposition, the LoE is unique for each Design Reference Mission and for each impact category, therefore assigned separately, producing a profile of certainty across the mission continuum that mirrors the resolution at which posture decisions are made. Together, the matrix position, the disposition, and the LoE constitute the formal scoring outputs recorded in every Risk Summary Package, and they are the means by which the agency operationalizes the evaluation of human system risk, and ultimately the basis for its progressive buydown, the structured reduction of a risk's likelihood and consequence as evidence accumulates and countermeasures are validated, a process examined in its own section later in this dissertation.

The assignment of a LoE draws on a structured hierarchy of evidence types that ranks sources by their directness and applicability to the human spaceflight context (7, 11). Eight evidence types are recognized, ordered from strongest to weakest. Type I, direct human spaceflight data, sits at the top because it is collected from astronauts in the operational environment itself. Type II, human ground-analog data, derives from terrestrial environments engineered to simulate aspects of spaceflight, such as bed-rest and isolation studies. Type III, terrestrial clinical and epidemiological data, draws on the broader human medical literature. Type IV, animal data, and Type V, cellular and tissue data, provide mechanistic evidence from non-human models. Type VI, computational and modeling data, contributes simulation-based inference. Type VII, expert opinion, represents the structured judgment of subject matter experts in the absence of stronger sources, and Type VIII, standards and requirements, reflects codified engineering and medical standards. The hierarchy treats Type I as the most applicable basis for spaceflight risk decisions and Type VII as

the weakest, with the remaining types ordered between them. The assignment of a LoE also incorporates a set of causal criteria adapted from the Bradford Hill considerations, of which the Board applies temporality, analogy, mechanism, reproducibility, specificity, and coherence, the subset best suited to the structure of spaceflight evidence (7, 13).

These eight types are defined operationally in the Methods, where each is described in full, and the distinct status of standards and requirements as the adjudicated synthesis of the other seven is taken up in the discussion of how a LoE is assigned (7).

Development and Assignment of the LoE

The LoE scale is not an informal label but a structured judgment whose assignment procedure is codified in the agency's governing Johnson Space Center risk management plan (7). The plan dedicates a formal guidance appendix to assigning the LoE, and that guidance is built explicitly on the causal considerations of Bradford Hill, whose criteria for distinguishing association from causation have been a foundation of epidemiologic causal reasoning since their introduction (13). The Board adapts the Hill considerations to the structure of spaceflight evidence, drawing in particular on temporality, biological plausibility and mechanism, analogy, coherence, reproducibility or consistency, and specificity, and it uses the degree to which a documented body of evidence satisfies these considerations to place a risk on the four-tier scale, so that a relationship supported by direct, temporally ordered, mechanistically coherent, and reproducible human evidence approaches a Strong rating while one resting on analogy or expert inference alone remains Weak or Speculative (7, 13). This grounding is what makes the scale a causal-evidentiary judgment rather than a simple count of citations, and it is the conceptual reason that documented evidence volume and assigned LoE need not move together, a possibility Project CASCADE examines directly.

The evidence that feeds this judgment is assembled and adjudicated by a defined team rather than by a single author. The agency assigns to each risk a Risk Custodian Team comprising an operational custodian and a research custodian, who together coordinate the evidence base from the operational and research perspectives, and, of particular relevance to an epidemiologic examination, an epidemiologist is assigned to each risk to support the gathering and analysis of the evidence and to ensure the quality of analyses drawn from astronaut surveillance data held in the Lifetime Surveillance of Astronaut Health program (7, 12). The team is expected to consult the standards organization and the agency biostatisticians during each update so that quantitative claims are reviewed and so that any new or revised standards are identified (3). This deliberative, multidisciplinary structure is important context for the study's central limitation, since the same process that assembles the evidence also assigns the score, and it situates the documented record as the product of expert adjudication rather than an independently compiled evidence ledger (3).

The documented record itself is standardized through formal templates, which is what makes a corpus-level computational examination feasible. When a risk is updated, the Risk Custodian Team works with the Risk Management Office to develop the information package in the agency's official Change Request kickoff template, and the resulting Risk Summary Package is recorded in a standardized format whose required fields, including the risk title and statement, the state of knowledge, the associated hazards, the contributing factors, the countermeasures, and the likelihood and consequence scores delineated by Design Reference Mission and impact category, are defined in the plan's risk summary template (3, 7). The five spaceflight hazards that originate every risk, namely altered gravity, ionizing radiation, isolation and confinement, the hostile closed environment, and the distance from Earth, are likewise defined in the governing plan and reproduced in the agency's evidence reporting, and they structure the taxonomy from which the

individual risks descend (7, 14, 15). Given that these templates fix where each kind of information appears and what it must contain, they impose the regularity that the dual-path extraction in this study exploits, and the standardized field structure is detailed in Appendix B.

A distinct position in this hierarchy is occupied by standards and requirements, which are not primary observational evidence but the adjudicated synthesis of the other evidence types. NASA's agency level human system standards are maintained by the Office of the Chief Health and Medical Officer, which assembles the prior evidence and experience base, develops or refines a defensible limit, and publishes it in the NASA Space Flight Human System Standard (16, 17). When a program is established, the applicable standards are levied as verifiable requirements through programmatic management and are dispositioned with data and rationale to certify the design and demonstrate flight readiness; once the program flies, crew health and performance data return to the standard owner to update the evidence base and, in turn, the standards themselves (18). A single standards and requirements citation therefore encodes a community vetted judgment rather than an individual study, which is why its evidentiary weight differs qualitatively from the primary data types beneath it (11, 19). In the eight-type hierarchy this is the reason standards and requirements, despite representing a vetted synthesis, are weighted below the primary observational types: their role is to codify established consensus rather than to supply new direct evidence, so they function as a derived rather than a primary source when the strength of a risk's evidence base is assessed.

From Scores to Decisions: Risk Disposition, Mitigation, and Human Risk Buydown

Human risk buydown is the central operational purpose of the entire risk management enterprise, and the scoring outputs recorded in a Risk Summary Package are the inputs that drive it. Risk buydown is the systematic process of reducing the uncertainty, severity, and likelihood of health and performance risks through evidence-based characterization, mitigation, and monitoring,

and it is a foundational objective of NASA's risk management framework, which holds that mission risks should be actively reduced to acceptable levels as data accumulate and countermeasures are validated (7, 20). The risk disposition is the formal expression of where each risk stands in that process, encoding a graded set of decisions that range from accepting a risk, to accepting it with monitoring, to accepting it with optimization, to requiring active mitigation, and to requiring mitigation with refinement of the governing standard (7). These dispositions are operationalized through the template sections that connect assessment to action, including the risk mitigation framework, the high-value risk mitigation targets, and the standards and requirements flow, which translate a risk's posture and LoE into countermeasure development priorities and into the medical and engineering standards that govern mission design (7).

The LoE is the confidence metric that gates these buydown decisions. Given that it expresses how much certainty the evidence warrants, it directly shapes which risks are judged to merit continued research investment and countermeasure development and which are accepted on the basis of existing evidence (9). The agency's research portfolio is aligned accordingly, with priorities set against the high-value risk mitigation targets agreed through the Board, so that the evidence judgments recorded in the packages influence the allocation of research resources and the guidance provided to program managers, mission planners, and agency leadership. Buydown is most consequential for long-duration and exploration-class missions, where medical evacuation is not possible and where incomplete or inconsistent evidence can compromise both crew safety and mission success (7, 9). In this setting the evidence base is the currency of buydown, and the LoE is the instrument through which that currency is converted into operational decisions. This decision-making role is what gives the relationship between documented evidence and assigned

LoE its practical weight, and it is the reason that relationship, previously unexamined across the portfolio, is the focus of the present study.

The scope of this research is deliberately bounded to the examination of whether the documented evidence aligns with the LoE scores and dispositions that underwrite risk buydown decisions, and whether the structural properties of the risk-evidence network reveal the portfolio's evidence-based buydown capacity, which is the dimension named in the study's central research question, stated in full later in this chapter. It does not design or validate countermeasures, set standards, or enact mitigation, all of which are operational activities that lie outside the defined scope and remain the province of the custodian teams and the Board. The broader implications of Project CASCADE's preliminary findings for risk buydown strategy, research prioritization, and resource allocation are developed in the Discussion, where they can be interpreted against the results rather than asserted in advance, which keeps the analysis aligned with the stated aims and research questions. This research explores the alignment but does not imply this research will direct operational strategy, rather provide the first portfolio-wide computational review of this evidence base, the study is exploratory in character: it is designed to characterize the documented evidence and to surface structural patterns and alignment tensions for expert review, rather than to deliver confirmatory tests of established hypotheses, and its findings are accordingly framed as a quantitative baseline and a transferable method rather than as definitive conclusions.

The Unexamined Relationship Between Documented Evidence and LoE

NASA's framework further recognizes a Quality of Evidence (QoE) consideration that weighs the methodological rigor and relevance of individual sources beyond their type alone (11). However, a formal appraisal of the methodological quality of each cited source lies outside the defined scope of the present study. Within the parameters of this analysis, the focus is restricted to

the relationship between the documented evidence base, characterized by its types, sources, and volume, and the LoE that the risk custodian assigns. Quality of Evidence is therefore acknowledged as a component of the agency's framework but is not itself assessed here, and it is noted as a dimension that future work could incorporate. In plain terms, this study measures what kinds of evidence a risk's documentation contains and how much of it, and whether that tracks the assigned score; it does not judge how methodologically strong each individual cited study is. A risk supported by many weak reports and a risk supported by a few rigorous studies are treated alike on the evidence-characterization measures used here. Whether the methodological quality of the underlying sources predicts the Level of Evidence is an important and distinct question that this analysis deliberately leaves open and recommends as a priority for future study.

Despite the central role that the LoE plays in NASA's risk decisions, the empirical relationship between the evidence documented in a Risk Summary Package and the LoE ultimately assigned to that risk has not previously been examined systematically across the risk portfolio. The absence of a prior systematic examination was confirmed through direct consultation with HSRB subject-matter experts during a preceding gap analysis. The framework specifies how evidence types are ordered and how scores are defined, but it does not establish, and no prior work has measured, whether risks supported by more, more diverse, or higher-ranked evidence in fact receive stronger LoE scores, nor whether the documented evidence aligns with the disposition decisions that follow. This absence is the gap that Project CASCADE addresses. By extracting and classifying the documented evidence systematically and relating it to the recorded scores across all active risks and mission profiles, the study provides the first empirical characterization of how the evidence base and the LoE correspond, establishing a baseline against which the consistency of the agency's evidence grading can be evaluated and future revisions assessed.

The reason this relationship warrants testing rather than assumption is that any system which grades evidence carries an intuitive expectation built into it: that a risk supported by more evidence, by more diverse evidence, or by more highly ranked evidence should, other things being equal, receive a stronger LoE, since that monotone logic is the implicit rationale for grading evidence at all (21). The expectation is seldom stated because it appears self-evident, yet whether it holds across a real risk portfolio is an empirical question that has not been answered. Project CASCADE treats this intuitive expectation as the hypothesis to be tested rather than a premise to be granted, so that a departure from it, including the absence of the expected association, becomes an informative result about how LoE scores are actually produced rather than a mere failure to find an effect.

Directed Acyclic Graphs, the DAG Narrative, and the Scope of Causal Analysis

Each Risk Summary Package represents the causal logic of its risk using a directed acyclic graph (DAG), a formal diagram in which nodes denote variables such as hazards, intermediate physiological mechanisms, and health outcomes, and in which directed edges denote the asserted direction of causal influence between them (22, 23). The term directed indicates that each edge points one way, encoding the direction of influence, and the term acyclic indicates that the graph contains no closed loops, so that no variable can be reached from itself by following the edges (23). These two properties make directed acyclic graphs a standard epidemiological tool for stating causal assumptions explicitly, identifying potential confounding, and reasoning about the variables on which a causal claim depends (23). Within a risk summary package, the DAG diagram and its accompanying narrative text make the custodian team's causal reasoning visible and reviewable. The full sequence by which a package is constructed, and an account of directed acyclic graphs as epidemiological causal-inference tools, are provided in Appendix B.

Among the standardized template features is the DAG Narrative, the prose section that accompanies the diagram and describes, in language, the contributing factors and causal pathways the diagram depicts. Given that the DAG Narrative renders the causal logic in text rather than in graph form, it is amenable to systematic text analysis, and Project CASCADE draws on it as one of the template features examined in the network and thematic components of the study, including the detection of instances in which one risk's narrative references another risk and thereby reveals cross-risk dependencies. A complementary line of inquiry, the quantitative analysis of the causal graph structure itself, meaning the nodes and directed edges of the diagram rather than its textual description, would add further value to the assessment of the portfolio. That analysis lies outside the scope of the present research, which is focused on the previously unexamined relationship between documented evidence and the LoE, and it is identified as a direction for future work that could extend the framework developed here (24).

The Value of Network Science and the Case for Computational Evidence

Examination

Two features of the corpus, the complete body of text assembled from the approved risk packages for computational analysis, shape the methodological approach that Project CASCADE adopts. The first is that the evidence supporting each risk is recorded in lengthy documents whose structure varies across risks and across time, though a standardized template was introduced across the 2020 to 2025 window, the complexity of the risks and earlier iterations of the submission process had permitted considerable variation in how material was reported, so that the same kind of evidence may appear in different sections, in different wording, and in different formats from one package to the next. Locating and comparing that evidence by manual reading across the full corpus would be neither reproducible nor scalable, and it would leave the criteria for what counts

as a given type of evidence implicit. Project CASCADE therefore applies natural language processing to extract and classify evidence systematically against the eight-type hierarchy, an approach that renders the examination reproducible, makes the classification criteria explicit, and allows the entire portfolio to be examined under a single consistent standard. The structuring of this heterogeneous documentary record into comparable, analyzable data is itself a central methodological contribution of the study, and the construction of the evidence corpus and classification lexicon is detailed in the Methods. While the ordinal eight-type ranking follows the agency's evidence hierarchy, the integer weights this study later attaches to those types to form a quality-weighted composite are an analytic device adopted for the present analysis; they are not prescribed by the governing risk management plan and are used here only as a heuristic.

The second feature is that the human system risks do not exist in isolation but share hazards, mechanisms, countermeasures, and mission-impact pathways, so that the portfolio is most accurately understood as a connected system rather than a list of independent entries. Network science provides the means to represent and analyze that system, quantifying the structural prominence of individual risks and evidence types, identifying clusters of risks that rest on similar evidence foundations and could benefit from coordinated review, and revealing connections that are not apparent when risks are read one at a time (25, 26, 27). Applied to the documented evidence base, these methods translate the agency's traditionally narrative assessments into reproducible, comparable, and interpretable structural measures. The specific network constructs, metrics, and topological frameworks used for this purpose, together with the formulas that define them, are developed in the section that follows.

Network Science as an Analytical Framework for Risk Architecture

Network science is the study of systems represented as nodes connected by edges, and of the structural properties that emerge from those connections (28). It has a long history in epidemiology and public health, where it has been used to model disease transmission, to identify influential actors within surveillance and organizational networks, and to characterize the structural determinants of population health (19, 29, 30). Its central contribution is the ability to move analysis beyond individual entities to the relationships among them, quantifying how influence, information, or risk is distributed and propagated across a system. This relational perspective is well matched to a risk portfolio in which the individual risks are not independent but share hazards, mechanisms, and evidence.

In Project CASCADE, network science is applied not to the internal causal diagram of a single risk but to the portfolio as a whole, by representing the human system risks and the evidence types that support them as a connected structure. Risks are linked to the evidence types documented in their packages, and risks are linked to one another through the evidence they share, through the overlap in the evidence types they cite, and through the similarity of their overall evidence profiles. Analyzed in this way, the network reveals features that are invisible when risks are read one at a time: the evidence types that function as common foundations across the portfolio, the risks that occupy structurally prominent or bridging positions, and the clusters of risks that rest on near-identical evidence bases and could therefore benefit from coordinated review. Two complementary traditions inform this approach, with social network analysis quantifying the relationships and prominence of specific nodes within a defined system and complex network analysis characterizing emergent, system-level properties such as clustering and modularity across the network as a whole (29, 31).

Network analysis offers methodological strengths that extend traditional risk evaluation beyond qualitative description. Rather than representing relationships only narratively, it applies mathematical metrics that quantify the strength and influence of each connection within a complex system, so that the prominence of an element, the efficiency of connection between elements, and the grouping of elements into clusters can all be expressed numerically and compared (19, 29). Metrics such as degree centrality identify the most connected elements of a system, and the resulting measures are reproducible and scalable in a way that narrative judgment is not (32). Applied to the human system risk portfolio, these methods quantify which evidence types are most broadly connected across risks, which risks occupy structurally prominent or bridging positions, and how risks group according to the evidence they share, converting a portfolio that is otherwise read one record at a time into a set of comparable structural measures. A further strength is adaptability. Given that the network is rebuilt from the documented evidence, new submissions can be incorporated and the structure re-evaluated without reconstructing the analysis from first principles, which supports a reproducible and updatable account of the portfolio over time.

Foundational Network Topologies and Their Metrics

Scale-Free Networks and the Identification of Hubs

A scale-free network is characterized by a power-law distribution of connectivity, in which a small number of highly connected nodes, or hubs, exert disproportionate influence while the majority of nodes maintain relatively few connections (33, 34). This pattern is widely observed in biological and sociotechnical systems, where a few elements concentrate influence. In the risk-evidence network, this topology directs attention to the evidence types that function as hubs because they are cited across many risks, such as direct human spaceflight data and expert consensus, and to the risks that draw on an unusually broad set of evidence types. The concept is

also useful for motivating why certain risks behave as cross-cutting contributors, since a risk shaped by a pervasive hazard such as radiation connects to multiple physiological domains and so occupies a structurally prominent position in the broader risk landscape. The prominence of a node is quantified principally through centrality. Degree centrality measures the number of direct connections a node maintains and, in normalized form, is expressed as

$$C_D(i) = k_i / (N - 1),$$

where k_i is the number of connections of node i and N is the total number of nodes, yielding a value between zero and one in which higher values indicate greater immediate connectivity (35). As a descriptive convention in network science, nodes with normalized centrality above roughly 0.80 are treated as highly influential and those below roughly 0.30 as structurally peripheral (36). Degree centrality captures direct connectivity, but it does not distinguish a node that merely has many connections from one that links otherwise separate parts of a network. Betweenness centrality addresses this by measuring the extent to which a node lies on the shortest paths between other nodes, identifying bridges and connectors, and is defined as

$$C_B(i) = \sum \sigma_st(i) / \sigma_st,$$

summed over all node pairs s and t , where σ_st is the number of shortest paths between s and t and $\sigma_st(i)$ is the number of those paths passing through i (35). In the risk-evidence network, a high betweenness value identifies an evidence type or a risk that connects regions of the portfolio that would otherwise be only weakly linked.

Small-World Networks and the Propagation of Influence

Small-world networks are distinguished by two properties, high local clustering and short average path lengths, which together allow influence to propagate efficiently across a system

despite dense local grouping (37, 38). The degree of local interconnection is quantified by the clustering coefficient, defined for a node as

$$C_i = 2E_i / (k_i (k_i - 1)),$$

where E_i is the number of connections among the neighbors of node i and k_i is the number of its direct neighbors. A high clustering coefficient indicates that a node's neighbors are themselves densely interconnected. The concept is well illustrated by behavioral health risks, where isolation-related stress, sleep disturbance, and cognitive decrement share underlying mechanisms and therefore form tightly clustered groupings, a pattern that recognizing as a cluster allows interventions to address several related concerns at once (39, 40, 41). The efficiency with which influence can traverse the network is captured by the average geodesic distance,

$$L = [1 / (N(N - 1))] \sum d_{ij},$$

summed over all node pairs, where d_{ij} is the shortest path length between nodes i and j and N is the total number of nodes. Short average distances indicate strong connectivity and the potential for rapid propagation across the system, whereas long distances reveal weak integration or structural bottlenecks. Interpreted for the risk-evidence network, short geodesic distances indicate that evidence types are broadly shared, so that risks are closely linked to one another through common evidence, while longer distances would indicate a more fragmented risk portfolio in which risks rest on more distinct evidentiary bases.

Community Structure and Modularity

Community structure refers to the tendency of nodes to form densely connected (42) clusters that are only sparsely linked to the remainder of the network (36). In the risk-evidence network, communities correspond to groups of risks that share common evidence foundations and may

represent functional subsystems amenable to coordinated review or bundled countermeasure development. The closed spacecraft environment provides a conceptual example, in that altered immune function, microbial shifts, and nutritional deficits are often reviewed independently yet may form a cohesive module whose recognition would justify integrated mitigation (43, 44). The strength of community structure is quantified by modularity, computed as

$$Q = (1 / 2m) \sum_{ij} [A_{ij} - (k_i k_j) / (2m)] \delta(c_i, c_j),$$

where A_{ij} is the adjacency matrix, k_i and k_j are the degrees of nodes i and j , m is the total number of edges, and $\delta(c_i, c_j)$ equals one when nodes i and j belong to the same community and zero otherwise, with higher values indicating more sharply defined communities (45). Detecting these communities requires a measure of how similar two risks' evidence profiles are. Cosine similarity provides this by treating each risk's profile of evidence types as a vector and computing

$$\cos(\theta) = (A \cdot B) / (|A| |B|),$$

which ranges from zero for risks with no shared evidence types to one for risks with identical profiles. Constructing an evidence-similarity network from these values, and then applying modularity-based community detection to it, isolates groups of risks with near-identical evidence bases, which is the basis on which the study identifies clusters of risks where coordinated review could improve portfolio efficiency. The strength of the resulting community structure is then assessed against a null model, as described below.

Triad Census and Network Motifs

Beyond node-level and community-level measures, the local architecture of the network is examined through a triad census, an enumeration of the configurations formed by sets of three nodes (35, 46). Triads are the smallest structures capable of expressing relational patterns beyond

a simple pair, and their frequencies characterize the local interaction tendencies of a network, revealing redundancy, closure, and consistency in how nodes associate (46). Originating in social network theory, triad analysis complements the global metrics by surfacing micro-level patterns that aggregate into the larger architecture, and configurations that recur more often than expected can signal meaningful interdependencies, as in the well-documented interrelation of stress, immune suppression, and sleep disruption (47, 48). In the risk-evidence network examined here, the triad census characterizes how risks and evidence types co-associate at the local level, for example whether particular evidence types tend to appear together across risks or whether certain risks consistently share their evidence neighborhoods. Read alongside centrality and modularity, the triad census provides a more granular account of the structural patterns that distinguish densely connected regions of the portfolio from sparsely linked ones.

Validating Network Structure

A central methodological concern in network analysis is distinguishing genuine structure from patterns that would arise by chance given a network's basic connectivity, and observed community structure in particular must be evaluated against an appropriate null model rather than accepted at face value (45). Project CASCADE tests the modularity of the evidence-similarity network against a configuration-model null, in which the network is repeatedly randomized while preserving each node's degree, generating a distribution of modularity values expected from random networks of the same connectivity. Observed modularity that exceeds this distribution supports the conclusion that the communities reflect real organization in the portfolio rather than artifacts of node connectivity. The relationships between the documented evidence and the LoE scores are evaluated using non-parametric statistical methods suited to ordinal data and to the

complete-census structure of the portfolio. The specific tests, the two-tier design, and the corrections applied for multiple comparisons are developed in the Methods.

Thematic Analysis in the Epidemiological and Health Sciences

Quantitative structural analysis characterizes how evidence is distributed and connected, but it does not by itself capture how evidence is described, framed, and reasoned about in the risk summary package. Thematic analysis provides the complementary qualitative method for this purpose. It is a systematic approach to identifying, organizing, and interpreting patterns of meaning, or themes (49, 50), within textual data, and it is applied in two principal modes: an inductive mode in which themes are derived from the data, and a deductive or directed mode in which a predefined framework of themes guides the coding (51). Within the epidemiological and health sciences, thematic and framework methods are increasingly used to analyze qualitative health data, institutional documents, and the rationale underlying decisions, and they align with the emphasis in implementation science on understanding how evidence is communicated and applied, not only what the evidence shows (30, 51). In this study, thematic analysis is applied in the deductive mode to the narratives of the Risk Summary Packages, using a codebook of themes drawn from the structural features of the corpus itself. Coding the narratives against these themes connects the qualitative documentation practices to the quantitative evidence and network findings, and it is the mechanism by which the synthesis aim integrates the results of the evidence analysis and the network analysis into a coherent interpretation.

The Systematic Review Framework: PRISMA and STROBE

The assembly and analysis of the evidence corpus are conducted as a systematic review and are reported against two established standards so that the methodology is transparent and

defensible to readers from outside the field. The Preferred Reporting Items for Systematic Reviews and Meta-Analyses, or PRISMA, governs what entered the analysis and how (52). Its four stages map directly onto the construction of the corpus: identification of every approved Risk Summary Package submitted across the study window, screening for approval status and conformance to the standardized template, determination of eligibility based on active risks carrying an approved posture, and the resulting included set of 93 packages spanning the 29 risks (53). Conducting the corpus assembly under the PRISMA structure ensures that the basis for inclusion and exclusion is explicit and reproducible rather than ad hoc. The Strengthening the Reporting of Observational Studies in Epidemiology standard, or STROBE, governs how the observational findings are reported (54). Framed epidemiologically, the 29 risks constitute the cohort, the eight evidence types are the exposures, and the validated LoE scores are the outcomes, with the descriptive characteristics of the cohort presented in an epidemiologic study-characteristics table and the analytic results reported in the structure that STROBE prescribes. Together the two standards render the design legible as a systematic review of observational evidence, with PRISMA governing the corpus at the front of the analysis and STROBE governing the findings at the back.

Natural Language Processing and How It Processes Text

Natural language processing (NLP) is the subfield of computer science and computational linguistics concerned with enabling computers to process, analyze, and derive structured information from human-language text (55, 56). Its necessity arises from a basic property of scientific and institutional records: the majority of the information they contain is expressed in unstructured free text, which methods designed for structured tabular data cannot analyze directly. NLP addresses this by transforming free text into structured representations that can be counted, classified, and compared, and it has become an established methodology in biomedical and health-

informatics research precisely because so much clinically and scientifically relevant information exists only in narrative form (55). The Risk Summary Package corpus is an instance of this general situation, in that the evidence supporting each risk is recorded predominantly in narrative prose rather than in structured fields.

An NLP system processes text through an ordered sequence of stages. The text is first acquired and ingested from its source documents; it is then preprocessed, which includes segmenting the document into analyzable units such as sections, sentences, and individual tokens through tokenization, and normalizing those units into a consistent form; and it is then subjected to the analytic task itself, which in a classification application means assigning spans of text to predefined categories on the basis of the terms they contain and the context in which those terms appear (55, 56, 57) NLP methods fall broadly into two families. Rule-based, or knowledge-based, systems apply human-curated dictionaries, patterns, and contextual rules, and their decision logic is explicit and inspectable (58). Statistical and machine-learning systems instead learn their decision criteria from annotated training data (59), which can extend coverage but renders the basis of any individual decision opaque (55). Project CASCADE employs a rule-based, lexicon-driven NLP approach, in which a curated dictionary of terms is matched against the segmented text under explicit contextual constraints. This approach was selected because the transparency and reproducibility of a rule-based system are essential to an examination of a governance corpus, where every classification must be traceable to the specific text or phrase and logical rule that produced it, and where the criteria must remain fixed and inspectable rather than learned and hidden. The processing stages summarized here, segmentation, contextual term matching, and classification against the eight-type scheme, are specified in operational detail in the Data Collection subsection of the Methods. In plain terms, for Project CASCADE this means the engine

first breaks each package into sentences and individual words, a step called segmentation; it then searches for evidence type keywords only where they occur alongside language signaling that evidence is being cited, which is what contextual term matching refers to; and it finally records each confirmed match under one of the eight evidence types, the classification step. A phrase such as “bed-rest analog study,” for instance, is counted as ground-analog human evidence only when the surrounding sentence shows it is being offered in support of the risk, not merely mentioned in passing.

The techniques assembled here are individually well established; the contribution lies in transferring and combining them for a governance-evidence corpus not previously examined computationally. Optical character recognition (OCR) is a mature method for recovering machine-readable text from scanned and image-based documents (60). Rule-based, lexicon-driven extraction of domain concepts from unstructured narrative underlies widely used clinical natural language systems such as cTAKES, which map free text onto controlled vocabularies in the same manner the present engine maps package narratives onto the eight evidence types (57). The application of NLP to safety-critical narrative reports is itself established practice: large repositories of aviation safety and incident reports, written, like risk summary packages, in acronym-dense, semi-structured prose, have been mined to classify occurrences and to surface latent risk and causal factors (61). Project CASCADE adapts this toolkit to NASA human system risk documentation, a domain in which it has not previously been applied. The evidence type categories and the vocabulary that defines them are not devised for this study but are drawn from the agency’s own JSC-66705 evidence framework, so that the keywords and the meaning of each evidence type are predetermined rather than invented here and are only mapped onto the documents by the engine. The contribution of Project CASCADE is that engine, which the author

designed and implemented in full, from the evidence-linking phrase logic and the canonical keyword dictionary through the multi-track extraction and deduplication pipeline. Because every classification rule is explicit and the lexicon is fixed and documented, the procedure is fully reproducible and can be transferred to other standardized governance corpora by substituting the domain vocabulary while retaining the same architecture.

Corpora, Lexicons, and Dictionary-Based Classification

The text that an NLP system analyzes is its ‘corpus’, a structured and bounded collection of documents assembled for systematic study, a concept foundational to corpus linguistics in which the corpus is the defined empirical basis from which all patterns are derived (62). The essential property of a corpus is that its composition is explicit, so that what it contains and how it was selected are stated in advance, which is what makes any analysis built upon it reproducible and what bounds the claims that can legitimately be drawn from it. In this study the corpus is the set of approved Risk Summary Packages assembled under the PRISMA framework described in the Methods, and its explicit definition is the foundation of the extraction’s reproducibility. A lexicon, in this context, is a curated set of terms organized into the categories that an analysis seeks to detect, and lexicon-based, or dictionary-based, classification applies that curated vocabulary to the corpus to assign text to predefined categories (63, 64). The defining property of the approach is that both the categories and the terms that signal them are specified in advance and remain inspectable, distinguishing it from machine-learning classification and yielding transparency at the cost of requiring careful curation and validation of the vocabulary. This is the operational form of directed content analysis, in which a predefined coding scheme, here the eight evidence types, is applied systematically and reproducibly across a defined corpus rather than allowing categories to emerge from the data (65).

Constructing such a lexicon involves more than listing keywords. A workable evidence lexicon pairs each category with two complementary components: the surface terms that name an evidence type, and the contextual or attribution phrases that establish whether those terms are actually being offered as evidence rather than merely mentioned. This is the same distinction that clinical natural language systems draw when they separate a concept from its assertion status, as in negation and context detection, where the algorithm decides not only that a finding is named but whether it is affirmed, negated, or hypothetical (66). In Project CASCADE the engine applies the analogous logic to evidence: a token such as “rodent” or “bed rest” is recorded under its evidence type only when it co-occurs with an evidence-linking phrase, a curated set of expressions such as “studies have shown,” “data from,” and “evidence indicates,” that mark the surrounding text as an evidentiary claim. This attribution layer is what prevents incidental mentions, method descriptions, and background prose from being miscounted as documented evidence, and it is the reason the lexicon is built as paired term and context rules rather than a flat word list.

This design follows established practice in health sciences text mining, where dictionary-based extraction has been used to populate disease registries, identify cases for epidemiological study, and surface findings that structured codes alone miss. Systematic reviews of these applications report that adding information extracted from free text materially improves case detection over coded data alone (67), and that lexicon- and rule-based information-extraction pipelines remain widely used in clinical research precisely because their decision criteria are transparent and auditable (68). Project CASCADE adapts the same corpus and lexicon architecture to a governance evidence setting: the corpus is the bounded set of approved Risk Summary Packages, the lexicon is the eight type evidence vocabulary paired with its attribution phrases, and the output is a reproducible, auditable mapping from narrative text to evidence categories that can

be ported to other standardized documentation corpora by substituting the domain vocabulary while retaining the same architecture.

Evaluation Against a Human Annotated Gold Standard

A defining methodological standard of NLP is that the output of an automated system is not assumed to be correct but is evaluated against a human annotated reference, termed the gold standard, because the accuracy of an automated classifier can be established only by comparison with expert human judgment (55, 69). Constructing a gold standard, in which human experts annotate a sample of the corpus according to the same scheme the system applies, and then measuring how closely the system's output agrees with those human annotations, is the standard paradigm for validating NLP systems in biomedical and information retrieval research, and it is what distinguishes a defensible automated analysis from an unverified one (69). The agreement between system and human annotations is quantified using established reliability metrics. The Cohen kappa coefficient measures categorical agreement corrected for the agreement expected by chance and is the conventional metric for classification reliability, interpreted against standard thresholds in which higher values denote stronger agreement (70, 71, 72). For annotation tasks in which the number of true negative cases is not well defined, as is common when marking spans of text, the F-measure, equivalent to positive specific agreement, is the appropriate reliability statistic, and the foundations of inter-coder agreement for computational linguistic tasks are established in the methodological literature of the field (69, 73). Selecting a chance corrected agreement coefficient rather than simple percent agreement is deliberate and is the recognized standard.

The annotation protocol was itself developed from a review of validation and annotation practices reported in prior NLP and content analysis work, so that the sampling design, the binary correct or incorrect judgment, and the treatment of disagreements followed established gold-

standard methodology rather than an ad hoc scheme (64, 65). In Project CASCADE the gold-standard sample comprised 287 classifications drawn by stratified sampling across the eight evidence types, a design chosen so that the sparsely represented types were evaluated in their own right rather than swamped by the dominant human spaceflight category. Agreement between the engine and the human reviewer reached a Cohen kappa of 0.923, with a 95 percent confidence interval of 0.879 to 0.968 (the agreement of the classifier with the single expert annotator, an accuracy benchmark against the gold standard rather than a between-rater reliability estimate). On the conventional Landis and Koch interpretation, in which values from 0.61 to 0.80 denote substantial agreement and values from 0.81 to 1.00 denote almost perfect agreement (71, 72), this places the engine in the almost perfect band and at the upper end of the reliability reported for validated dictionary and rule-based extraction systems in the clinical and epidemiological literature (67, 68). The complete validation design and the per-type agreement results are reported in the Results.

A growing body of work has begun to explore using large language models (LLMs) themselves as automated judges, in place of human annotators, to evaluate the outputs of other automated systems, an approach often termed LLM as a judge that can lower the cost and increase the speed of evaluation (74). That approach is deliberately not adopted here. The validation in this study rests on a single domain expert rather than a model-based judge for reasons specific to a governance corpus. A language model used as a judge can introduce errors of its own, including hallucinated justifications, statements that are fluent and plausible but not grounded in the text, and the biases inherited from its training data, such as the position, verbosity, and self-enhancement effects that a model cannot reliably detect in itself (74). Holding the gold standard to human expert judgment keeps the reference standard independent of the kind of system being

evaluated and anchors every accepted classification in a person's reading of the specific passage, which is the practice recommended for constructing reliable annotated references in computational linguistic work (69, 73). Model assisted annotation is a promising direction for scaling validation of this kind in future work, but it falls outside the scope of the present analysis, whose claims depend on a reference standard that is transparent, independent of the system under test, and free of that system's characteristic failure modes.

The Selection of the 2020 to 2025 Study Window

The analysis is bounded to risk documentation approved between 2020 and 2025, a window selected because it represents the period in which the Human System Risk Board's documentation and LoE scoring practices became substantially standardized. This interval followed the agency's redesign of its evidence grading framework and coincided with intensified planning for the Artemis lunar campaign and subsequent exploration missions, a period of heightened activity in risk classification updates, revisions to the scoring system, and decisions about which risks apply to which Design Reference Missions (7, 11). Concentrating the analysis on this window aligns it with the era in which the standardized Risk Summary Package template was consistently applied, which is the condition that makes systematic, comparable extraction across the portfolio feasible. Earlier periods were characterized by more heterogeneous formatting, fewer updates, and less consistent documentation of the rationale underlying risk decisions, and their inclusion would have introduced variability attributable to documentation practice rather than to the evidence itself. Restricting the corpus to this window therefore reduces a source of selection bias by emphasizing an interval of consistent documentation practice and scoring logic across the active risk portfolio.

The window also has a longitudinal character that the analysis exploits: across these years, most risks were revised and resubmitted one or more times, so that the corpus comprises 93

approved package submissions distributed across the 29 active risks. This repeated measures structure permits examination not only of the cross-sectional state of the evidence base at a point in time but also of how documentation and scoring evolved across successive approved submissions, which is a dimension of the evidence architecture that a single time point analysis could not address, thus, increasing evidence and the overall number of observations available for analysis. The full set of variables, their definitions, and their measurement levels is catalogued in the epidemiological study characteristics table in Appendix H (Table H1).

Accordingly, every result reported here reflects the evidence as documented in the Risk Summary Packages approved within the 2020 to 2025 window, and not the entirety of the evidence base that may exist for each human system risk. Earlier package revisions and predecessor documentation approved before this window may contain additional citations and supporting material that contribute to a risk's evidence base but that fell outside the scope of the present corpus; characterizing that earlier material would be a valuable extension and the results produced through this analysis are then treated here as exploratory rather than confirmatory, because the corpus captures the standardized modern record rather than the complete historical evidence base, and the analysis is therefore positioned to characterize the documented evidence and generate hypotheses rather than to adjudicate them. The findings should therefore be read as a faithful account of how evidence was represented and scored in the standardized modern risk summary package, rather than as a census of all evidence ever assembled for these risks.

The Project CASCADE Contribution

Project CASCADE constitutes the first systematic computational evaluation of the evidence architecture underlying NASA's human system risk decisions. It integrates four components within a single reproducible analytical pipeline: the systematic extraction and classification of

documented evidence against the published eight evidence type hierarchy, the statistical evaluation of how that evidence relates to the assigned LoE scores, the multilayer network modeling of the relationships among risks, evidence types, and Design Reference Missions (75, 76), and the thematic analysis of how evidence is documented across the portfolio (77, 78) By bringing these components together over the complete census of active risks and their approved submissions, the study addresses the previously unexamined relationship between the documented evidence and the scoring it is meant to support, and it characterizes the structural properties of the risk evidence network that bear on the portfolio's evidence based buydown capacity, which are the questions named in the research question and pursued through the specific aims.

The contribution is both substantive and methodological. Substantively, the study establishes an empirical baseline of how the evidence base and the LoE correspond across the portfolio, against which the consistency of the agency's evidence grading can be evaluated and future revisions assessed. Methodologically, the framework it develops, namely systematic evidence extraction, classification against a published evidence hierarchy, network synthesis, and thematic integration, is portable beyond the spaceflight context to any domain that grades expert reviewed evidence, including drug-safety review, occupational-health standard-setting, and pandemic-preparedness assessment, as well as other high-consequence operational domains in which human risk must be managed from incomplete evidence, such as military operational medicine and deep-sea or hyperbaric work, analogs developed further in the Discussion. The study is positioned as a pilot-scale, exploratory analysis of a novel governance corpus, and its findings are framed accordingly, as a reproducible characterization of documented evidence patterns intended to inform the refinement of risk documentation and evidence grading practice rather than to prescribe operational decisions, which remain the province of the risk custodian teams and the Board.

Project CASCADE is best understood as an exploratory, hypothesis generating study rather than a confirmatory one. Its central contribution is a reproducible analytic framework, applied here to a defined corpus of human system risk evidence, and the findings it produces are preliminary and, in several places, secondary to the primary record from which they are derived. The study is deliberately scoped as a pilot: its purpose is to demonstrate that the evidence to decision relationship across the risk portfolio can be made explicit, measurable, and reproducible, and to establish whether that approach yields interpretable signal worth pursuing at greater depth. Several lines of inquiry that fall outside this scope, among them finer-grained examinations of data provenance and collection fidelity, formal causal modeling of inter-risk dependencies, and the reconstruction of individual adjudication decisions, were considered in framing the analysis and are noted throughout as directions for future work rather than as claims of the present study. Where the data could not support a confirmatory conclusion, that boundary is stated explicitly, and the limitations of working from documented evidence are acknowledged rather than minimized. Read in this light, the value of Project CASCADE lies not only in the granular, portfolio wide picture it produces of how evidence is documented and scored, a level of detail not previously available, but equally in its demonstration of a transferable method whose assumptions are transparent, whose steps are repeatable, and whose outputs can be extended, challenged, or refined as the underlying risk landscape evolves.

Public Health Significance

The Importance of Spaceflight Health Risk Assessment

The future of human spaceflight, including the Artemis lunar program, the Gateway orbital station, and eventual multi-year expeditions to Mars, will expose astronauts to the five recognized hazards of spaceflight and to the more than 20 human health and performance risks linked to them (79, 80). As missions extend toward sustained lunar operations and interplanetary transit, the demands placed on crew health risk architecture grow well beyond what low Earth orbit operations required, because the margin for evidentiary uncertainty narrows as the distance from definitive medical care increases (3). Crews who are days or months from return require risk decisions built on a transparent and defensible evidence foundation, which makes the quality and consistency of the agency's evidence evaluation a determinant of crew safety rather than a documentary formality. The Human System Risk Board manages the systematic assessment of these risks through evidence review, causal diagramming, and LoE scoring, and its decisions directly determine which risks receive research investment and countermeasure development and which are accepted on the basis of existing evidence (7, 9). Given that those decisions allocate finite resources and shape mission design, the reliability of the evidence judgments that underlie them is a matter of public health consequence for the astronaut population, which constitutes a small, highly screened occupational cohort whose longterm health surveillance raises the same methodological challenges studied in occupational epidemiology, where exposure → outcome inferences must frequently be drawn from small samples and incomplete longitudinal evidence. A systematic evaluation of how the documented evidence relates to the assigned scores therefore addresses a question of direct importance to the protection of crews across all future human spaceflight endeavors, and even reflected in workers and communities in extreme environments.

Project CASCADE as a Public Health Contribution

From a public health standpoint, the central value of Project CASCADE is to make visible, across the NASA's entire risk portfolio and under a single consistent standard, where the evidence is strong, where it is thin, and where documentation practices diverge from the scores they support. The evidence supporting each risk is necessarily heterogeneous. Given that each risk reflects a distinct physiological system and a distinct set of spaceflight exposures, no single documentation format can render the evidence directly comparable across risks, and the kinds of studies that can be brought to bear, from terrestrial clinical data to spaceflight specific observation, differ from one risk to the next according to what the demanding and largely irreproducible spaceflight environment allows. This visibility is not available from reading risk records individually, and it converts a body of expert reviewed but heterogeneously documented evidence into a structured, analyzable form from which portfolio level patterns can be discerned. In doing so, the study establishes an empirical baseline against which future revisions to the evidence base can be measured.

The analysis is also positioned to locate the evidence gaps that matter most for exploration. Given that the directly applicable human evidence is concentrated in low Earth orbit operations and thins systematically toward Mars missions, a portfolio level characterization can identify precisely where the evidentiary foundation is weakest relative to the missions being planned, and whether the documented evidence volume and diversity track the LoE scores those risks receive. Identifying those gaps supports more informed allocation of risk reduction research, including by the specific type of evidence that is thin, since the eight evidence types examined here are not an arbitrary count but the source categories the agency's JSC risk management plan uses to characterize the evidence behind each score, spanning direct human spaceflight data, ground

analog evidence, terrestrial clinical evidence, animal evidence, cellular and molecular evidence, computational and modeling evidence, expert opinion, and standards and requirements more transparent communication between researchers and mission planners, and ultimately better protected crews, which are the public health ends that the agency's risk architecture is intended to serve.

For the scientific and operational relevance, this study contributes a reproducible methodology for evaluating expert graded evidence at the level of an entire risk portfolio, combining systematic natural language extraction, classification against a published evidence hierarchy, network analysis, and thematic synthesis. This approach complements traditional epidemiologic reasoning with a systems level account of how evidence is produced, weighted, and organized, and it does so in a form that is transparent and repeatable rather than dependent on the judgment of any single reviewer. The methodology responds to a relative scarcity of approaches that bridge qualitative context and quantitative precision in risk evaluation under uncertainty, particularly at the scale of an entire risk portfolio. By characterizing where documentation practices align with the assigned scores and where they diverge, the analysis identifies opportunities to refine the evidence collection and documentation processes that feed the Board's decisions. These contributions are framed as informative to practice rather than prescriptive of it, consistent with the bounded scope of the study, and their operational implications are developed against the results in the Discussion.

Transferability and Broader Public Health Impact

NASA maintains one of the most comprehensive and publicly accessible bodies of evidence on space related health risks of any space agency, and it functions as a global reference for engineering risk architecture in human spaceflight (9, 81). As international and commercial

spaceflight programs expand crewed operations, the methodological frameworks and evidence practices established by the agency are positioned to serve as a model for how the broader spaceflight community assesses, communicates, and mitigates human risk. A reproducible method for examining the evidence base contributes directly to that role. Beyond spaceflight, the framework developed in this study is transferable to other evidence-based risk governance systems that grade expert-reviewed evidence under conditions of limited data, including, but not limited to, drug-safety review, occupational-health standard-setting, and pandemic-preparedness assessment. The common structure across these domains, in which heterogeneous evidence is appraised, graded, and translated into decisions, is precisely the structure that the methods address. The broader public health impact of Project CASCADE therefore lies as much in its demonstration of a portable analytic approach to evidence examination as in its specific findings about the spaceflight risk portfolio.

Given that the analytic pipeline is fully reproducible, a fixed corpus, a deterministic and gold-standard-validated evidence lexicon, pre-specified tests with multiplicity control, and network structure assessed against a configuration-model null, it constitutes a portable method rather than a single-portfolio examination. The structure it exploits, a fixed roster of hazards, a standardized evidence hierarchy, and expert-assigned postures recorded in narrative documents, recurs across occupational epidemiology wherever small, highly selected workforces face severe, data sparse hazards in extreme environments (82). The same extract > classify > network examination approach could be applied to multiple research in harsh conditions, including commercial and saturation diving, where circulatory, respiratory, decompression, and thermal hazards are managed under standards driven safety regimes; to submarine and other isolated, confined, and extreme operations, whose circadian and behavioral stressors parallel those of spaceflight; and to deep-sea

and subterranean industry, where emerging activity is pushing quantitative occupational risk assessment into data-poor regimes (83, 84) In each setting the evidence is heterogeneous, unevenly documented, and adjudicated by expert panels, where the conditions under which the present contribution, making the evidence > score relationship explicit, comparable, and reproducible, is of value to occupational and environmental epidemiology and not to space medicine alone. A particular feature of this setting deserves emphasis. Expert consensus is itself one of the eight evidence types, and in a domain where direct human data are scarce it frequently carries much of the weight behind a score, so a risk may be rated on the strength of expert judgment as much as on empirical measurement. Given that the same deliberative process that weighs the evidence also assigns the score, examining the evidence at a granular level, type by type and package by package, is what makes it possible to see when a score rests primarily on consensus rather than on direct data, which is precisely the distinction that a portfolio level grade can obscure (3, 21).

Research Question, Specific Aims, and Hypotheses

Overarching Research Question

This study is organized around a single overarching research question: how do the types, sources, and volume of human spaceflight human system risk evidence documented in NASA Human System Risk Board risk summary package submissions, across the 29 active human system risks and the 93 approved submissions spanning 2020 to 2025, align with the LoE scoring and the Board's risk disposition decisions across Design Reference Missions, and what do the structural properties of the resulting risk evidence network reveal about the organization, connectivity, and evidence based buydown capacity of NASA's spaceflight human health risk architecture? Thus assessing whether the evidence documented for NASA's human system risks truly reflects the

agency's own LoE scoring influencing risk decisions across mission types, and what the connections between risks and their evidence reveal about where the system can be strengthened to protect crew health and occupational safety as humanity reaches further into space. The question is pursued through three specific aims, the first addressing the extraction and evaluation of the evidence, the second the structural modeling of the portfolio, and the third the synthesis of both into interpretable patterns and recommendations. The analytic workflow that links the research question, the evidence corpus, and the three aims is summarized in Figure B1 (Appendix B).

Aim 1: Systematic Evidence Extraction and LoE Evaluation

Aim 1: To conduct a systematic review of NASA Human System Risk Board human system risk summary packages submitted between 2020 and 2025, using NLP to extract, classify, and quantify the documented evidence base and to evaluate how that evidence aligns with the assigned LoE scoring across the 29 active human system risks, with corpus assembly following the PRISMA framework and the observational analysis reported in line with STROBE guidance for observational studies (53, 54).

This aim applies an NLP pipeline to the full corpus of approved packages, extracting and classifying citations of the eight evidence types and counting them at both the section level, for assessment of documentation completeness, and the package level, for analysis of the association between evidence and scoring. The LoE values recorded in the structured posture tables are validated against the agency's scoring definitions through a flagging system that distinguishes valid, erroneous, and missing entries. The attribution of phrases to evidence types is carried out computationally by a deterministic rule engine that applies word boundary matching, contextual gating that inspects the surrounding text before accepting a match, and a source over mechanism precedence rule, so that every assignment is reproducible and traceable to the specific triggering

phrase. The association between evidence profiles and LoE scores is then evaluated using non-parametric methods appropriate to ordinal data. This aim evaluates the risks by evidence types diversity and volume, identifies instances in which the evidence appears misaligned with the disposition, and informs the selection of two contrasting complexity risks for the network modeling in Aim 2, with the contrasting risks selected by a composite complexity score that combines each risk's evidence volume, the number of distinct evidence types it documents, and how evenly its evidence is distributed across those types.

The hypotheses associated with this aim are the following. Together they interrogate the same question, whether the documented evidence predicts the assigned LoE, from several complementary angles: the quantity of evidence, its diversity across types, the presence of the most direct evidence, its graded quality, and how it varies across mission profiles. Testing the relationship from these different directions makes the overall finding more defensible than any single test would be, because a consistent result across measures that each capture a different facet of the evidence base is far less likely to be an artifact of how the evidence was counted. H1a: risks with greater total evidence volume have stronger, that is lower numbered, LoE scores. The LoE is recorded on a four-tier scale running from 1 for Strong evidence to 4 for Speculative, so a lower number denotes greater certainty; where evidence is quality weighted, the eight evidence types are assigned descending weights from 8 for direct human spaceflight data down to 1 for the least direct type, a heuristic ordering defined for this analysis to reflect the relative directness of each type rather than a weighting prescribed by the agency's risk management plan. This hypothesis is evaluated by Spearman rank correlation, a measure of whether two quantities tend to rise and fall together once each is ranked from lowest to highest, which suits ordinal scores and does not assume a straight-line relationship. H1b: risks with greater evidence diversity, meaning more of the eight

evidence types represented, have stronger scores independent of total volume, evaluated by Spearman rank correlation on the diversity count. H1c: risks citing direct human spaceflight data are more likely to carry strong or moderate scores than risks without such evidence, evaluated by the Fisher exact test, which assesses whether two categorical conditions, here the presence of direct human spaceflight evidence and a strong or moderate score, co-occur more often than would be expected by chance in a small sample. H1d, in two parts: risks that triangulate across more distinct evidence types receive stronger scores, evaluated by the Kruskal-Wallis test, a rank-based comparison of whether several groups, here the score strata, differ in their typical evidence breadth without assuming a normal distribution; and evidence breadth declines systematically from low Earth orbit toward Mars missions, evaluated by stratified Spearman correlation of an evidence-breadth metric against mission rank within the operations and long-term health outcomes separately. H1e: risks whose evidence is rated more strongly across the triangulated evidence streams carry stronger scores, evaluated by Spearman rank correlation, serving as the bridge to the mission-stratified analysis.

For each of the eight evidence types, the detection cues the classifier keyed on, an illustrative package sentence, and the resulting classification decision are documented in Table D1 (Appendix D), which records the worked examples that anchor every assignment to the specific cue that produced it.

Aim 2: Multilayer Network Topology Modeling

Aim 2: To model the multilayer network topology of the human system risk portfolio and of two contrasting complexity risks, using structural metrics and triad census to quantify the prominence, connectivity, and architectural patterns of risks and evidence types within NASA's evidence framework.

This aim constructs risk-evidence networks at three layers, each answering a different structural question. The first is the within risk layer, which maps how the evidence types interconnect inside each of the two contrasting risks selected for close study, testing what gives a complex risk its dense internal structure and a simple risk its sparseness. The second is the cross-portfolio layer, which connects every risk to the evidence types it documents, with edges reflecting shared evidence, evidence type overlap, and evidence profile similarity, testing which evidence types act as hubs across the portfolio and which risks group into communities built on a common evidentiary foundation. The third is a comparative layer that positions the two selected risks within the whole portfolio, testing how the most and least complex risks differ in their structural role. Examining the portfolio at these three resolutions is deliberate: any single layer answers only part of the structural question, and reading the same architecture from the level of one risk, from the level of the whole portfolio, and from the comparison between them guards against conclusions that hold at one scale but not another. Questions of causation or mechanism beyond documented co-citation remain outside the scope of this structural analysis. Multilayer visualization, the full triad census, and topology metrics including degree centrality, betweenness centrality, and the clustering coefficient characterize the structure, with interpretation focused on identifying hub risks, bridging evidence types, and over represented structural patterns that mark where the portfolio's evidence is concentrated and where its risks are most interdependent.

The hypotheses associated with this aim are the following. Where the first aim asks whether evidence predicts the score, this aim turns to the shape of the evidence base itself, treating the risks and the evidence types as a network and asking how they are connected. H2a: risks of greater composite complexity, a single summary measure that combines how much evidence a risk documents, how many distinct evidence types it draws on, and how evenly that evidence is spread

across those types, exhibit higher degree and betweenness centrality, reflecting their role as hubs connecting multiple evidence types. Degree centrality counts a node's direct connections and betweenness centrality captures how often it lies on the shortest path between other nodes, so a risk high on both sits at the structural center of the portfolio. H2b: risks with similar evidence profiles, measured by cosine similarity, a score from zero to one that compares two risks by the pattern of evidence types they cite rather than the sheer quantity, so that two risks built on the same mix of types score near one regardless of how much each documents, cluster together into identifiable communities that share common evidence foundations. H2c: cross-risk references in the narrative text reveal dependency structures that evidence profile similarity alone does not capture, so that mechanistically related risks reference one another even when their evidence profiles differ. Together these hypotheses move from the portfolio viewed as a whole to its internal architecture: where a conventional review sees the forest, the network analysis traces how the individual trees are linked, surfacing shared foundations and dependencies that a risk by risk reading would miss, and examining the structure from these complementary angles guards against reading meaning into connections that are an artifact of how the network was built.

Aim 3: Thematic Synthesis and Recommendations

Aim 3: To synthesize the outputs of Aims 1 and 2 through thematic analysis aligned with the network metrics and a package completeness assessment, in order to identify patterns in evidence documentation, LoE scoring consistency, and risk mitigation prioritization across NASA's human system risk architecture.

This aim integrates the quantitative outputs of the preceding aims, namely the evidence type distributions, the scoring validation flags, the network centrality and community results, and the triad profiles, with a deductive thematic analysis of how evidence is documented across the

packages. In plain terms, this aim steps back from the individual measurements produced by the first two aims to ask what recurring patterns they reveal across the portfolio: where the documented evidence is thin, where the assigned scores sit uneasily with that evidence, and where the structure of the risk network suggests that evidence development could be shared across related risks. The thematic codebook is drawn from the structural features of the corpus itself, including the weighting of evidence types along the 8-1 hierarchy, the linking of evidence types within a risk's documentation, the structural patterns recovered by the network analysis, the structure of the standardized template and the sections in which evidence is located, the provenance of the evidence within the documents, and the diversity of evidence types represented. A package completeness and robustness assessment evaluates section presence, evidence density, the use of figures and tables, and the consistency of submissions across the 2020 to 2025 study window. The aim identifies where documentation practices align with the assigned scores and where they diverge, and it consolidates the findings into data informed recommendations.

The hypotheses associated with this aim are the following, each framed as a directional association between a documentation feature and a LoE or structural outcome across the portfolio. Where the first two aims test what the evidence is and how it is connected, this aim asks whether how completely and how independently each risk is documented relates to its score. H3a: risks with higher package completeness, meaning a larger proportion of the expected template sections present, higher word counts, and more references cited, carry stronger, that is lower numbered, LoE scores, with completeness measured for each package against the 34-section HSRB Risk Summary Package template and examined alongside the scoring analysis. H3b: cross-risk dependency language, narrative text in which the documentation of one risk explicitly references another, appears in fewer than half of the packages, indicating that risks are predominantly

documented in isolation despite sharing mechanisms, countermeasures, and mission impact pathways, evaluated through thematic coding for cross-risk dependency (85). H3c: risks with greater evidence type concentration, measured by a concentration index that approaches one when a single evidence type dominates a risk's documentation and falls toward zero when the eight types are evenly represented, carry weaker scores, evaluated through the concentration analysis. Taken together, these hypotheses ask whether the form of the documentation, how complete it is, whether it acknowledges related risks, and how evenly it draws on the evidence types, tracks the strength of the score it supports. The complete thematic codebook, giving the definition, identification cues, and prevalence of each narrative theme, is provided in Appendix D (Table D5).

Methods

Study Design

This study is a systematic, computational, mixed methods analysis of the documented evidence base underlying NASA's human system risk decisions, conducted as a census of the approved submissions in the active risk portfolio between 2020 and 2025 rather than as a sample of them. The design integrates four components within a single reproducible pipeline: a systematic review and natural language extraction of the documented evidence, a statistical evaluation of the alignment between that evidence and the assigned LoE scores, a network analysis of the structural relationships among risks and evidence types, and a deductive thematic analysis of how evidence is documented (86). The analysis is observational and retrospective, drawing on the risk summary packages authored by risk custodians and approved through the established Human System Risk Board review process in the course of normal operations, and the study is framed against the PRISMA standard for the assembly of the corpus and the STROBE standard for the reporting of the observational findings (53, 54). The study is explicitly positioned as a pilot scale, proof of concept transfer of NLP and network science into the occupational and aerospace epidemiology of human spaceflight, and, consistent with established guidance on pilot studies, its quantitative findings are reported as preliminary, with the network and thematic components treated as exploratory and hypothesis generating (87, 88, 89).

One feature of this records-based design shapes the reading of every alignment result and is flagged here at the outset. Within a single package, the documented evidence and the assigned LoE are not produced by independent parties. The risk custodian who compiles the evidence in a package is also the person who assigns its LoE and writes the rationale that cites that evidence, and the Board reviews the completed package rather than generating the evidence or the score

separately. The two quantities this study relates, the documented evidence and the score, therefore share a single authorship rather than being drawn from independent sources, so some degree of alignment between them is expected by construction. What the study measures is how far the documented evidence actually tracks the score across the portfolio, not whether two independent processes happen to agree, which is why a weak or absent association is especially informative: it indicates that even evidence and scores produced by the same author do not move together in the simple way an outside reader might assume. This non-independence leaves the descriptive aims intact, since the study characterizes the documented record as it exists, but it bounds any causal interpretation of the evidence to score relationship, and its full consequences are developed in the Limitations. Naming it now ensures that the alignment findings are understood from the first as properties of a jointly produced record rather than as a test of an evidentiary input gathered apart from the score.

The research was conducted under a Space Act Agreement between the NASA Johnson Space Center and The University of Texas Health Science Center at Houston School of Public Health, agreement number SA-SA-22-36464, which established the terms of access to the risk documentation and the scope of the collaboration. The analysis used the approved Risk Summary Package records maintained by the Human System Risk Board, supplemented by publicly available NASA evidence reports and the published descriptions of the agency's risk management framework. The computational analysis was performed in a controlled environment using the documents made available under the agreement and subject matter expert consultation with the Human System Risk Board was established and available throughout the study to clarify or confirm data statements and package-level information where needed. No astronaut identifiable

health data were accessed; the records analyzed are risk level documentation of evidence and scoring, not individual medical data.

Study Subjects and Units of Analysis

The study population is the complete portfolio of active human system risks governed by the Human System Risk Board, comprising 29 risks at the time of analysis. Given that two risks included independently scored sub-risks within their submitted risk summary packages, with Non-Ionizing Radiation separated into laser and solar exposure and Toxic Exposure separated into high, medium, and low concentration tiers, the 29 risks then correspond to 32 distinct risk posture sub-tables. The corpus of documentation analyzed comprises 93 approved Risk Summary Package submissions distributed across these risks and spanning the years 2020 through 2025, with most risks revised and resubmitted between two and four times across the window, which gives the corpus its longitudinal, repeated measures structure.

The unit of analysis differs by tier of the design, and the distinction is fundamental to the interpretation of the results. In the portfolio level analyses, the unit is the risk, with one summary outcome per risk derived across its mission profiles, yielding a maximum of 29 units, one for each active risk. In the mission-stratified analyses, the unit is the risk by Design Reference Mission observation, so that each risk contributes one observation per mission profile for which it carries a scored posture, yielding up to 227 scored operations observations and up to 192 scored long term health observations across the portfolio. In plain terms, the portfolio level analyses treat each risk as a single data point, whereas the mission stratified analyses break each risk into its separate mission-specific postures, so the evidence to score relationship can be examined at a finer resolution. Two risks, Carcinogenesis and Urinary Retention, were retained for the evidence extraction and thematic components but excluded from the LoE association analyses. The basis

for this exclusion is documentary rather than substantive: within the 2020 to 2025 study window, the approved risk summary packages for these two risks did not record a finalized LoE in any Design Reference Mission and outcome cell, so no valid score was available to anchor the evidence to scoring analyses. These two packages appear to have been moving through review and approval during the study window, but a disruption to that process, arising from broader institutional changes outside the control of this analysis and of the Board, interrupted finalization before a score was recorded. This absence reflects the state of the adjudication record at the time of analysis, a LoE determination that had not yet been finalized for these two risks, rather than any deficiency in their underlying evidence. Both are, in fact, well documented: Carcinogenesis is among the most quantitatively instrumented of all human system risks, supported by the peer-reviewed NASA Space Cancer Risk model and an extensive radiation-epidemiology base, and Urinary Retention carries a documented evidence base spanning multiple evidence types (90). Because the LoE association analyses require a valid assigned score against which the documented evidence can be related, a risk with no recorded score cannot enter those analyses without imputation; the two risks were therefore held out of the scoring tier while being retained in full for the evidence extraction and the thematic and network components, where an assigned score is not required. This treatment is consistent with the study's central finding that a LoE encodes mission applicability and deliberative judgment beyond the documented evidence alone, and it is corroborated by the supplementary analysis reported in the Discussion section, which found that the evidence record could not mechanically reconstruct a score for these two risks. Their exclusion from the scoring analyses reduces the portfolio level analytic set to 27 overall scored risks while preserving the full corpus for the extraction and documentation analyses.

Sample Size and Statistical Power

The study analyzes a complete census of the active risk portfolio rather than a sample, so the number of units is fixed by the size of the portfolio and cannot be expanded. This census structure has a direct and consequential implication for statistical power. With 29 risks at the portfolio level, the design has approximately 80 percent power (91) to detect only large effects, corresponding to a Spearman correlation of approximately 0.48 in magnitude, under the Bonferroni-corrected significance threshold described below. Small and moderate associations would not be reliably detected at this sample size. This constraint is not a deficiency to be corrected but a fixed property of examining a finite portfolio, and it is the principal reason the portfolio level findings are interpreted with explicit attention to the distinction between a true absence of association and an inability to detect one. The minimum detectable effect was computed for the Spearman correlation under the corrected threshold and is reported with the corresponding results. The mission-stratified tier, in which the unit is the risk by mission observation, provides substantially more observations, up to 227, and therefore improved power for the analyses conducted at that resolution, which is one motivation for designating the mission-stratified analysis as the primary test of the evidence-to-scoring relationship. The 29-risk count, moreover, constrains only the portfolio level correlation tests. The study is genuinely multi-level: it draws on the deduplicated evidence extracted from every narrative, the LoE scores recorded for each Design Reference Mission across both the operations and long term health tracks, and the network of connections among risks and evidence types. The mission-stratified, per-mission, and network analyses therefore rest on this fuller structure rather than on 29 points alone, so the design is best understood as a layered examination of the evidence base in which the portfolio level correlation is only the most aggregated view, and

the power limitation described here applies to that aggregated view rather than to the analysis as a whole.

Given that the portfolio size is fixed at 29 risks, the analysis is powered only for large effects: at this sample size, a two-sided test at the conventional five percent level attains approximately 80 percent power to detect a Spearman correlation of about 0.48 or larger, and smaller effects could go undetected (91). This constraint is the reason every non-significant portfolio result is interpreted as the absence of a strong, monotone association rather than as proof that no association of any size exists, and it is the explicit justification for the pre-specified equivalence test and Bayes-factor analysis, which are the tools by which an informative null is distinguished from a result that is merely underpowered. In plain terms, with only 29 risks the portfolio-level test can reliably detect only a strong relationship; a weak or moderate one could be present yet escape detection. A non-significant result therefore means the data do not show a strong association, not that no association of any size exists, which is precisely why the pre-specified equivalence test and Bayes-factor analysis are used to tell an informative null, one that actively supports the absence of a meaningful effect, apart from a result that is simply too underpowered to conclude anything.

Data Sources and Provenance

The primary data source is the set of approved Risk Summary Packages, presentation format documents that record, for each risk, the causal diagram and its narrative, the Likelihood by Consequence matrix, the per-mission posture tables, the Overall Assessment of the Evidence, the State of Knowledge, the supporting metrics, the standards and requirements flow, and the high-value risk mitigation targets. Across all eight Design Reference Missions and for both the operations and long term health outcomes, the posture tables record the Likelihood by Consequence rating, the resulting posture, the disposition, the LoE, and the supporting rationale,

while the narrative sections and the text embedded in figures and tables carry the documented evidence itself across the eight evidence types. Although the packages follow a common template, the sections are populated with varying completeness across risks, and they are read alongside the supporting agency documentation that governs them, principally the JSC-66705 Human System Risk Management Plan (7). Each package contains both structured content, principally the posture tables, and unstructured content, principally the narrative prose and the text embedded within figures and tables. To capture both, the pipeline processes each package along two parallel paths, a structured ingestion path and a natural language path, and records the provenance of every extracted data point so that each classification is traceable to its location in the source document. The dual-path design ensures that the structured scoring data and the narrative evidence are each handled by methods appropriate to their form, and that the two can be reconciled against each other.

Beyond the Risk Summary Packages themselves, the mission, analog, and human-research context that anchors the evidence is housed in the NASA Life Sciences Portal (NLSP) (92), which aggregates the Life Sciences Data Archive and the Lifetime Surveillance of Astronaut Health program (12) and exposes them through an Explore Data repository, a Standard Measures collection, and a structured data request workflow; the mission and analog inventories used to build the recognition catalogs were requested through this portal, and de-identified human-research holdings there correspond to the Type I and Type II evidence the engine attributes in the narrative. In short, the portal is the central archive from which the mission names, analog studies, and human research records that the engine recognizes in the package text are drawn.

Corpus Construction and the PRISMA Selection Process

The corpus was assembled under the PRISMA framework so that the basis for inclusion and exclusion is explicit and reproducible. In the identification stage, every Risk Summary Package submission approved across the study window was enumerated. In the screening stage, submissions were screened for approval status and for conformance to the standardized template. In the eligibility stage, packages were retained when they corresponded to an active human system risk carrying an approved posture within the window. The Risk Summary Package records were provided directly by the Human System Risk Board under the Space Act Agreement, in their approved presentation format, rather than retrieved from a public repository. The included set comprised the 93 approved submissions spanning the 29 risks that constitute the analytic corpus. The analytical pipeline and the architecture of the assembled corpus are summarized in Figure 1a.

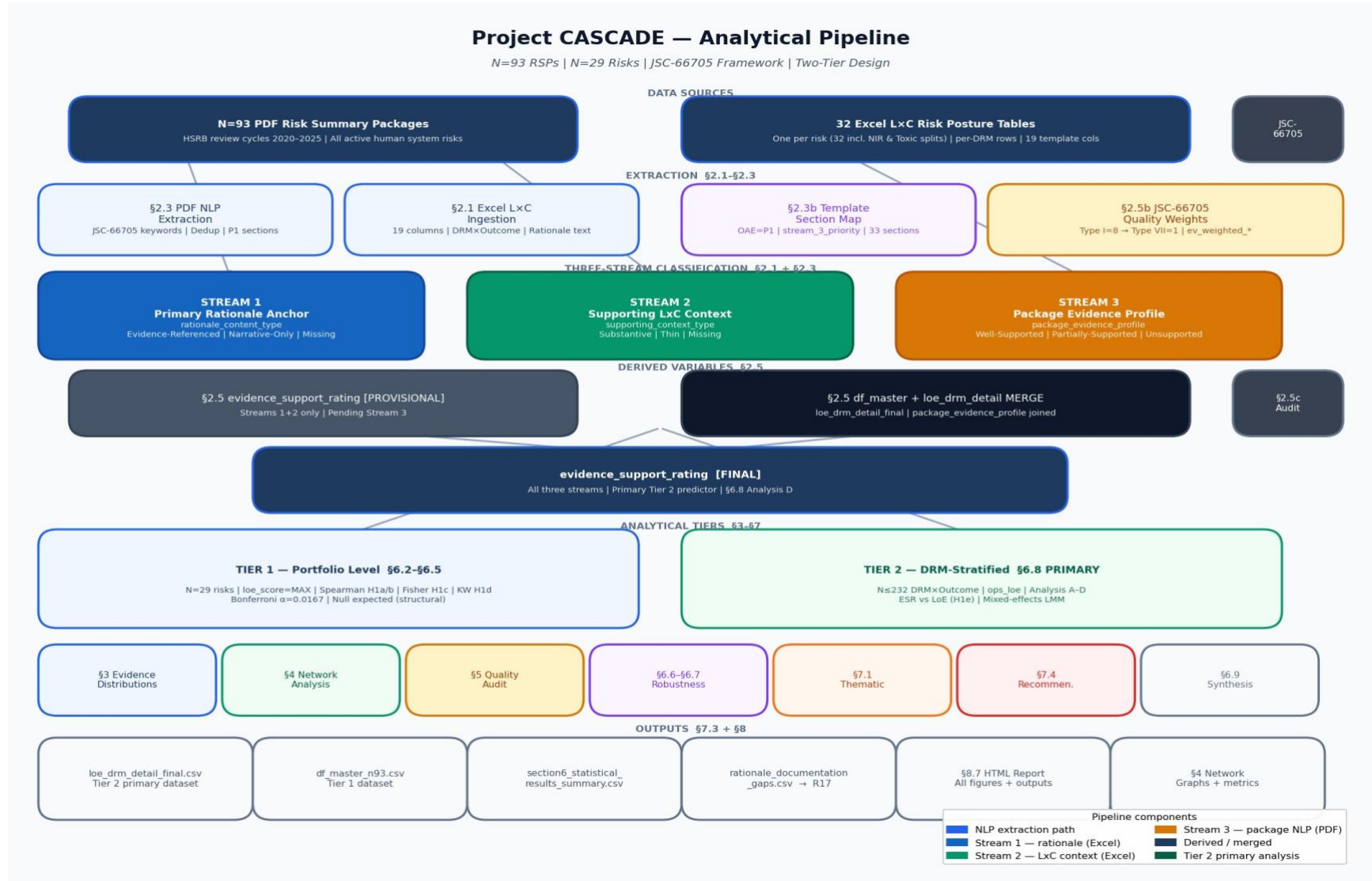


Figure 1a. The Project CASCADE analytical pipeline and the architecture of the assembled corpus, from dual-path ingestion of the approved Risk Summary Packages and the structured posture tables, through the three-stream classification and derived variable construction, to the two analytical tiers and their outputs. A larger, navigable version of this pipeline is available in the interactive dashboard (Appendix F).

Structured Ingestion and LoE Validation

The structured ingestion path parses the posture tables per risk, which follow a standardized schema of 19 columns that records, for each risk by mission by outcome combination, the likelihood and consequence scores, the resulting posture, the disposition, the LoE, and the supporting rationale. Each posture table distinguishes the two impact categories, In-Mission Operations and Long Term Health, so that a single risk contributes separate scored entries for operations and for long-term health across the Design Reference Missions for which it is assessed. Parsing this structured content yields the LoE values that serve as the study's primary outcome and the disposition values used in the alignment analyses. The posture tables analyzed are those from each risk's most recent approved submission within the study window. Where a package recorded its likelihood and consequence data in an older posture format rather than the current standardized table, the scored entries were retained and normalized into the same 19-column schema, so that every risk is represented by one consistent, standardized set of posture tables. The standardized template structure, including the Likelihood by Consequence matrix that these posture tables capture, is described in the Background and provided in full in Appendix B.

Given that the extracted scores drive every downstream analysis, each LoE entry was validated against the agency's scoring definitions through a flagging system that classified entries as valid, erroneous, missing, or not applicable. A valid entry carries a recognized score within the defined 1-4 range with a consistent label. A missing entry is one for which no score was recorded where one was expected. A not applicable entry corresponds to a risk by mission combination that the agency does not score, as occurs for the four risks that carry no long-term health pathway, namely Crew Egress, Extravehicular Activity Induced Hand and Shoulder Overuse, Electrical Shock, and Team. An erroneous entry is one in which the recorded value violates the scoring

schema, either by falling outside the defined range or by exhibiting a mismatch between the numeric score and its text label. To make the scheme concrete, a valid entry records, for instance, a score of 2 labeled Moderate; a missing entry is a posture cell left blank where a score was expected; a not applicable entry is a long term health cell for a risk such as Crew Egress that has no long term health pathway; and an erroneous entry is one such as a range entered where a single score was required, for example a cell recorded as 2-3, or a cell scored Strong but labeled with the numeral 2. Across the corpus of 473 expected posture entries, 419 were valid, for an overall coverage of 88.6 percent, with operations coverage of 89.7 percent and long-term health coverage of 87.3 percent; 40 entries were missing, 14 were erroneous, and 39 were not applicable (Figure 3). The 14 erroneous entries comprised 7 range entries, in which a span such as 2-3 was recorded where a single score was required, and 7 numeric-to-label mismatches, the latter consisting of 3 entries scored as Moderate but labeled with a value of 3, 2 scored as Strong but labeled 2, and 2 scored as Weak but labeled 2. These flagged entries were handled conservatively in the analyses, and their enumeration is itself one of the study's findings about documentation consistency. Erroneous and missing entries were excluded from the LoE mapping and from the downstream correlation, network, and complexity analyses rather than imputed, a deliberately conservative choice whose effect on coverage and on the interpretation of the null results is revisited among the limitations.

For clarity these counts sit within a complete grid of 512 possible cells, the 32-risk posture sub-tables each scored across 8 Design Reference Missions for both the operations and long-term health outcomes, so that a single risk contributes up to 16 posture cells (8 missions $\times$ 2 outcomes) and the 32 sub-tables yield 512 in total. Removing the 39 not applicable cells leaves the 473 applicable cells that serve as the coverage denominator, of which 253 are operations cells and 220

are long term health cells; the 419 valid scores, the 40 documentation gaps, and the 14 erroneous entries are partitioned within those two tiers. The per-cell detail underlying every total is held in the per-mission LoE detail table (`loe_drm_detail.csv`) and its wide coverage matrix companion, both of which can be explored through the interactive dashboard described in Appendix F, so each figure above can be traced to the specific risk, mission, and outcome that produced it. In plain terms, this cell-by-cell accounting was done so that every coverage figure reported in the study can be traced back to the exact posture cells that produced it, which keeps the evidence base auditable rather than asserted.

The Natural Language Processing Engine

The natural language path extracts and classifies the documented evidence from the unstructured content of each package using a rule-based, lexicon-driven system whose decision logic is explicit and inspectable (the analysis code, derived datasets, dashboard, and Jupyter notebook are described in the data and code availability statement in Appendix F). The engine processes each package by segmenting it into analyzable units, matching curated terms against those units under defined contextual constraints, and classifying the resulting matches into the eight evidence types. The design prioritizes transparency and reproducibility over the broader but opaque coverage of a learned classifier, a system that infers its own classification rules from labeled training examples rather than following rules specified in advance and whose decisions cannot always be traced to a single explicit reason, because every classification in an examination of a governance corpus must be traceable to the specific text and rule that produced it.

The Evidence Type Lexicon

Classification is governed by a curated lexicon that maps a controlled vocabulary onto the eight evidence types. In NLP, a lexicon is a curated list of terms paired with the category each term signals, so that when a listed term appears in the text the system can assign the surrounding passage to the corresponding category; here it links each curated term to one of the eight evidence types. The lexicon comprises 117 canonical keywords (with composition and extraction provenance detailed in Appendix D), expanded through an alias structure to 514 surface forms that capture spelling variants, abbreviations, and morphological variation, so that a single underlying concept is recognized across the heterogeneous wording of the corpus. The keywords and aliases were derived from the agency's own descriptions of the evidence types and from the recurring terminology of the packages themselves, and each is assigned to exactly one evidence type, which preserves the interpretability of every match. The lexicon is the operational definition of the eight-type scheme for the purpose of this study, and its full composition is reported in the appendices so that the classification is reproducible and examinable.

The meaning of each evidence type and the sources that qualify under it were framed by the agency's JSC risk management plan and by risk custodian documentation, which define the evidence categories used in Risk Summary Package adjudication; these sources governed both the controlled vocabulary and the inclusion rules built into the corpus, so that the lexicon reflects the categories the agency itself applies rather than an external taxonomy. These governing documents include the agency's Human System Risk Management Plan, JSC-66705, which defines the evidence categories and the LoE scale (7); the NASA Spaceflight Human-System Standard, NASA-STD-3001, which sets the health and performance requirements that the standards and requirements evidence type points to (16); and the mission and analog inventory exported from

the NASA Life Sciences Portal, which was used to verify crewed and analog mission names (92). Each is listed in the references and, where applicable, detailed in the appendices.

Mission and Analog Catalogs

To recognize references to specific missions and ground analogs within the narrative, the engine draws on two catalogs both compiled from official NASA mission and analog records (92). The crewed mission catalog enumerates 261 missions organized into 13 mission families, with 66 aliases capturing the varied ways missions are named in the text. An alias is an alternative surface form for the same mission, recorded so that the engine recognizes a mission however the narrative happens to name it. The International Space Station, for example, appears as ISS, as Space Station, and as International Space Station; Apollo missions appear with Arabic or Roman numerals, as in Apollo 11 and Apollo XI; and Space Shuttle flights appear both by their mission designation, such as STS-135, and by descriptive phrasing. Capturing these aliases is what allows a single mission to be counted consistently across packages whose authors named it differently, and the range of forms encountered points to a future opportunity for the agency to standardize how missions are named in the risk documentation. The analog catalog enumerates 25 ground-analog environments organized into 15 groups, with 64 aliases. The same principle applies to the ground analogs. NASA documentation refers to a given analog by an acronym, a spelled-out name, or a descriptive phrase, so the analog catalog records these variants as aliases, for example HERA for the Human Exploration Research Analog, NEEMO for the NASA Extreme Environment Mission Operations undersea analog, and head-down-tilt bed rest described in several ways, so that each analog is recognized regardless of how a package phrases it. These catalogs allow the engine to attribute narrative references to the appropriate mission or analog context, which supports the attribution of

human spaceflight and ground-analog evidence to its source and contributes to the mission-stratified analyses.

These catalogs were derived from the mission and analog inventory maintained by the NASA Life Sciences Portal (NLSP), the Human Research Program resource that consolidates the Life Sciences Data Archive (LSDA) and the Lifetime Surveillance of Astronaut Health (LSAH) program and provides human spaceflight and ground-analog research data dating to 1961; the inventory was obtained through the portal’s Explore Data repository and its formal data request process as of March 2026 (92). Given that the analysis window is bounded at 2020 to 2025, missions accessioned after that request period fall outside scope, while every mission that could have been cited as evidence within the window is represented. Risks that were undergoing review or preparation for approval during the window but were never granted an approved posture are not included, since the analysis is restricted to approved Risk Summary Packages. The alias structure captures the varied ways a single mission is named in the narrative; Gemini missions, for instance, appear in both Roman and Arabic forms (Gemini IV and Gemini 4), and many programs carry shorthand or acronym variants, so that numeral and abbreviation iterations resolve to a single canonical mission.

Three Track Matching and Contextual Constraints

Term matching proceeds along three tracks that together balance recall against precision. The first track performs direct matching of canonical keywords and their aliases against the segmented text. In other words, the engine first looks for the lexicon’s exact terms and their known variants wherever they appear in the text, which captures the clearest and most direct evidence references. The second track resolves mission and analog references through the catalogs. The third track captures multi-word and contextually qualified expressions that single keyword matching would

miss. To control false positives, matches are accepted only when the matched term and its qualifying context fall within a defined proximity, operationalized as a 200-character acceptance window, and a hybrid sentence boundary rule constrains the window to respect sentence structure so that a term is not associated with context belonging to an unrelated sentence (66). In plain terms, a keyword is credited as evidence only when its supporting context sits nearby, within the same sentence, with a 150-character fallback window applied only to very short segments, so that a term is not tied to wording that belongs to an unrelated statement. This combination of a proximity window and a sentence boundary constraint is the principal mechanism by which the engine favors precision, accepting that some genuine but distantly phrased references will be missed in order to ensure that accepted matches are well supported by their immediate context. Favoring precision in this way means the engine would rather miss a few genuine but distantly phrased references than count weakly supported ones, which keeps the evidence totals trustworthy as inputs to the later analyses. The system does not perform negation detection, a limitation acknowledged in the Limitations. The three-track structure was chosen to balance breadth against accuracy: direct matching and catalog resolution maximize recall for clearly named evidence and missions, while the multi-word track together with the proximity and sentence boundary constraints holds precision high. One capability the engine deliberately omits is negation detection, the automatic recognition that a statement denies rather than asserts something, as in the phrase no evidence was found (66). Given that negated evidence statements are rare in these adjudicated packages, where custodians document what supports a risk rather than what does not, and that reliable negation detection adds substantial complexity and introduces its own errors, the design favors a transparent and inspectable rule set; the residual chance that a negated mention is counted is acknowledged

among the limitations. The complete extraction and classification pipeline, from ingestion through deduplication and weighting, is depicted in Figure 1b.

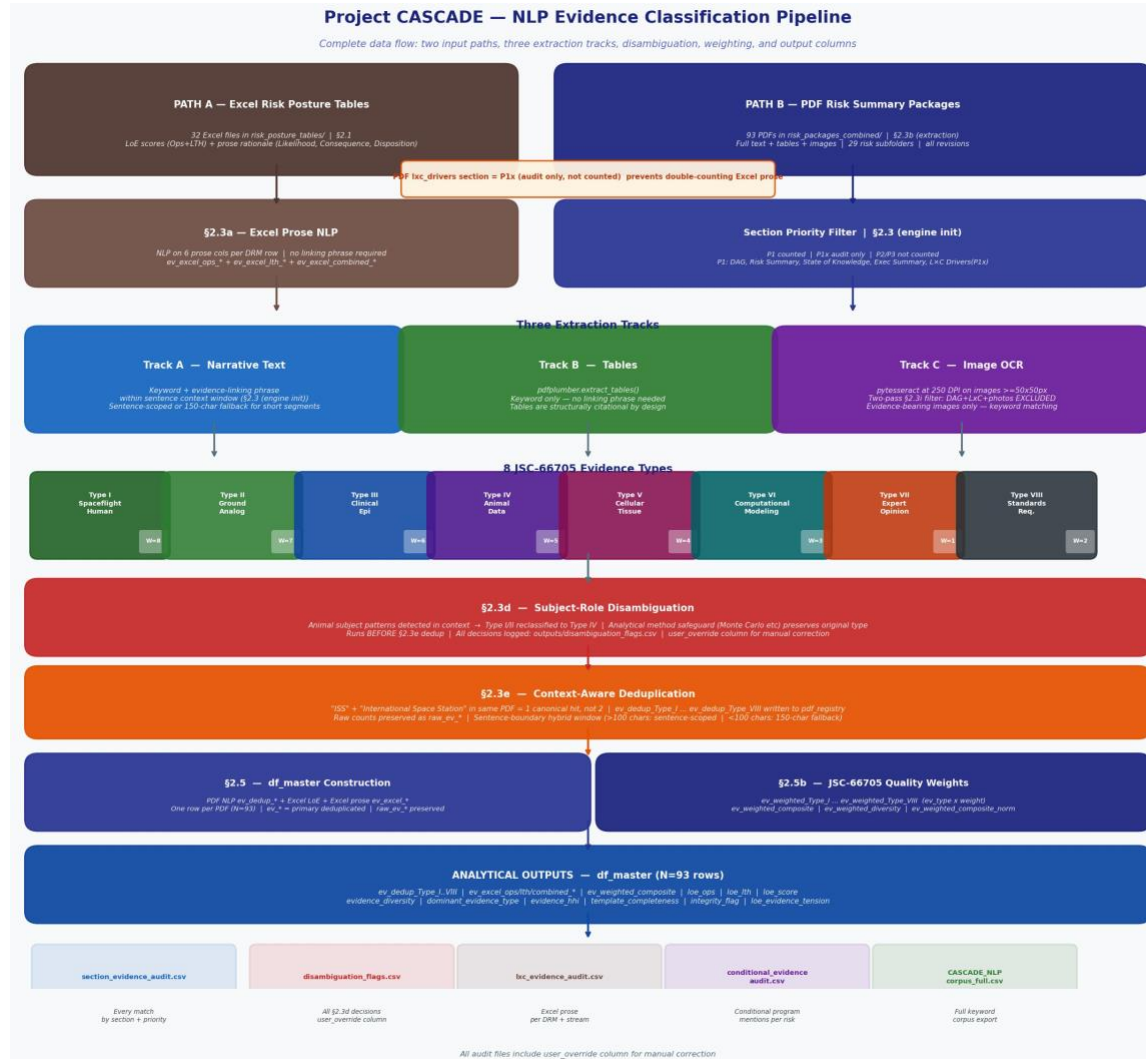


Figure 1b. The Project CASCADE natural language evidence-classification pipeline: the two ingestion paths (Excel risk-posture tables and PDF Risk Summary Packages), the three extraction tracks (narrative text, tables, and image OCR), subject-role disambiguation, context-aware deduplication, the eight JSC-66705 evidence type weights, and the resulting analytical outputs. The figure is intentionally dense; a full-resolution, zoomable version is browsable in the interactive dashboard (Appendix F) for closer inspection.

Deduplication, Image Handling, and Subject Role Disambiguation

Several processing steps protect the integrity of the match counts. Duplicate matches arising from repeated phrasing within a package are reconciled so that the same underlying citation is not counted multiple times (93). This reconciliation prevents a single citation that an author repeats across slides from inflating a risk's apparent evidence, which would otherwise bias the volume measures toward verbose packages rather than evidence rich ones. The step is deliberately limited to exact repeats of the same canonical citation within a package, since aggregating too aggressively across distinct mentions would risk discarding genuinely separate evidence and obscuring true differences between risks (93). A substantial share of the evidence in these packages is presented graphically, in photographs of experiments and in figures and tables, rather than in prose, so the pipeline must read images as well as text to avoid undercounting that evidence. Treating figures and tables as a data source in this way mirrors practice in scientific information extraction, where key findings frequently appear in non-textual form, and it points to a recommendation that future submissions standardize how graphical evidence is provided, since the presentation format package is uniquely demanding to parse. Text embedded within figures and tables, which carries a substantial share of the evidence references, is recovered through optical character recognition so that evidence presented graphically is not lost (60). Optical character recognition is a mature and widely validated technique for converting text rendered as an image into machine readable form, and its fidelity here was verified by manual spot checking against the image-audit record, so that graphically presented evidence is captured to the same standard as the prose (60). A subject role disambiguation step, a determination of which risk an evidence statement is actually about, distinguishes instances in which a risk is the subject of a statement from instances in which it is merely mentioned in the context of another risk, which prevents the misattribution of evidence

across risks (94). A dedicated disambiguation examination reviewed the corpus for instances in which a risk reference might be incorrectly attributed, and it returned zero flagged cases, indicating that the subject-role attribution was clean across the analyzed packages. For example, a statement in the Bone Fracture package that names Renal Stone only to draw a contrast would have its evidence credited to Bone Fracture rather than to Renal Stone, so that a risk mentioned only in passing is not miscounted as the subject of the evidence.

The image pipeline can be quantified directly. Across the 93 packages the extractor recovered 17924 embedded images, of which 4290 carried text recoverable by optical character recognition at 250 dots per inch. These 4290 images were passed through the two-pass categorizing filter, which sets aside decorative photographs, DAGfigures, and the likelihood by consequence posture matrices because those are captured through the structured path; 2872 images were retained as evidence bearing and counted, 878 were flagged for manual review, and the references surviving the gate contributed 1210 accepted matches, the 13.3 percent image-derived share noted above. Mining figures, tables, and their embedded text from research PDFs, and filtering out non-evidential graphics, is an established task in scholarly document processing, where automated figure extraction has been applied at the scale of millions of papers (95). Following the same logic, the two-pass filter sets aside decorative photographs and the figure types already captured through the structured path, so that only evidence bearing graphics are counted. The per-image category, the recovered text, and the review flags are recorded in the image-audit table (image_audit_full.csv) identified in Appendix F, which is the record against which the optical-character-recognition fidelity was manually checked. Two deliberate scope decisions are recorded in that audit table. First, the Likelihood by Consequence matrices in the Risk Posture section were not read from images, since those scores were taken from the standardized posture tables of the

single most recent approved scoring for each risk. Second, the causal diagram was used only in its narrative form: as submitted in the presentation-format package, the diagram does not carry its underlying code or full node and edge information, so the DAG narrative was analyzed while the network itself was not, and reconstructing that network for formal causal inference is identified as a future recommendation.

Three Evidence Streams and Triangulation

The engine derives three complementary streams of evidence from each package. The first is the structured posture table content, which yields the scores and dispositions, read from the fixed fields of the agency's standardized posture template so that every package's entries are captured on identical terms rather than according to how an individual author happened to phrase them. The second is the narrative classification, which yields the counts of the eight evidence types from the prose. The third is the optically recovered content from figures and tables. These three streams are triangulated, in the methodological sense of combining multiple sources and methods to corroborate a finding, so that a classification supported across streams is treated as more robust than one resting on a single stream (96, 97). Triangulation across the three streams is the basis on which the study integrates structured and unstructured evidence into a single characterization of each package. Each stream on its own draws on methods established outside this setting, NLP in clinical and biomedical text mining and quantitative content analysis in the social sciences (57); their combination into a single pipeline for expert graded risk evidence is what is novel here, and this study represents the first computational analysis undertaken for the project.

The Master Dataset and the Quality Weighting Scheme

The outputs of the three streams are consolidated into a master dataset in which each package contributes its evidence type counts, its structured scores and dispositions, and its derived metrics.

Consolidating the three streams into a single master dataset gives every package one harmonized record, which is what makes the package the unit of analysis and lets the evidence counts, scores, and derived metrics be related to one another within and across risks without re-parsing the source documents for each test. From the evidence type counts, a quality-weighted composite is computed by weighting each evidence type according to its position in the hierarchy, which is the agency's qualitative ordering of the eight evidence types from the most direct to the least direct form of evidence. The weighting assigns a value of 8 to Type I, direct human spaceflight data, descending by one for each step down the hierarchy through Type VII, expert opinion, which receives a value of 1, while Type VIII, standards and requirements, is assigned a value of 2 to reflect its codified but indirect character. This 8-1 (98) weighting scheme is the author's own analytic device, constructed to translate the agency's qualitative ordering of evidence types into a quantitative composite; the agency supplies only the ordering of the types, not numeric weights. Given that the choice of weights is an analytic decision rather than an agency standard, its influence on the results is examined directly through a weighting-sensitivity analysis reported in the Results. Weighting simply means letting more probative evidence count for more than less probative evidence when the types are combined, so that a composite can surface a relationship between evidence quality and scoring that a flat count of all evidence, treating a single expert opinion the same as direct human spaceflight data, might otherwise mask.

This weighting is a heuristic device for exploratory variable construction and forms no part of NASA's risk management process. It does not appear in the agency's risk plan, in the LoE rubric, or in any operational scoring procedure, and no risk disposition depends on it; it was introduced solely so that this analysis could ask whether a quality-sensitive aggregate of the documented evidence behaves any differently from a simple count. Its role is therefore that of an

exploratory analytic variable rather than a measurement of evidentiary worth, and it is read in that spirit throughout.

The numeric increments the scheme assigns are likewise not empirically calibrated. The one-unit spacing that separates each evidence type from the next, from 8 at Type I down to 1 at Type VII, expresses only the agency's qualitative ordering of the types and carries no claim that direct human spaceflight evidence is, for example, 8 times as probative as expert opinion; the steps could equally be set to any other monotone spacing, including equal weights that count every type once, without contradicting anything the agency specifies. The same caution applies to the LoE values themselves, whose integers are ordinal labels rather than measured quantities, so that the distance between adjacent levels is not metric. For this reason, the primary analyses use rank-based, non-parametric methods that depend only on ordering and not on the spacing between values, and the weighting-sensitivity analysis reported in the Results re-runs the principal test under an equal-weight scheme precisely to confirm that the heuristic increments do not drive any finding.

The Extraction Examination

The completeness and provenance of the extraction were examined across the corpus, to confirm that the pipeline recovered evidence comprehensively and to record where in each document that evidence was found. The engine accepted 9098 evidence matches across the 93 packages. The provenance of these matches was distributed across the document structure, with 14.9 percent drawn from the canonical template sections, 25.5 percent from tables, 13.3 percent from optically recovered image text, and 46.3 percent from non-canonical narrative locations, the last reflecting the substantial share of evidence recorded outside the standardized sections. This provenance distribution both validates the necessity of processing the unstructured content, since nearly half of the evidence lies outside the canonical sections, and quantifies the degree to which

documentation departs from the standardized template, which is itself a finding relevant to the documentation-consistency aim. In plain terms, this matters because it shows the engine is not simply reading the tidy template sections and stopping there; almost half of the evidence it accepted sits in non-standard locations, so a pipeline that ignored the unstructured text would have missed a large share of what the packages actually document.

The provenance distribution, that is, the breakdown of where in each document the accepted evidence matches were found, can equivalently be read as a section-mapping result, which is where the unmapped, or unknown, sections are quantified. Every accepted match carries a section label assigned by matching the heading above it against the canonical template, and across the accepted matches that label resolved to a recognized template section for 53.7 percent and remained unmapped, recorded as unknown, for the remaining 46.3 percent. An unmapped label does not weaken the match: the canonical term, its linking phrase, and the verbatim heading text are retained in every case, so an unknown-section match is a fully attributed evidence reference where the heading simply matched no template pattern. The unmapped share is concentrated in the ‘rationale’ sections, which custodians record as compact tables rather than prose headings and which therefore escape heading detection. The per-match section label, the raw headers, and the mapped-against-unknown split are stored in the section-evidence audit table (`section_evidence_audit.csv`) and carried into the master dataset, so the unknown count is fully auditable rather than a silent loss. Retaining unmapped matches rather than discarding them is a deliberate balance: discarding them would under-count genuine evidence recorded outside the template, while forcing every match into a template section would over-correct by imposing a structure the document does not actually have. Keeping each match fully attributed but labeled unknown preserves the real evidence while

still quantifying the departure from the template, the same recall-versus-precision trade that governs information-extraction evaluation generally (69).

Package completeness against the standardized template was measured for every submission. Each of the 93 packages was scored for the proportion of the 30 scoreable template sections that its narrative headings populated. On a strict heading-recognition basis the mean completeness was 25.5 percent, and on a table-adjusted basis that credits rationale content recorded in tables it was 35.1 percent, the latter being the figure carried into the master dataset; 81 of the 93 packages, or 87 percent, fell below 50 percent heading completeness. This figure reflects real variation in how packages were prepared: some retained portions of an older structure while moving only the updated sections into the current template, and others added sections beyond the template or used headings that did not match its names, so a package could carry more sections, fewer sections, or differently labeled sections than the standardized template anticipates. Recognition varied sharply by section: countermeasure descriptions were detected in 80 percent of packages and the DAGnarrative in 56 percent. These detection rates are lower than the underlying content alone would imply, in part because some submissions included only the template sections carrying new evidence and left unchanged sections out of the resubmission; an absent section therefore does not necessarily mean the package lacked that content, only that submission practice was not uniform across the corpus. The three rationale sections, the narrative justifications recorded for the likelihood rating, the consequence rating, and the overall evidence assessment, were detected in close to zero percent because they are authored as tables rather than as headed prose. This low template adherence is not a deficiency of the extraction but a measured property of the corpus, and it is the same structural feature that produces the large unmapped-section share; the per-package

scores are stored in the master dataset in the `template_completeness` field and summarized in the package-completeness figure.

Human-Annotated Gold Standard

The Gold-Standard Principle: Reliability Versus Validity

An automated classifier that assigns labels by matching words cannot be trusted on the strength of its internal logic alone; its output must be checked against human judgment (69). Two distinct properties have to be established, and they are easily confused. The first is reliability, whether the engine returns the same answer when it is run repeatedly or under small changes to its settings. The second is validity, whether those answers are actually correct. A classifier can be perfectly reliable and still be systematically wrong, so reliability is necessary but not sufficient (73). In this study the two properties were established by separate procedures: reliability was assessed by an intra-pipeline stability analysis that re-ran the classifier under perturbed parameters and measured how well its own outputs agreed across runs, whereas validity was established by comparing a sample of the engine’s classifications against an independent human-annotated reference, a stratified sample of the engine’s accepted classifications that a domain expert judged by hand. That reference is the gold standard, the field-standard method for validating a natural language-processing system, in which human experts label a sample of the same material using the same scheme and the machine is judged by how closely it agrees with them (55, 69, 73). Model-based alternatives, such as using a large language model as an automated judge, were considered and deliberately set aside in favor of human expert annotation, for the reasons set out earlier: a governance corpus demands a reference standard that is transparent and independent of the kind of system under test, and a model judge can introduce hallucinated justifications and inherited biases that were not planned for here. Anchoring every accepted classification in an expert’s

reading of the specific passage is also what builds the context mapping of evidence that the downstream measures rely on.

The Annotation Protocol and Its Basis

The annotation protocol followed established practice in content analysis and computational linguistics rather than being devised ad hoc. A stratified sample of the engine’s accepted matches was drawn, with a target of 40 matches per evidence type across the eight types, for a final validated sample of 287 annotated instances. Stratifying by type rather than drawing a simple random sample is deliberate: a random sample of a corpus dominated by a few common evidence types would leave the rarer types with too few cases to evaluate, whereas a fixed quota per type yields a stable accuracy estimate for every type, including the sparse computational and expert-consensus categories that the first-pass classifier handled poorly. The entire analysis was implemented as Python code organized into a single Jupyter notebook, using Jupyter features in JupyterLab and JupyterNotebook, in which the processing stages are arranged as sequential, individually executable cells. Running the notebook in order reproduces the full pipeline, from ingestion of the source packages through extraction, classification, validation, and the statistical and network analyses, and the chronological cell order allowed each data-cleaning step to be inspected and verified before the next stage consumed its output. The source Risk Summary Packages, together with the catalogs and the lexicon, were held as protected files on a secured local drive; these inputs were read into the notebook environment, where only the analysis code, its intermediate datasets, and the resulting outputs and deliverables were exposed for review, so that the sensitive source material remained separated from the analytical workspace. This structure is what makes the analysis transparent and re-runnable, and it is documented in full in the data and code availability statement in Appendix F. 40 per type also exceeds the minimum of roughly 30

cases per category that the methodological literature treats as the floor for a stable agreement estimate (64, 65). Each sampled instance was presented to the reviewer with the matched keyword, the surrounding sentence of context, and the risk and document section from which it was drawn; the reviewer recorded whether the engine's assigned evidence type was correct and, when it was not, the type that should have been assigned, together with a free-text note. Given that a single domain expert performed the annotation, the human labels serve as the reference standard against which the machine is measured, the design appropriate to a single-analyst computational content analysis (64, 73). The annotation was substantially more effort-intensive than a simple correct-or-incorrect tally suggests. Adjudicating many instances required consulting subject-matter experts on how a passage should be read and learning the specifics of each risk, since mapping a phrase to an evidence type often depended on understanding what kind of study or data the surrounding text actually described. Two recurring judgments illustrate why human review was necessary. First, a single phrase can legitimately carry more than one evidence type, and the reviewer recorded these secondary types rather than forcing a single label, as when a sentence reports an immune finding measured in flight and so draws on both spaceflight and cellular evidence. Second, the keyword matcher can be misled when two cues appear together: a statement describing immune-cell measurements collected from crew aboard the International Space Station contains both a cellular cue, such as T cell, and a spaceflight cue, such as ISS, and a first-pass match on the cellular term alone would mislabel a spaceflight observation as cellular evidence. Human annotation with context mapping resolved these cases by assigning the evidence type that the surrounding text actually supports. Representative annotated instances, including both correct classifications and characteristic false positives, are shown below. Illustrative human-annotation examples from the

gold-standard validation sample: three correct classifications and two false positives, with the basis for each reviewer judgment.

Matched keyword	Context excerpt	Engine type	Reviewer	Basis for the judgment
bed rest	Recovery was tracked across a head-down-tilt bed rest study, an established spaceflight ground analog.	II: Human Analog	Correct	Head-down-tilt bed rest is a recognized ground analog for spaceflight.
t cell	In-flight assays documented reduced T cell function in crew relative to their preflight baseline.	V: Cellular/Molecular	Correct	An immune-cell functional measure is cellular and molecular evidence.
preflight	In flight, crew bone density fell more than 10 percent below each member's preflight measurement.	I: Human Spaceflight	Correct	A pre- and in-flight crew measurement is direct spaceflight evidence.
antarctica	HDT 90d Bed; NEEMO; Antarctica; Space Shuttle; ISS ...	II: Human Analog	Incorrect	Keyword matched a column header listing analog settings, not an evidence statement. More info needed to determine evidence.
sme	SME (Subject Matter Expert)	VII: Expert Consensus	Incorrect	Keyword matched an acronym definition in a glossary, not an expert-consensus finding. More info needed to determine evidence.

'Basis for the Judgement' was used to build context and rule-based mapping to link attribution context to keywords from lexicon

Implementation in the Analytical Notebook

The procedure was built directly into the analytical Jupyter notebook, with the pipeline implemented in Python and its libraries and extensions inventoried in the data and code availability statement in Appendix F, so that it is transparent and re-runnable. A dedicated validation step draws the stratified sample from the master table of accepted matches, exports it to a spreadsheet, a tabular review file with one row per sampled match, carrying the keyword, the sentence of context, the risk, and the section for each instance, and provides adjacent columns in which the reviewer records the correct-or-incorrect judgment and the corrected type. The scored spreadsheet, the same file once the reviewer has entered the judgments, is then read back into the notebook, where a second step computes, for each evidence type, the precision, the proportion of the engine's

assignments of that type that the reviewer confirmed, and an overall Cohen kappa coefficient comparing the engine's labels to the human labels across the full sample. Cohen's kappa is used in preference to simple percent agreement because it discounts the agreement expected by chance, so that a high value reflects genuine concordance rather than the coincidence of two raters both favoring a common category (70). Cohen's kappa is the standard measure of inter-rater agreement across medical, behavioral, and computational-linguistics research, where it is used precisely to certify that a coding or classification scheme reproduces expert judgment rather than chance, which is why it is adopted here as the headline indicator of classifier validity (72). Conducting the validation inside the same Jupyter notebook, in the Python environment that runs the classifier, ties the reported accuracy to the exact version of the classifier that produced the analysis, and re-running the classifier regenerates both the sample and its evaluation.

How Classification Accuracy Propagates to the Findings

This validation is central rather than a formality because the engine's evidence type counts are not the final result but an input to it: they serve as the independent variable in the evidence-to-LoE correlations, and they determine the type-share composites and the structure of the evidence networks. Classification error therefore does not merely look untidy; it propagates. Given that the error is non-differential, unrelated to the LoE score being predicted, its expected effect on the correlations is attenuation, a bias toward zero, which means a noisy classifier can mask a real association but cannot manufacture a false one. In plain terms, the validation does two jobs at once. By comparing the engine's labels against an expert's reading of the same passages, it measures how often each evidence type is assigned correctly, and it does so type by type rather than only in aggregate, so the analysis knows which counts are solid and which are noisier. Because any remaining misclassification is unrelated to the score being studied, its only effect is to blur a real

relationship toward zero rather than to create a false one, which is what lets a weak or null finding be read as conservative rather than as an artifact of the instrument. Per-type precision identifies which type counts are trustworthy enough to interpret, and the overall kappa establishes the reliability band of the instrument as a whole; together they govern how strongly the downstream findings may be stated. Taken together, these checks make the classifier an instrument with known and reportable error characteristics rather than an opaque black box, which is what allows its evidence counts to carry weight in an analysis whose conclusions bear on governance decisions.

Reliability Benchmarks and Interpretation

Two complementary conventions anchor the interpretation. Cohen's kappa is read against the bands of Landis and Koch, in which values from 0.00 to 0.20 indicate slight agreement, 0.21 to 0.40 fair, 0.41 to 0.60 moderate, 0.61 to 0.80 substantial, and 0.81 to 1.00 almost-perfect agreement; a kappa of 0.61 or above is the conventional minimum for a defensible measurement instrument, and 0.81 or above is regarded as near-perfect (70, 71, 72). The Landis and Koch bands provide the standard, widely cited interpretation of how much agreement a kappa value represents, translating an otherwise abstract coefficient into descriptive categories from slight to almost-perfect; the same bands are used across medical, behavioral, and computational-linguistics research to judge whether a coding or classification scheme is reliable enough to trust, which is why they are applied here to characterize the classifier. Krippendorff's alpha supplies the second, more demanding convention for drawing conclusions from coded data: a value of 0.80 or above licenses firm conclusions, a value between 0.667 and 0.80 supports only tentative ones, and values below roughly 0.667 are not considered reliable enough to base substantive claims upon (64). Furthermore, Krippendorff's alpha is a reliability statistic for coded data that, like kappa, corrects for the agreement that would occur by chance, but generalizes across any number of coders and

category types; reporting both holds the validation to a recognized and demanding standard rather than to an informal impression that the coding looked consistent (64).

These thresholds set the strength of the language used in reporting rather than acting as a single pass-or-fail gate. Where the overall kappa reaches 0.80 or above, the evidence type findings are stated directly; where it falls between 0.667 and 0.80, they are framed as tentative and exploratory; and were it to fall below that range, the analysis would be restricted to the evidence types whose per-type precision is high, direct human spaceflight, animal, and cellular evidence validated most cleanly, or the type-share composite would be reported as descriptive only, with the correlation results carrying an explicit measurement-error caveat and a sensitivity analysis confined to the reliable types. Tying the strength of the stated conclusions to a measured reliability threshold, rather than reporting every finding with uniform confidence, follows standard guidance in content analysis and inter-coder reliability, where the agreement coefficient is treated as a precondition governing how firmly coded results may be interpreted (71, 73). The completed validation confirmed this: across the final, fully scored sample of 287 matches the overall Cohen kappa was 0.923 (95% CI 0.879 to 0.968), which falls in the almost-perfect band of Landis and Koch, and per-type precision ranged from 90 percent for expert-opinion text to 100 percent for the clinical, animal, cellular, and computational types, as summarized in the validation results below; the wording of the evidence type conclusions follows the rule stated here.

The rules that decide whether a keyword match counts as evidence, and as which evidence type, were not specified once and applied blindly. They were developed and then locked through an iterative human-annotation procedure. In each round, a stratified sample of classified matches was drawn across all eight evidence types, targeting at least 40 per type so that no category rested on too few examples (64). A trained reviewer judged every sampled match in context, recording

whether the assigned type was correct, supplying the correct type where it was not, and noting the reason. Those notes, rather than the analyst's prior assumptions, drove each successive refinement: terms the reviewer judged non-evidential were moved to the denylist; the requirement for an explicit attribution or finding signal was tightened wherever bare descriptors had slipped through; the source-over-mechanism precedence was added after the reviewer flagged cellular findings that were really spaceflight or analog data; and guards for hedged, forward-looking, causal, and list-only language were introduced in direct response to specific misclassifications. Refinement continued only until agreement stabilized at a high level across every type, at which point the lexicon and the acceptance gates were frozen and the final accuracy was measured on a fully scored validation sample. Locking the mapping in this way anchors the classifier to what trained human reviewers accept as evidence rather than to an a priori keyword list, and it is the basis on which the downstream volume and complexity measures can be trusted.

On the final, fully scored validation sample of 287 matches stratified across the eight evidence types, the classifier agreed with the human reviewer on 96.2 percent of cases, corresponding to a Cohen's kappa of 0.923 with a 95 percent confidence interval of 0.879 to 0.968 (70). On the Landis and Koch (1977) scale this falls in the almost perfect band, the highest of the conventional categories and well above the 0.80 level commonly treated as excellent agreement (71, 72). Agreement was uniformly high across categories rather than carried by one easy type: Types III, IV, V, and VI reached 100 percent, Types I and II reached 95 percent, Type VIII reached 92.5 percent, and Type VII, historically the most difficult because expert-opinion language is easily confused with governance and process text, reached 90 percent. Furthermore, this validation matters because every substantive result in the study, namely the relative complexity of each risk and the relationship between evidence volume and the assigned LoE, depends on the evidence

types having been assigned correctly. The first pass of the classifier fell short of this standard, agreeing with the reviewer on fewer than 80 percent of cases, and the climb to almost-perfect agreement came through several rounds of refinement. The most consequential factor was context: early versions credited a type whenever its keyword appeared, so that a sentence describing a spaceflight experiment that merely mentioned cells was counted as cellular evidence when the operative context was spaceflight data. Adding contextual gating and the source-over-mechanism precedence in the Python pipeline, so that a match is accepted only when the surrounding text supports the type and the most direct source prevails, was the core change that resolved this class of error. A classifier that systematically mislabeled evidence would invalidate those analyses no matter how carefully the counts were aggregated afterward. Establishing near-perfect, category-balanced agreement before any substantive analysis therefore licenses interpreting the volume and complexity measures as reflecting the evidence base itself rather than classifier error. The aim was a defensible operating point rather than perfect agreement. Tightening the gates toward 100 percent correctness would have discarded legitimately attributed evidence and lost real data, so the engine was tuned to a transparent boundary, well above conventional thresholds yet not claiming to be error-free, that draws on prior literature to map evidence phrases reliably within a stated margin (69).

The small number of remaining disagreements, 11 of the 287 reviewed matches, were concentrated in a single well-characterized pattern rather than scattered across the categories: a legitimate evidence type term appearing inside a list or in passing, without surrounding text sufficient to establish that it functioned as evidence. It is worth noting that evidence is not documented uniformly across packages: some record points as bullet lists, others as narrative prose, and others as photographs of graphs or tables, often in combination and without a single

standard practice, so a measure of flexibility in the matcher also reflects the genuine heterogeneity of the source documents. Examples included a bare subject-matter-expert or working-group label, the Human Integration Design Handbook named within a list of facilities and tools, a historical mission named only in passing, and a preflight reference occurring in an astronaut-selection note. These cases were frequently ambiguous to the human reviewer as well, who at times could not determine from the extracted fragment whether the term denoted evidence at all, and they therefore reflect the same fragmentary, optical character recognition derived context described earlier rather than a systematic classifier failure. They bound the precision attainable on this corpus. A stricter context gate capable of removing them was evaluated during development but was found to suppress the recall of legitimately attributed evidence in other categories, and so was not imposed. A separate and smaller source of discrepancy is confined to the multi-label annotation: when two distinct sources are listed with only spaces between them rather than an explicit conjunction or delimiter, the secondary source is not jointly tagged. Given that each evidence type is scanned for independently, this affects only the secondary-label display and not the underlying type counts, which already register each source separately.

A reasonable question is how a gold standard built by a single expert annotator can be trusted, and the answer turns on what kind of judgment the task requires. The annotation here is not subjective opinion coding, where several raters are pooled to average out personal interpretation; it is the application of a fixed, published codebook, in which each passage is mapped to one of eight evidence types whose boundaries are defined in advance by the agency's own evidence hierarchy. The question for each instance is therefore whether the engine's label is correct against those definitions, not how one reader happens to feel about it, and correctness against an objective definition is something a single qualified expert can establish. Reliability, the separate property of

returning the same answer under repetition, was not left to the annotator at all; it was measured independently by re-running the classifier under perturbed settings, so the human reference was asked only to settle validity. Specialization is the decisive consideration. Reading whether a sentence reports head-down-tilt bed rest as a spaceflight analog, an in-flight cellular assay, an animal model, or a standards citation requires fluency in spaceflight physiology, in the eight-type hierarchy, and in the specific risks; a panel of non-expert annotators would introduce more error than it removed, since the boundary cases that matter are exactly the ones that demand domain knowledge to resolve. The annotator was, moreover, not working unaided: ambiguous instances were adjudicated in consultation with subject-matter experts and against the published definitions, and the attribution rules were locked through repeated rounds of expert review before the final accuracy was measured, so that the standard reflects a stabilized expert reading rather than one rater's momentary call. Two further properties bound the consequences of any residual error. The validated agreement is itself high and balanced across all eight types, a Cohen kappa of 0.923 in the almost-perfect band, which is the direct evidence that the instrument is trustworthy; and since any remaining misclassification is unrelated to the LoE being predicted, its only statistical effect is to blur a real association toward zero, so an imperfect annotator can weaken a finding but cannot manufacture one. Furthermore, for a baseline-establishing study of this kind, a single domain-expert reference is a standard and accepted design, and the planned addition of a second independent annotator, or an annotation team, to report a formal inter-rater reliability coefficient is named as the immediate next step that would convert this validated single-expert standard into a multi-rater one.

Reproducibility and Computational Environment

The entire pipeline was implemented in Python within a Jupyter notebook environment, using established open-source libraries for data manipulation, statistical computation, network analysis, and optical character recognition, with the last performed by the Tesseract engine (60) (99, 100). The analysis is fully scripted, so that the corpus, the lexicon and catalogs, the extraction rules, and the statistical and network procedures together constitute an executable specification that can be re-run to reproduce every reported result. This reproducibility is a deliberate design objective, both because it is required for an examination whose conclusions bear on governance decisions and because it allows the framework to be re-applied as new submissions are approved, supporting an updatable account of the portfolio over time.

The specific computational stack is reported here for reproducibility. Data manipulation used pandas and NumPy; the statistical analyses used SciPy for the rank correlations, the Fisher exact test, and the Kruskal-Wallis test, and stats models for the linear mixed-effects models; the penalized logistic regression used the Firth procedure; and the classification-agreement statistic used scikit-learn, which also supplied the cosine-similarity computation for the evidence-profile network, the principal component analysis used for ordination, and the random-forest and leave-one-out routines used in the score-reconstruction sensitivity analysis. Each of these statistical and computational methods is defined in plain language at the point where it is applied, with its purpose, its inputs, and the interpretation of its output given alongside the corresponding analysis. Network construction, projection, and community detection used UCInet and NetworkX, including its greedy modularity-maximization routine, the established software packages for social-network analysis and the algorithm that groups nodes into communities by iteratively merging the clusters that most increase the network's modularity, a measure of how cleanly it

separates into groups (100, 101). Visualizations were produced with MuxViz and Matplotlib, including its patch and grid-specification modules for the custom schematic figures, with Plotly used for the interactive charts embedded in the deliverables, and image handling used the Python Imaging Library. Optical character recognition used the Tesseract engine as noted above. The network diagrams were arranged with a force-directed, spring-model layout, which positions each node as though connected risks were joined by springs while all nodes repel one another, so that densely linked risks settle close together and the larger structure becomes legible (102, 103). Each of these is a standard, openly available library used here in its published form rather than modified for this study, so the full computational stack can be reproduced from the package versions recorded in Appendix F.

Statistical Analysis Plan

The Two-Tier Design and the Primary Outcome

The statistical analysis follows a pre-specified two-tier design. The first tier is the portfolio-level analysis, in which the unit is the risk and the analysis asks whether characteristics of a risk's documented evidence predict the LoE assigned to it. The second tier is the mission-stratified analysis, in which the unit is the risk by mission observation and the analysis asks how the LoE and the evidence base behave across the Design Reference Mission continuum. The mission-stratified tier is designated the primary analysis of the evidence-to-scoring relationship because its larger number of observations affords greater power, while the portfolio tier provides the higher-level summary of alignment across the portfolio. This ordering reflects a substantive point about the data, not merely a question of sample size. Risks are not directly comparable to one another at the portfolio level, because a single Design Reference Mission can affect different risks in very different ways, and the operations and long-term health outcomes of the same risk can diverge

sharply across the mission continuum. Examining the LoE and the evidence base within the risk by mission structure, separately for the operations and long-term health tracks, therefore both enlarges the number of observations and respects these differences, whereas a portfolio-level summary that collapses each risk to a single value cannot fully reflect how mission context shapes the evidence. The Design Reference Missions and their differing demands on each risk are described in the background so that this distinction is clear to the reader.

For the portfolio-level analysis, the primary outcome for each risk is its maximum, that is worst-case, LoE across the Design Reference Missions for which it is scored. The maximum was chosen deliberately over the median, mode, or minimum. Given that the LoE is numerically inverted, with higher numbers denoting weaker evidence, the maximum identifies the mission context in which a risk's evidentiary support is weakest, which is the context of greatest operational concern and the appropriate evidence floor for a safety-oriented examination. The median or mode would obscure the worst case by averaging over mission contexts of unequal consequence, and the minimum would report only the best-supported context, which is the least informative for risk governance. An effect of the maximum aggregation is that the strongest scores are eliminated for risks whose support weakens in any mission context, which is reflected in the portfolio distribution of the outcome. Examining the full portfolio, rather than a handful of individual risks, is what makes the exercise informative in the first place: only a census reveals whether the relationship between documented evidence and assigned score is a portfolio-wide regularity or a risk-by-risk idiosyncrasy, and only the complete set of Design Reference Missions, across both the in-mission operations and long term health outcomes, exposes where evidentiary support is strongest and weakest along the mission continuum. This portfolio-wide, exploratory framing is precisely what

the worst-case aggregation serves, surfacing the decision-relevant evidence floor for every risk so that systematic patterns, and their absence, can be detected and tested rather than assumed.

Given that this worst-case aggregation removes within-risk variation by construction, a natural question is whether any portfolio-level null it produces is an artifact of the aggregation rather than a substantive result. It is not. The two tiers are built to answer different questions: the portfolio tier asks whether a risk's worst-case evidentiary support, the decision-relevant floor, tracks the quantity, quality, or diversity of what the risk documents, while the mission-stratified tier asks how support varies along the mission continuum, so that a null at the portfolio level and a gradient at the mission level are complementary answers to distinct questions rather than a contradiction. The worst-case score, moreover, is not a statistical convenience but the operationally meaningful quantity, the level at which a risk's evidence is weakest across the missions it must cover, so its independence from documented volume is a finding about the decision-relevant outcome and not a by-product of collapsing the data. The mission tier subsequently recovers precisely the variation the collapse sets aside, which is why both analyses are reported rather than either one alone.

All primary hypotheses, the corresponding tests, the multiple-comparison correction, and the primary outcome definition were specified before the analyses were conducted, in order to avoid hypothesizing after the results are known and the inflation of false-positive findings that selective post hoc analysis produces (104). The exploratory analyses, including the per-evidence type correlations, the network analyses, and the link-prediction analysis, are explicitly labeled as exploratory and are interpreted as hypothesis-generating rather than confirmatory. These tests were framed in consultation with Human System Risk Board subject-matter experts, whose discussion of where the LoE process might be unevenly applied helped establish the analytic gap this research

addresses and grounded the questions in a practical concern within the agency. This separation of pre-specified confirmatory tests from exploratory analyses is maintained throughout the reporting of the results.

The Linking Framework and Variable Definitions

The analysis links the documented evidence to the assigned scores through a defined set of variables. The exposures are the characteristics of the documented evidence: the total evidence volume, computed as the count of accepted evidence matches; the evidence diversity, computed as the count of the eight types represented with at least one match; the quality-weighted composite, computed under the eight-to-one weighting; the presence of direct human spaceflight evidence, coded as a binary indicator; and the evidence type concentration, computed as a concentration index defined below. The outcomes are the validated LoE scores, analyzed as the worst-case score per risk in the portfolio tier and as the per-mission score in the mission-stratified tier, together with the disposition values used in the alignment assessment. Within each package the evidence was extracted in full and read as a whole, since a custodian documents the evidence base for a risk across the entire package rather than restating it for every mission; the LoE, by contrast, is recorded specifically for each Design Reference Mission and separately for the operations and long term health outcomes, and these scores were triangulated against the rationale in the risk posture table. The mission-stratified tier exploits this structure by treating each risk-by-mission-by-track score as an observation. Not every risk carries every combination: some risks are not scored for missions their nature excludes, as the surface-dust risk has no low-Earth-orbit Design Reference Mission score, and some are scored on only one track, as the Team risk is scored for operations but carries no long-term health pathway. Where a posture cell contained an internally inconsistent entry, such as a value paired with the wrong label, a compound entry, or a blank where a score was expected,

it was treated as erroneous or missing and excluded rather than inferred. This exclusion is a deliberate limitation: it lowers the apparent volume of LoE scores, but it keeps the analysis an accurate portrayal of what was actually submitted, an effect revisited among the limitations. Covariates in the mission-stratified models include the Design Reference Mission, the submission year, and a composite risk-complexity measure.

A consequential feature of this linking framework is that its two sides are recorded at different levels of granularity, and the analysis is built to respect that asymmetry. The evidence characteristics on the exposure side are extracted at the level of the whole Risk Summary Package: the natural language engine reads the entire package as a single document and returns one evidence type profile per risk, because the documented evidence is written for the risk as a whole and is not partitioned in the source between the operations and the long term health outcome tracks. The scored quantities on the outcome side, by contrast, are recorded per outcome and per mission: the risk custodian assigns a LoE, a likelihood, a consequence, and a supporting rationale, together with the Board's disposition, separately for the operations track and for the long term health track, and separately for each Design Reference Mission in which the risk is evaluated. The two tracks capture different time horizons of harm. The operations track concerns risks to crew health and mission performance during a mission, where an effect can compromise the crew or the mission objectives in real time, whereas the long-term health track concerns chronic or latent outcomes that may not appear until long after a mission ends. The agency monitors them separately because a risk can be well controlled operationally yet still carry a substantial long term health consequence, or the reverse, so a single combined score would obscure which horizon drives the posture. The analysis therefore anchors a single whole-package evidence profile to the family of per-outcome, per-mission scores and rationales that the same package carries, relating the one

evidence characterization a risk presents to each of the distinct evidentiary judgments recorded against it. Put simply, each risk offers a single description of its evidence base, and that one description is compared against every separate score the package assigns across its missions and its two outcome tracks. This asymmetry is the reason the portfolio tier reduces each risk to a single worst-case score while the mission-stratified tier preserves the per-outcome, per-mission resolution, and it is why the evidence type counts do not change with the outcome track even though the LoE and its rationale do.

Statistical Tests

The relationship between an ordinal or continuous evidence characteristic, with the evidence volume, the evidence diversity, and the quality-weighted composite treated as continuous counts and the presence of direct human spaceflight evidence as a binary indicator, and the ordinal LoE was evaluated by the Spearman rank correlation (105) (the formal definitions of the statistical and network measures are collected in Appendix C), a non-parametric measure of monotonic association appropriate to ordinal data and robust to non-linearity and to the small portfolio sample (106). In practice, this measures whether risks with more, or more varied, documented evidence also tend to receive higher LoE scores. It works on ranks rather than raw values, asking whether the ordering of risks by evidence matches their ordering by score, which suits a four-point ordinal scale and a small portfolio in which a few extreme values could otherwise dominate. A coefficient near zero indicates that evidence and score move largely independently, a positive value that they rise together, and a negative value that they move in opposite directions. The coefficient is defined as

$$\rho_s = 1 - (6 \sum d_i^2) / (n(n^2 - 1)),$$

where d_i is the difference between the ranks of the two variables for observation i and n is the number of observations, yielding a value between minus one and plus one. 95 percent confidence intervals for the Spearman coefficient were obtained through the Fisher z -transformation throughout. A rank-based coefficient is the appropriate choice here because the LoE is an ordinal grade rather than an interval measure, and the hypothesized evidence-to-score relationship is monotonic rather than necessarily linear; the Spearman coefficient therefore makes no distributional assumptions and is robust to the small portfolio and to outliers, which is why it is preferred over the Pearson product-moment correlation that would impose the linearity and interval scaling the data do not satisfy (105, 106). The association between a binary evidence indicator and the ordinal score was additionally examined through the point-biserial correlation, which quantifies how strongly the presence or absence of direct human spaceflight evidence relates to the score, testing whether risks that document that most direct evidence type tend to be scored differently from those that do not,

$$r_{pb} = [(M_1 - M_0) / s_n] \cdot \sqrt{(p/q)},$$

where M_1 and M_0 are the mean scores for the two groups defined by the indicator, s_n is the standard deviation of the score, and p and q are the proportions of observations in the two groups. In this analysis the two groups are the risks that document direct human spaceflight evidence and those that do not, the two means are their average LoE values, and the coefficient summarizes how far apart those averages sit relative to the overall spread of scores. A value near zero would therefore indicate that documenting this evidence type makes little difference to the score a risk receives.

The association between the presence of direct human spaceflight evidence and a strong-or-moderate score was tested by the Fisher exact test, which is exact for the small, sparse two-by-two

table that the census produces. The Fisher exact test is used in place of a chi-square test because the table is small and sparse, computing the exact probability of an association at least as extreme as the one observed rather than relying on a large-sample approximation. The odds ratio expresses the strength and direction of that association: it is the odds of a strong-or-moderate score among risks documenting direct human spaceflight evidence divided by the same odds among those that do not, so a value of one indicates no association, a value above one that the evidence type accompanies higher scores, and a value below one the reverse. Reporting the odds ratio with its confidence interval conveys not only whether an association is statistically detectable but how large it is and in which direction it runs. Where the odds ratio was estimable, it was computed as the cross-product ratio and accompanied by a Woolf-type confidence interval,

$$OR = (a d) / (b c), \text{Var}(\ln OR) = 1/a + 1/b + 1/c + 1/d.$$

In the portfolio, every risk documented at least some direct human spaceflight evidence, which produced a degenerate two-by-two table with an empty cell, quasi-complete separation, and an undefined maximum-likelihood odds ratio. To obtain a finite, interpretable estimate under separation, Firth penalized logistic regression was applied, which augments the log-likelihood with the Jeffreys-prior penalty,

$$\ell^*(\beta) = \ell(\beta) + \frac{1}{2} \log |I(\beta)|,$$

where $\ell(\beta)$ is the ordinary log-likelihood and $I(\beta)$ is the Fisher information, a correction that removes first-order bias and yields finite estimates when ordinary estimation fails (107, 108). Two Firth models were fitted, one with a binary indicator of human-spaceflight-evidence presence and one with the continuous count of such evidence, the latter testing whether a greater quantity of direct human evidence is associated with stronger scoring. Firth penalized regression is the standard remedy when a predictor perfectly separates the outcome, as happens here because every

risk documents at least some direct human spaceflight evidence: ordinary logistic regression would return an infinite coefficient, whereas the penalty produces a finite, interpretable estimate and a usable confidence interval. Its role in this analysis is therefore to let the human-spaceflight-evidence association be quantified at all, rather than reported only as an undefined or degenerate result. Across this section the choice of each test follows from the epidemiological character of the data, a complete census of a small occupational cohort measured on ordinal grades with sparse cells and non-normal distributions, for which exact, rank-based, and penalized methods are the standard tools and large-sample parametric tests would be inappropriate.

Whether evidence diversity differs across the ordinal score groups was tested by the Kruskal-Wallis (109) H-test, the non-parametric analog of one-way analysis of variance for comparing a continuous or ordinal measure across more than two independent groups (110). It is appropriate here because the LoE groups are ordinal and the per-group samples are small, conditions under which the normality assumptions of analysis of variance do not hold. In this analysis it asks whether the diversity of documented evidence differs systematically among risks grouped by their assigned score, so that a significant result would indicate that at least one score group tends to document a broader or narrower range of evidence types than the others. The statistic is

$$H = [12 / (N (N + 1))] \sum (R_j^2 / n_j) - 3 (N + 1),$$

where N is the total number of observations, n_j is the number in group j , and R_j is the sum of ranks in group j . The accompanying effect size was the eta-squared, a measure of how much of the variation in evidence diversity is explained by the score grouping, on a scale from zero, where the score groups do not differ, to one, where group membership accounts for all the variation (110), derived from H , with a bootstrap confidence interval. Reporting the effect size alongside the test guards against over-reading a difference that is statistically detectable but small in magnitude,

which matters where the score groups are of uneven and modest size. Concentration of a risk's evidence across the eight types was summarized by the Herfindahl-Hirschman index, The Herfindahl-Hirschman index is a standard concentration measure drawn from economics, where it gauges how concentrated a market is among its firms; applied to evidence it captures whether a risk's documentation is dominated by a single type or spread evenly across many, complementing the simple diversity count by reflecting the balance, and not only the number, of evidence types present (111).

$$HHI = \sum s_i^2,$$

where s_i is the share of a risk's evidence accounted for by type i , so that the index approaches its minimum when evidence is spread evenly across types and equals one when a single type accounts for all of a risk's evidence.

Three pre-specified tests constitute the primary confirmatory family for Aim 1, the Spearman correlation of evidence volume against the score, the Spearman correlation of the quality-weighted composite against the score, and the Fisher exact test of human-spaceflight-evidence presence against strong-or-moderate scoring. These three were designated confirmatory in advance because each addresses the central question of Aim 1 through a different lens, the quantity of evidence, the quality-weighted composite, and the presence of the most direct evidence type, so that the conclusion does not depend on any single way of operationalizing the evidence base. All remaining analyses, including the per-type correlations, the diversity and concentration tests, and the network measures, are treated as exploratory and hypothesis-generating rather than confirmatory, and questions the observational design cannot resolve, such as the causal direction of any association, are explicitly out of scope. To control the family-wise error rate across this family, the probability of at least one false-positive across the three tests considered together,, a Bonferroni correction

was applied, a deliberately conservative adjustment that divides the significance threshold evenly among the tests so that the chance of any spurious positive across the family stays at the nominal level, dividing the nominal two-sided alpha of 0.05 by the three tests to yield a corrected threshold of

$$\alpha^* = 0.05 / 3 = 0.0167.$$

All confidence intervals are reported at the 95 percent level and all tests are two-sided because the analysis does not presume the direction of any association in advance, so that a relationship in either direction would be detected. The exploratory analyses were evaluated against the nominal 0.05 threshold and are labeled as such. Given that a non-significant result in an underpowered census cannot by itself distinguish a true null from an undetectable effect, the principal portfolio correlation was additionally subjected to an equivalence test using the two one-sided (112) tests procedure, with a region of practical equivalence defined as a correlation magnitude below 0.10, a band below which any true association is treated as too weak to matter in practice, on the rationale that a correlation of that magnitude explains less than one percent of variance and is practically negligible (113, 114). The equivalence test evaluates whether the observed effect is small enough to be declared practically equivalent to zero, complementing the conventional test of difference from zero. In this analysis a significant equivalence result would license the stronger statement that evidence volume and the assigned score are, for practical purposes, unrelated, rather than only the weaker statement that no relationship was detected. It was paired with a complementary Bayes-factor approximation derived from the Bayesian information criterion A Bayes factor quantifies how much more probable the observed data are under one hypothesis than under the other, and the Bayesian information criterion provides a practical approximation to it that needs no prior to be specified and is widely used for exactly this purpose across the behavioral and biomedical sciences

(115, 116), which expresses the relative support the data lend to the null and the alternative hypotheses, so that a retained null can be read as positive evidence for the absence of a meaningful association rather than merely as an absence of evidence for one. Together, these procedures address the central difficulty of a small census. A p-value answers a narrow question, namely how surprising the data would be if there were truly no association, and it is informative when a study has enough observations to detect an effect of the size that would matter. With only 29 risks, however, an ordinary non-significant p-value is ambiguous, because it cannot separate a genuine absence of association from a real effect the study was simply too small to detect. The equivalence test and the Bayes-factor approximation are reported alongside the p-value precisely to resolve that ambiguity: the first asks whether the effect is small enough to be treated as null, and the second weighs the relative support for the null against the alternative, so that a retained null can be stated as positive evidence of no meaningful relationship rather than as mere silence. This is why the analysis does not rely on the p-value alone, while still reporting it for transparency and comparability.

Mission-Stratified Primary Analyses

At the mission resolution, the per-mission LoE was first summarized across the Design Reference Mission continuum to characterize the gradient of evidentiary support, and the association between evidence and score was evaluated within each Design Reference Mission by Spearman correlation. The systematic behavior of the LoE across the continuum was then modeled with a linear mixed-effects model that accommodates the repeated-measures structure of the corpus, in which each risk contributes multiple mission observations. The choice of a linear mixed-effects model was made a priori, from the established treatment of repeated-measures and longitudinal data in the biostatistical literature: when the same units are observed under several

conditions, the observations within a unit are correlated, and a model that ignores this dependence understates the uncertainty of its estimates. The standard remedy, adopted here, is to add a unit-specific random effect, so that the random intercept for risk absorbs each risk's baseline tendency and the fixed effects are estimated against the within-risk variation that remains (117, 118, 119). The fixed-effect structure was specified to match the questions the analysis asks rather than chosen by data-driven search: the evidence term carries the core evidence-to-score association, the Design Reference Mission term the mission gradient, their interaction whether that association changes along the continuum, and the year and complexity terms adjust for temporal drift and for differences in intrinsic risk complexity that might otherwise confound the comparison. The model is

$$\text{LoE}_{ij} = \beta_0 + \beta_1 \text{Evidence}_{ij} + \beta_2 \text{DRM}_j + \beta_3 (\text{Evidence} \times \text{DRM})_{ij} + \beta_4 \text{Year}_{ij} + \beta_5 \text{Complexity}_i + u_i + \varepsilon_{ij},$$

where LoE_{ij} is the LoE for risk i under mission j , Evidence_{ij} is the documented evidence measure, DRM_j indexes the Design Reference Mission along the continuum, the interaction term tests whether the evidence-to-score relationship varies across the continuum, Year_{ij} is the submission year, Complexity_i is the composite risk-complexity covariate, u_i is a random intercept for risk that captures the repeated-measures dependence, and ε_{ij} is the residual (117) (118, 119, 120). The Design Reference Mission main effect tests whether the LoE shifts systematically across the continuum, and the interaction tests whether evidence predicts the score differently at different points along it. The model was fitted separately for the In-Mission Operations and Long Term Health outcomes. These models are the means of testing, rather than assuming, that the evidence-to-score relationship holds across missions and outcome tracks. Fitting them separately for operations and long-term health lets the analysis confirm whether a pattern

seen in one track also appears in the other, instead of presuming the two behave alike. A complementary degradation analysis examined, which tracks how far the evidentiary basis weakens toward the more demanding end of the mission continuum, for the risks scored at both the low-Earth-orbit and Mars ends of the continuum, the change in LoE across that span and its association with the quantity of direct human spaceflight evidence. The behavior of evidence breadth across the continuum, where breadth refers to the number of distinct evidence types a risk documents, was evaluated by stratified Spearman correlation of an evidence-breadth metric against mission rank, conducted separately for operations and long-term health. This asks, separately for each outcome track, whether risks tend to document a wider or narrower range of evidence types as missions move from low-Earth orbit toward Mars, so that a negative correlation would indicate the evidentiary foundation narrows precisely as mission demands grow. The supporting computations are implemented in the analytical notebook and the interactive dashboard described in Appendix F.

In this analysis the two ranked variables are the documented evidence measure and the assigned LoE. For the portfolio test of volume, the evidence variable is the total count of context-accepted evidence matches summed to one value per risk, and the score variable is the worst-case LoE taken as the maximum across the eight Design Reference Missions. To put it plainly, the evidence-volume variable is how many accepted evidence citations a risk's package contains, and the LoE variable is the agency's four-point rating of how well supported the risk is, summarized per risk as its strongest mission-level score. Each variable is converted to a rank from one to 27, that is, across the 27 of the 29 parent-level risks that carry a scored worst-case LoE, the two unscored risks (Carcinogenesis and Urinary Retention) being excluded, with tied values assigned the average of the ranks they span, and the difference d_i is taken between a risk's two ranks. A

worked example makes the logic concrete: the Immune risk, among the most heavily documented in the portfolio, carries only a Moderate worst-case LoE and so sits below the most strongly scored risks, while several sparsely documented risks carry high scores. Because the two rankings diverge in this way, the rank differences are large rather than near zero, and it is the accumulation of such mismatches that holds the coefficient near zero. The resulting value of $+0.046$ implies that the two rankings share well under one percent of their variance, since the squared coefficient is only 0.002 , which is the formal sense in which documented volume and the assigned score move almost independently. In plain terms, a risk can sit near the top of the evidence ranking while sitting only in the middle of the score ranking, and each such mismatch pulls the correlation toward zero. The coefficient therefore summarizes, across all 29 risks at once, how closely the ordering by documented evidence tracks the ordering by assigned score.

The variable entering the Fisher and Firth analyses is a binary indicator, set to one when a risk documents any Type I direct human spaceflight evidence and zero otherwise, cross-tabulated against a binary score indicator set to one when the LoE is Strong or Moderate. Both variables are reduced to a yes-or-no form so that the test addresses a single decision-relevant question, whether documenting any of the most direct evidence type is associated with a strong-or-moderate rating, and because the exact and penalized methods used here are designed for the small, sparse two-by-two table that this dichotomy produces. In this corpus the resulting two-by-two table is degenerate, meaning one entire row of the table is empty, because the human-spaceflight indicator equals one for all 27 scored risks: the entire population falls in the evidence-present row, the evidence-absent row is empty, and the cell that would capture risks with no human spaceflight evidence and a strong score contains no observations, which is why the maximum-likelihood odds ratio is undefined and the Fisher p-value is exactly 1.000. The per-risk evidence-presence values behind

this degenerate table are tabulated in Appendix G. The Firth correction, a standard small-sample adjustment to logistic regression, adds the Jeffreys penalty to the log-likelihood, a term derived from the Jeffreys prior that gently pulls the estimate back from the extremes (107), which removes the first-order bias, the systematic tendency of ordinary maximum-likelihood estimates to be too large in small or sparse samples, and yields a finite estimate, a defined and reportable odds ratio in place of the infinite value that ordinary logistic regression returns, even under this complete separation; applied here it returns an odds ratio of 0.717 with a p-value of 1.000, confirming in a numerically stable way what the degenerate table already implies, namely that the presence of human spaceflight evidence is universal and therefore cannot discriminate among scores.

The variable tested by the Kruskal-Wallis H-test is evidence diversity, the integer count of the eight evidence types for which a risk documents at least one accepted match, and the grouping variable, the categorical variable by which observations are sorted into groups for comparison, is the ordinal Level of Evidence. The scale defines four grades, but only three are populated at the worst-case portfolio level, since no risk carries a Strong worst-case score: nine Moderate, 13 Weak, and five Speculative risks. Each risk's diversity value is replaced by its rank across all 27 risks, the ranks are summed within each score group to form R_j , and the group sizes n_j are nine, 13, and five; the statistic compares the average rank within each group against the grand mean rank, so that a large H would indicate that more diverse risks concentrate in the stronger score groups. The accompanying eta-squared expresses the proportion of rank variance attributable to score group, with a bootstrap confidence interval, and the non-significant result indicates that the breadth of evidence types a risk documents does not separate the score strata.

Each term in the mixed-effects model maps to a specific analysis variable. The outcome is the per-mission operational LoE for risk i under mission j , that is, the score recorded for that single

risk and mission cell, drawn from the posture-level master table rather than the worst-case collapse, so that all valid risk by missioncells contribute. The evidence term is the standardized documented-evidence measure, the quality-weighted composite expressed as a z-score, that is, recentered to a mean of zero and rescaled so that one unit equals one standard deviation, so its coefficient is read per standard deviation; the mission term is an ordinal index running from zero for short-duration low Earth orbit to seven for long-duration Mars-planetary missions; the interaction term is the product of the standardized evidence and the mission index; submission year and risk complexity enter as covariates; and the random intercept for each risk absorbs the correlation among the eight observations contributed by the same risk, so that the repeated mission cells are not treated as independent. Interpreted in these units, the significant mission coefficient means that each step along the mission continuum raises the expected LoE value by a fixed increment regardless of evidence, which is the model-based statement of the observed climb from a mean of 1.39 at short low Earth orbit to 2.68 at long Mars-planetary missions, while the non-significant interaction means that this climb has the same slope whether a risk documents little or much evidence, which confirms in the mission-stratified model the same independence of assigned score from documented volume that the portfolio correlations show.

The confirmatory tests of Aim 1 were framed as formal null-hypothesis tests. For each evidence dimension, the null hypothesis stated that there is no monotone association between the documented evidence measure and the assigned LoE, and the alternative stated that such an association exists; the family of three primary tests was evaluated against a Bonferroni-corrected significance threshold of 0.0167. The Bonferroni correction guards against false positives that arise from testing several hypotheses at once: dividing the conventional five percent level by the three primary tests keeps the chance of any spurious significant result across the whole family near five

percent, at the cost of being deliberately conservative (121). Given that a study of this size has limited power to detect small effects, so that with only 27 to 29 scored risks even a real but modest association can fail to reach significance, a limitation quantified by the formal power analysis reported with the results (91), the analysis plan specified in advance that a failure to reject the null would not be read as proof that no association exists, but would instead be characterized through its effect size and confidence interval and adjudicated with two complementary procedures: a two-one-sided-test equivalence analysis, which asks whether the observed effect is small enough to be practically negligible, and a Bayes-factor approximation which quantifies how strongly the data favor the null relative to the alternative. The mission-stratified mixed-effects model was specified to test two further nulls at once, namely that the LoE does not change across the Design Reference Mission continuum and that the strength of any evidence-to-scoring relationship does not vary by mission, the latter carried by the evidence-by-mission interaction term.

Network Analysis Procedures

The network analyses construct and characterize the risk-evidence structure of the portfolio. A bipartite structure, a network containing two distinct kinds of node with links running only between the kinds and never within them, here risks on one side and evidence types on the other (25, 26), linking risks to the evidence types they document was projected onto the set of risks to produce a risk-by-risk network in which an edge reflects shared evidence, with edge weights, the numerical strengths attached to those links, derived from the cosine similarity of the risks' evidence type profiles, a measure that treats each risk's documented mix of evidence types as a vector and scores how closely two such vectors point in the same direction, near one when two risks share a similar evidence profile and near zero when they share little (63). Principal component analysis, which re-expresses many correlated measures as a few uncorrelated summary

dimensions, was used to summarize the evidence-profile space and to inform the construction of the similarity structure (122). Community structure was detected using the greedy modularity-maximization algorithm, which partitions the network to maximize the modularity defined in the Background (101). Node prominence was characterized by degree, betweenness, and closeness centrality, the last measuring the inverse of a node's average distance to all other nodes. The local architecture was summarized by the triad census, a count of the configurations that every possible trio of risks can form; where the community detection shows the broad stands of the forest, the triad census inspects the individual trees. An exploratory link-prediction analysis, which flags pairs of currently unconnected risks that the network's structure suggests are most likely to be linked, examined which currently unlinked risks are structurally most similar, as a hypothesis-generating indication of risks that might share evidentiary foundations not yet jointly documented. The statistical significance of the observed community structure was assessed against a configuration-model null that preserves each node's degree while randomizing its connections, with 1000 randomizations generating the null distribution of modularity against which the observed value was compared. All network analyses are reported as exploratory. They are framed this way because the portfolio is a single, complete census of one agency's risks rather than a sample drawn for hypothesis testing, so the structures they surface, the clusters of shared evidence, the central risks, and the gaps between them, are best read as a map of where evidentiary foundations are concentrated or thin that generates hypotheses for targeted follow-up rather than confirmed claims about a wider population.

The network metrics reported here, including degree, density, and modularity-based community structure, are the standard descriptors of the network-science literature and are implemented across the established dedicated software and packages in the field, such as UCINET

and Gephi; NetworkX was selected as the implementation for this study because it integrates directly with the Python analytic pipeline, which keeps the network construction reproducible from the same scripted workflow that produces the rest of the analysis.

Thematic Analysis Procedures

The thematic analysis applied a deductive, directed-content-analysis approach, a qualitative method in which text is coded against a codebook fixed in advance from prior theory rather than one allowed to emerge from the data, to the package narratives, using a codebook of themes derived from the structural features of the corpus rather than allowed to emerge inductively (51, 65). The codebook comprised the following themes: explicit acknowledgment of evidence gaps, the status of countermeasures, the maturity of mechanistic knowledge, operational impact, temporal progression of the risk across missions, cross-risk dependency, evidence-referenced rationale, narrative-only rationale, and explicit acknowledgment of specific evidence gaps. Each package narrative was scanned for the presence and frequency of language corresponding to each theme, and the prevalence of each theme across the corpus was tabulated. The cross-risk dependency theme, which credits language in which one risk's narrative explicitly refers to another, was used to test whether risks are documented in isolation or with explicit reference to one another, which bears directly on the hypothesis that mechanistically related risks are nonetheless documented separately. The thematic results were integrated with the evidence and network findings to produce the synthesis that Aim 3 specifies.

Two features of this procedure bound the strength of the thematic claims and are stated for transparency. The coding was deductive and keyword-anchored, so it shares the precision-over-recall character of the evidence extraction: a theme was credited where its language appeared in a recognizable form, which favors specificity at the cost of missing idiosyncratically phrased

instances, so that the reported prevalences are best read as conservative lower bounds. Furthermore, the coding was applied programmatically from a fixed codebook rather than by several independent human coders, a formal inter-rater reliability coefficient was not computed; the reproducibility of the coding rests instead on the determinism of the rule set, which applies identical criteria to every package. A human double-coding of a corpus sample, yielding a chance-corrected agreement statistic, is the appropriate next step to establish the reliability of the cross-risk dependency prevalence on which the central qualitative finding depends, and it is identified here as such.

Mission-Impact Consequence Categories

The consequence dimension of the posture matrix is interpreted in the analysis with reference to the two operational consequence categories that the agency distinguishes for mission-impact assessment, the loss-of-mission category, which concerns outcomes that would compromise mission objectives, and the loss-of-crew category, which concerns outcomes that would threaten crew survival. These categories frame the interpretation of the posture and disposition values and inform the consequence-related elements of the documentation analysis, while the LoE remains the primary outcome of the statistical analyses.

Robustness and Validation

Several analyses evaluate the robustness of the findings to analytic choices. A weighting-sensitivity analysis recomputed the principal portfolio correlation under alternative weighting schemes, including an equal-weight scheme, to determine whether the eight-to-one weighting drives the result. A deduplication-sensitivity analysis recomputed the correlation on the raw, non-deduplicated match counts to confirm that the deduplication step does not alter the conclusion. A

matching-threshold analysis varied the minimum number of hits required for a type to count toward diversity. The configuration-model null described above validates the network community structure against chance. Together these analyses establish whether the reported patterns are stable across reasonable variation in the analytic pipeline. The outcome of the weighting-sensitivity analysis is shown in Appendix E (Figure E15): the evidence-to-LoE correlation stays near zero and non-significant under the primary eight-to-one weighting, an equal weighting, and a compressed three-to-one weighting. In plain terms, the finding that documented evidence does not predict the assigned score does not depend on how the evidence types are weighted; rescaling the weights leaves the portfolio-level null essentially unchanged.

Bias and Confounding

The principal threat to the interpretation of the evidence-to-scoring relationship is that the documented evidence and the assigned score are not independent. Two distinct threats are relevant here: confounding, in which a third factor influences both the documented evidence and the assigned score and so induces an association that is not causal, and information bias, in which the way the data are generated or recorded distorts the measured relationship; neither is removed by a larger sample, and both must be addressed through study design and by making the assumed causal structure explicit (23, 123, 124). The risk custodian subject-matter experts assigned to each risk compile the evidence record and the assigned LoE together within a single submission package, which the Board then reviews and approves; the score and the supporting evidence are developed and presented jointly, as one coupled deliverable, rather than the custodians fixing a score for each mission and operational or long term health track in isolation and then assembling evidence to match it. Since the documentation and the score are produced together rather than captured independently, the two are coupled by construction and cannot be treated as separate assessments.

This coupling is a form of information bias that no statistical adjustment can fully remove, because it arises from the data-generating process itself, and it is treated as the most consequential limitation of the study and developed at length in the Limitations. The analytic relationships among the exposures, the outcome, and the principal sources of bias and confounding were represented in an analytical directed acyclic graph that makes the assumed structure explicit and guided the selection of covariates for the mission-stratified models (23, 123). Mission applicability is a particularly important common influence, since the missions for which a risk is scored shape both the evidence assembled and the score assigned, and it is represented accordingly in the analytic structure and addressed through the mission-stratified design.

Human Subjects Considerations

This study does not constitute human-subjects research. The determination that the work is not human-subjects research was reached through the National Aeronautics and Space Administration institutional review process, and the institutional review board of the University of Texas Health Science Center likewise reviewed the activity and recorded it as not constituting human-subjects research. The records analyzed are risk-level documentation of evidence and scoring produced by the Human System Risk Board in the course of its normal governance activities, not individual astronaut medical data, and no astronaut-identifiable health information was accessed. The protected health data maintained under the agency's surveillance programs were not used; where those programs are referenced, it is only as validation infrastructure within the agency's framework, not as a data source for this analysis. The work was conducted under the Space Act Agreement identified in the Setting, which governed access to the risk documentation. The evidence information analyzed consists of corpus-level prevalence and frequency measures

of how risks are documented, not records pertaining to individuals, and it contains no personal identifiers.

Operational Definitions of the Eight Evidence Types

For reproducibility, the eight evidence types were operationalized as follows for classification. Type I, direct human spaceflight data, comprises observations collected from astronauts in the spaceflight environment, including in-mission physiological and clinical measurements and references to specific crewed missions. Type II, human ground-analog data, comprises observations from terrestrial environments engineered to simulate spaceflight, including bed-rest, isolation, and confinement analogs enumerated in the analog catalog. Type III, terrestrial clinical and epidemiological data, comprises human medical evidence drawn from the broader clinical and population-health literature. Type IV, animal data, and Type V, cellular and tissue data, comprise mechanistic evidence from non-human in-vivo and in-vitro models respectively. Type VI, computational and modeling data, comprises simulation-based and in-silico inference. Type VII, expert opinion, comprises the structured judgment of subject-matter experts. Type VIII, standards and requirements, comprises codified engineering and medical standards. These operational definitions follow the evidence type language of the Human System Risk Management Plan (JSC-66705 Rev B), so that the classification reflects the agency's own categories rather than an externally imposed scheme (7). Each lexicon term maps to exactly one of these types, and the mapping is reported in the appendices.

The detection of the lower-tier types, those carrying the least weight in the JSC-66705 hierarchy, namely expert opinion and standards, follows the same lexicon-driven logic. Type VII, expert opinion, is identified through cues that signal professional judgment in the absence of primary data, such as references to expert opinion, expert consensus, subject-matter-expert

judgment, professional or clinical judgment, working group or consensus panels, and best-practice statements; Type VIII, standards and requirements, is identified through references to the governing documents, including NASA-STD-3001, the Spacecraft Maximum Allowable Concentrations, and the system requirements documents. Given that the eight-type hierarchy and its ordering by directness and applicability are taken directly from the Human System Risk Management Plan (JSC-66705 Rev B), the classification mirrors the same evidentiary ranking the Board itself applies, and the lowest-weight types are detected conservatively so that a passing rhetorical mention does not inflate a risk's evidence profile. For the two human evidence types in particular, the rosters of crewed missions and of ground-analog missions were not compiled by the analyst: a report enumerating all crewed spaceflights and all analog missions was exported from the NASA Life Sciences Portal (NLSP), the agency's repository of human-research investigations and data from spaceflight and ground-analog studies, and adopted as an agency-confirmed keyword set, so that a reference to any documented mission is matched against an authoritative roster rather than a hand-assembled list (92).

Posture table Standardization and Data Schema

The per-mission posture tables were normalized to a standardized 19-column schema during ingestion so that records from packages of differing vintage and formatting could be compared on a common basis. The schema, the fixed set of fields to which every record is mapped, records the risk identity, the mission profile, the impact category, the likelihood and consequence scores, the posture, the disposition, the LoE, the rationale text, and the associated provenance and validation flags. Normalizing to this schema is what permits the structured scores to be analyzed across the portfolio under a single consistent definition and what makes the validation flagging described above possible.

Results

Systematic Evidence Extraction and LoE Evaluation

The Documented Evidence Landscape

The corpus comprised 93 approved Risk Summary Package submissions across the 29 active risks, with a mean of 3.2 submissions per risk across the study window. The per-risk evidence and LoE values that underlie the portfolio analyses are tabulated in Appendix G. Averaged across the 93 submissions, the packages were substantial documents of 53.2 pages and 31.9 references each. Each package drew, on average, on 3.4 of the eight evidence types, so that a typical risk documented between three and four distinct kinds of evidence rather than the full range; this count of distinct types present is the measure referred to throughout as evidence diversity. The mean quality-weighted evidence composite was 583.1, a single per-package score that sums a package's evidence citations after weighting each one by the quality of its evidence type under the JSC-66705 hierarchy, so that one piece of direct human spaceflight evidence contributes more than one expert-opinion citation; the measure is defined in full in the methods and is reported here as the average over the 93 submissions. Read together these figures describe an evidence base that is voluminous but narrow, substantial in length and citation count yet concentrated in a few evidence types. Mean template completeness, measured as the share of standardized sections present, was 35.1 percent, which quantifies the substantial departure of the corpus from full conformance to the standardized template and corroborates the provenance finding that nearly half of all evidence matches were recovered from non-canonical locations in the documents.

Across the eight evidence types, two types were universal: every risk documented Type I direct human spaceflight evidence and Type VII expert opinion, each present in 100 percent of

risks (Table 2). Type VIII standards evidence was present in 93% of risks, Type III terrestrial clinical and epidemiological evidence in 86%, Type VI computational evidence in 83%, Type II ground-analog evidence in 66%, Type IV animal evidence in 52%, and Type V cellular evidence in 48%. Evidence diversity ranged from four to eight types with a median of six. The universality of Types I and VII establishes a shared evidentiary foundation across the entire portfolio, against which risks differentiate principally by the volume and concentration of their evidence rather than by the kinds of evidence they cite. The distribution of evidence across types and risks is shown in Figure 2, and the coverage and validity of the structured LoE entries is shown in Figure 3.

A direct product of this extraction was an audit of the scoring records themselves. Of the 473 posture cells that should have carried a Level of Evidence, 419, or 88.6 percent, were complete and correctly formatted, while 40 were left blank and 14 contained formatting errors. The errors were of two kinds: seven cells recorded a range such as '2-3' where a single score was required, all of them in the Hypoxia tables, and seven paired a numeral with a mismatched word grade, such as a cell marked Moderate but carrying the numeral 3. Two risks, Carcinogenesis and Urinary Retention, carried no usable Level of Evidence at all and were excluded from the scored analyses, leaving 27 scored risks. These are small counts, but because the same records inform crew-safety decisions, the presence of recoverable formatting errors in the scoring fields is itself a documentation-quality finding and a concrete target for template enforcement.

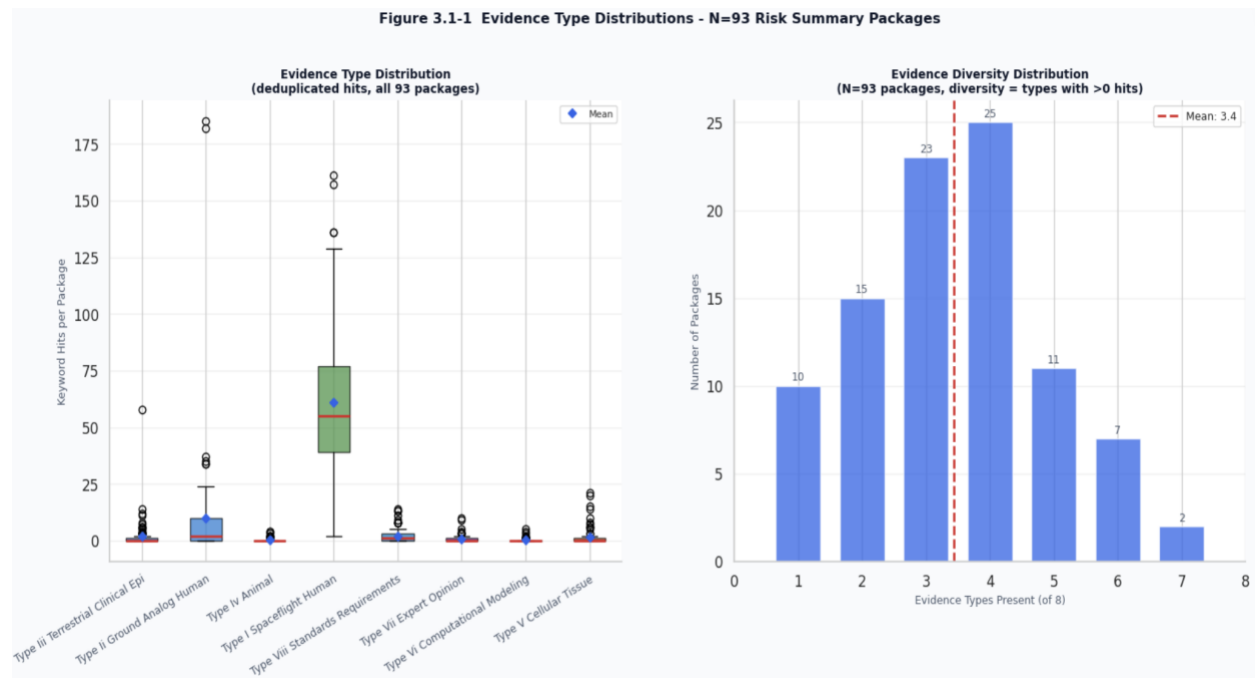


Figure 2. Distribution of documented evidence across the eight evidence types and the 29 risks, showing the universal presence of Type I and Type VII and the graded prevalence of the remaining types.

Figure 2.2 (3 of 3) - Every LoE score by risk x DRM x outcome (512 cells)
green = valid (1-4) . orange E = error . red M = missing/gap . grey . = N/A (not applicable)

	Ops (In-Mission Operations)								LTH (Long-Term Health)							
	LEO-S	LEO-L	Lun-OS	Lun-OL	Lun-SS	Lun-SL	Mars-P	Mars-L	LEO-S	LEO-L	Lun-OS	Lun-OL	Lun-SS	Lun-SL	Mars-P	Mars-L
Aerobic	1	2	1	2	3	3	2	4	2	2	2	2	3	3	2	4
Behavioral Health	2	2	2	2	2	2	3	3	3	3	3	3	3	3	3	3
Bone Fracture	1	1	1	2	2	2	2	3	1	2	E	2	E	2	E	E
CO2	2	2	3	3	3	3	3	3	2	2	2	2	2	2	2	2
Carcinogenesis	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M
Cardiovascular	3	3	3	3	3	3	3	3	1	1	1	1	1	1	1	3
Crew Egress	1	1	2	2	2	M	2	M	-	-	-	-	-	-	-	-
DCS	1	1	1	1	2	2	1	2	1	1	1	1	1	1	1	1
Dust	-	-	1	2	1	2	-	3	-	-	1	2	1	2	-	3
Dynamic Loads	1	1	1	1	1	2	1	1	1	1	1	1	1	2	1	2
EIHSO	1	1	2	2	2	2	2	2	-	-	-	-	-	-	-	-
EVA	1	1	1	1	2	2	2	3	1	1	1	1	2	2	2	3
Electrical Shock	1	1	2	2	2	2	2	2	-	-	-	-	-	-	-	-
Food and Nutrition	2	2	2	1	2	2	1	1	2	2	2	2	2	2	1	1
Hearing Loss	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Hypoxia	2	2	2	2	2	2	E	E	3	E	3	E	3	E	E	E
Immune	1	1	2	2	2	2	2	2	1	1	2	2	2	2	2	2
Medical	1	2	2	3	2	E	E	E	1	2	2	3	2	3	3	3
Microhost	1	1	2	2	2	2	3	3	1	1	2	2	2	2	3	3
Muscle	1	2	1	2	3	3	2	4	2	2	2	2	3	3	2	4
NIR Risk LASERS	1	1	1	1	1	1	3	3	1	1	1	1	1	1	3	3
NIR Risk SUNLIGHT	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Pharmacy	1	1	2	3	3	3	3	4	1	1	1	3	3	3	3	4
Renal Stone	1	2	1	2	1	2	3	3	1	2	1	2	1	2	3	3
SANS	1	2	1	2	1	2	2	4	2	4	2	4	2	4	4	4
Sensorimotor	1	1	2	2	3	3	2	3	4	4	4	4	4	4	4	-
Sleep	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Team	1	1	3	3	3	3	3	2	-	-	-	-	-	-	-	-
Toxic Exposure Risk HIGH	1	1	1	1	1	M	2	3	1	M	1	1	1	1	2	3
Toxic Exposure Risk LOW	1	1	1	1	1	M	2	3	1	M	1	1	1	1	2	3
Toxic Exposure Risk MEDI	1	1	1	1	1	M	2	3	1	M	1	1	1	1	2	3
Urinary Retention	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M	M

Figure 3. Coverage and validity of the structured LoE entries across risks and Design Reference Missions, distinguishing valid, missing, not-applicable, and erroneous entries.

Table 1. Documented evidence landscape across the corpus of 93 approved submissions.

Characteristic	Value
Approved package submissions	93
Active risks (posture sub-tables)	29 (32)
Mean submissions per risk	3.2
Mean pages per package	53.2
Mean references per package	31.9
Mean evidence diversity (of 8 types)	3.4
Mean quality-weighted composite	583.1
Mean template completeness	35.1%
Accepted evidence matches	9,098
Structured LoE coverage (valid/expected)	419/473 = 88.6%

Table 2. Prevalence of the eight evidence types across the 29 risks.

Type	Evidence category	Present in
I	Direct human spaceflight	100%
VII	Expert opinion	100%
VIII	Standards and requirements	93%
III	Terrestrial clinical / epidemiological	86%
VI	Computational / modeling	83%
II	Human ground analog	66%
IV	Animal	52%
V	Cellular / tissue	48%

Portfolio-Level Alignment of Evidence and LoE

At the portfolio level, the worst-case LoE was distributed across 27 scored risks as nine Moderate, 13 Weak, and five Speculative, with no risk carrying a Strong worst-case score. The absence of Strong scores at the portfolio level is a direct consequence of the worst-case aggregation, since a risk retains a Strong classification only if its evidence is strong in every mission context, and no risk met that condition across the full continuum.

The three pre-specified confirmatory tests of Aim 1 were all non-significant under the Bonferroni-corrected threshold of 0.0167 (Table 3). Total evidence volume did not predict the worst-case LoE, with a Spearman correlation of +0.046, a 95 percent confidence interval from

−0.340 to +0.419, and a p-value of 0.818, so the null hypothesis was retained. The quality-weighted composite likewise did not predict the score, with a Spearman correlation of +0.054, a confidence interval from −0.333 to +0.425, and a p-value of 0.790. The Fisher exact test of human-spaceflight-evidence presence against strong-or-moderate scoring could not be computed in the usual way because every one of the 27 scored risks documented at least some direct human spaceflight evidence, producing a degenerate two-by-two table with an empty cell, quasi-complete separation, and an odds ratio that is undefined, with an exact p-value of 1.000. The substantive meaning of this degeneracy is itself a finding: the presence of direct human spaceflight evidence is not a differentiating variable for the LoE, because it is universal across the portfolio.

The independence of documented volume and the assigned score is most clearly seen in the individual risks. Counting raw matches across all of a risk's submissions, prior to the deduplication used in the correlation, Cardiovascular is the most heavily documented risk in the portfolio, with 1198 matches spanning all eight evidence types, yet it carries a worst-case LoE of Weak; Muscle, the second most heavily documented at 734 matches, carries the weakest possible classification of Speculative. By contrast, Electrical Shock rests on only 48 accepted matches across four evidence types, roughly twenty-fifth of the Cardiovascular risk's volume, yet carries a stronger worst-case score of Moderate, and several other lightly documented risks are scored at the same Moderate level. These are not anomalies but the typical pattern that produces the near-zero correlation: the quantity of evidence a risk has accumulated is not what determines the worst-case score it receives.

Firth penalized logistic regression, applied to obtain finite estimates under separation, confirmed the absence of association. The model using a binary indicator of human-spaceflight-evidence presence returned an odds ratio of 0.717 with a p-value of 1.000, and the model using the continuous count of such evidence returned an odds ratio of 1.165 with a p-value of 0.9251; neither

was significant at the corrected threshold, and the direction of the point estimates, both below one, runs counter to the hypothesis that more direct human evidence accompanies stronger scoring. Evidence diversity also did not differ significantly across the score groups. The Kruskal-Wallis's test returned an H-statistic of 0.448 on two degrees of freedom with a p-value of 0.7995 and an eta-squared effect size of 0.000, with a bootstrap confidence interval from 0.000 to 0.253. The group means ran counter to the hypothesized direction, with the Moderate group averaging 3.89 types, the Weak group 4.38, and the Speculative group 4.00, so that the weakest scored risks did not document the least diverse evidence. As a bridge to the mission-stratified analysis, the modal evidence-support rating was negatively associated with the score, with a Spearman correlation of -0.359 , a confidence interval from -0.660 to $+0.043$, and a p-value of 0.0783, a relationship in the expected direction but not significant at the nominal level.

Table 3. Primary confirmatory tests of portfolio-level alignment (Tier 1; N = 27 scored risks; Bonferroni $\alpha = 0.0167$).

Hypothesis	Test	Effect	95% CI	p	Decision
H1a volume	Spearman	$\rho = +0.046$	$-0.340, +0.419$	0.8180	Retain H0
H1b weighted	Spearman	$\rho = +0.054$	$-0.333, +0.425$	0.7900	Retain H0
H1c SF presence	Fisher exact	OR undefined	–	1.000	Retain H0
H1c (Firth, binary)	Firth logistic	OR = 0.717	–	1.000	Retain H0
H1c (Firth, count)	Firth logistic	OR = 1.165	–	0.9251	Retain H0
H1d diversity	Kruskal-Wallis	H = 0.448, $\eta^2 = 0.000$	–	0.7995	Retain H0
H1e modal ESR	Spearman	$\rho = -0.359$	$-0.660, +0.043$	0.0783	Retain H0

Stated in hypothesis-testing terms, the portfolio-level analysis failed to reject the null hypothesis of no evidence-to-scoring association on every evidence dimension examined. Furthermore, the null was retained for total evidence volume, for the quality-weighted composite, and for evidence diversity, none of which reached the corrected threshold; it was retained for the presence of direct human spaceflight evidence, where the test was degenerate because that evidence type is universal across the scored portfolio; and it was retained for each of the eight

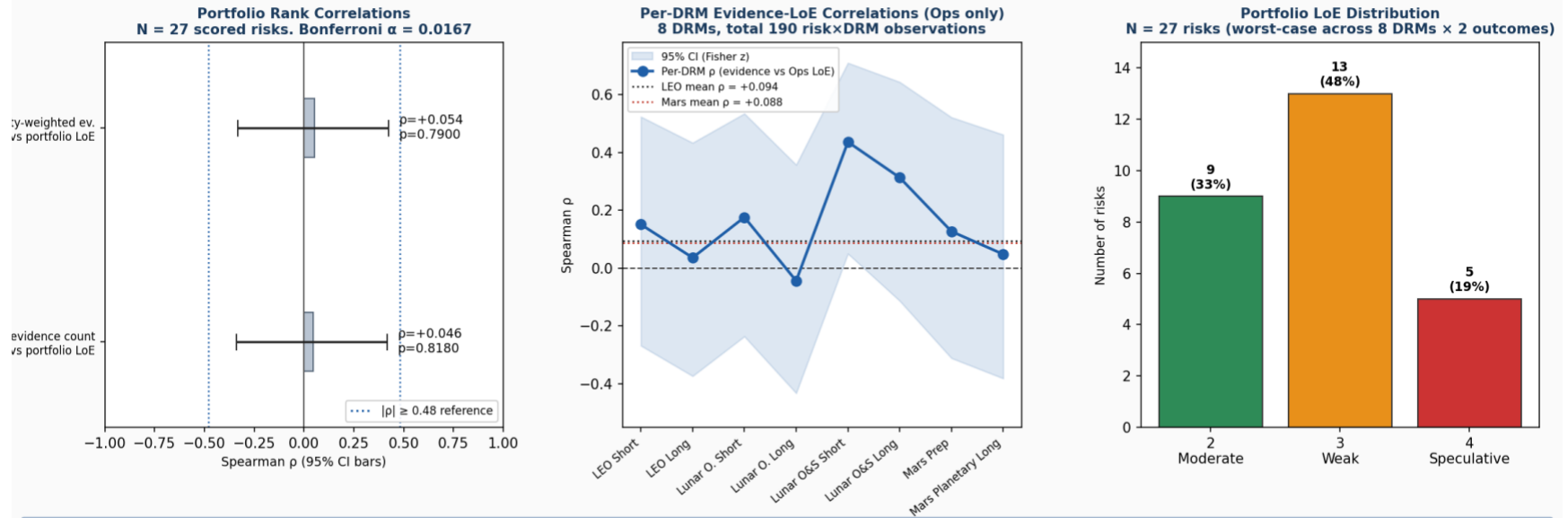
evidence types taken individually, whose per-type correlations were uniformly small, ranging from a positive $+0.226$ for ground-analog evidence to a negative -0.189 for standards and requirements evidence, with none significant. This consistent non-rejection across volume, quality, breadth, type, and the three-stream support rating is interpreted with deliberate caution. With 27 scored risks the confidence intervals are wide, spanning from a small negative to a moderate positive association, so the result establishes the absence of a strong, monotone relationship between documented evidence and the worst-case score rather than proof that the relationship is exactly zero. The pre-specified equivalence test and Bayes-factor analysis, which ask whether the near-zero correlation is small enough to count as practical equivalence and how strongly the evidence favors the null respectively, reported in the portfolio-level statistical summary, were included precisely to bound this interpretation and to distinguish an informative null from a merely underpowered one.

The exploratory per-evidence type correlations, which repeat the volume test separately within each of the eight evidence types, were uniformly small and non-significant, with positive coefficients only for ground-analog evidence at $+0.226$ and terrestrial clinical evidence at $+0.081$, and negative coefficients for the remaining types, the largest in magnitude being standards and requirements evidence at -0.189 and animal evidence at -0.130 , none of which reached significance. In plain terms, no single evidence type, whether direct human spaceflight data or any other, shows even a moderate tendency to accompany a stronger or weaker score, so the null portfolio result is not concealing a strong effect confined to one type. Given that every primary portfolio test was non-significant in an underpowered census, the principal correlation was subjected to an equivalence test. The two one-sided tests procedure, with a region of practical equivalence defined as a correlation magnitude below 0.10 , returned a lower-bound test with z of

0.749 and p of 0.227 and an upper-bound test with z of -0.275 and p of 0.3918, for an overall equivalence p-value of 0.3918 and a 95 percent confidence interval from -0.340 to $+0.419$. The equivalence test was therefore inconclusive: the confidence interval extends beyond the region of practical equivalence, so the data can neither demonstrate a non-zero association nor confirm practical equivalence to zero. The power at the observed correlation was only 2 percent, and 80 percent power at this sample size would require a correlation magnitude of at least 0.48. The portfolio-level analysis is thus best read not as a demonstration that evidence and scoring are unrelated, but as a demonstration that simple counts of documented evidence do not determine the worst-case LoE, with the study underpowered to resolve associations smaller than large.

The portfolio findings were robust to analytic choices. Recomputing the principal correlation under an equal-weight scheme returned essentially the same coefficient of $+0.046$ with a p-value of 0.8180, and the coefficient varied by only 0.015 across the full set of weighting schemes examined, confirming that the eight-to-one weighting does not drive the result. Recomputing on the raw, non-duplicated match counts returned a coefficient of $+0.124$ with a p-value of 0.5374, confirming stability to the deduplication step. Evidence concentration, summarized by the Herfindahl-Hirschman index (a standard measure of concentration that rises toward one when a risk's citations cluster in a single evidence type and falls toward zero when they are spread evenly across types, 127), averaged 0.727 across the 93 packages, with 86 percent of packages exceeding 0.5, and reached its maximum of 1.0 for the Pharmacy risk and reached its maximum of 1.0 for the Pharmacy risk, which documented a single dominant evidence type and stands as the clearest concentration outlier in the portfolio. The portfolio-level statistical summary is shown in Figure 4.

Statistical Summary — Portfolio Tests, Per-DRM Trajectory, LoE Distribution



Method Panel 1 shows portfolio-level Spearman rank correlations between evidence and Level of Evidence across N=27 scored risks, with Bonferroni-adjusted threshold. Panel 2 shows per-DRM Spearman correlations from the primary analysis (Ops outcome only, N=190 risk×DRM observations). Panel 3 shows the distribution of portfolio-level Level of Evidence across the 27 scored risks, the maximum (worst-case) across all 8 DRMs and both Ops/LTH outcomes. Confidence intervals via Fisher z transformation.

Figure 4. Portfolio-level statistical summary of the primary confirmatory tests, effect sizes, confidence intervals, and the equivalence-testing result.

Mission-Stratified Behavior of Evidence and LoE

The mission-stratified analysis, designated the primary test of the evidence-to-scoring relationship for its greater power, revealed the structural signal that the portfolio summary masks. The mean LoE weakened monotonically across the Design Reference Mission continuum (Table 4), rising from 1.39 for short-duration low Earth orbit, to 1.58 for long-duration low Earth orbit, 1.65 and 1.89 for short and long lunar-orbital missions, 1.91 and 2.14 for short and long lunar-orbital and surface missions, 2.18 for Mars-preparation missions, and 2.68 for long-duration Mars-planetary missions. Since higher numbers denote weaker evidence, this monotone ascent is the quantitative expression of an evidence base that is strongest for the near-Earth missions from which most direct human data derive and weakest for the most demanding exploration-class missions.

Table 4. Mean LoE across the Design Reference Mission continuum (operations outcome; lower is stronger).

Design Reference Mission	Mean LoE
LEO Short	1.39
LEO Long	1.58
Lunar Orbital Short	1.65
Lunar Orbital Long	1.89
Lunar Orbital and Surface Short	1.91
Lunar Orbital and Surface Long	2.14
Mars Preparation	2.18
Mars Planetary Long	2.68

The linear mixed-effects model confirmed that this gradient, the systematic weakening of scores from low Earth orbit toward Mars, is statistically robust. For the operations outcome, the model estimated an intercept of 1.976 with a 95 percent confidence interval from 1.818 to 2.134, a significant Design Reference Mission main effect with a p-value below 0.001 indicating that the LoE weakens systematically toward Mars, and a non-significant evidence-by-mission interaction

with a coefficient of +0.003, a standard error of 0.038, and a p-value of 0.9282, with a confidence interval from -0.072 to $+0.078$. The parallel model for the long term health outcome estimated an intercept of 2.145 with a confidence interval from 1.840 to 2.449, again a significant mission main effect with a p-value below 0.001, and a non-significant interaction with a coefficient of -0.018 and a p-value of 0.6728. The significant main effect together with the non-significant interaction has a precise interpretation: the LoE declines along the mission continuum regardless of how much evidence a risk documents, so that the gradient is driven by the diminishing applicability of evidence to distant missions rather than by differences in documentation volume.

Expressed as hypothesis tests, the mission-stratified model reversed the portfolio verdict for one term and upheld it for the others. The null hypothesis that the LoE is constant across the Design Reference Mission continuum was decisively rejected, with a mission main effect significant at a p-value below 0.001 in both the operations and the long term health models, corresponding to the monotone weakening from a mean of 1.39 at short-duration low Earth orbit to 2.68 at long-duration Mars-planetary missions. Furthermore, by contrast, the null hypothesis that the evidence-to-scoring relationship does not vary by mission was not rejected, the evidence-by-mission interaction remaining non-significant in both models, with a coefficient of +0.003 and a p-value of 0.9282 for operations and a coefficient of -0.018 with a p-value of 0.6728 for long-term health. The joint pattern, a rejected mission effect alongside a retained interaction, carries a single coherent meaning: the LoE falls along the mission continuum irrespective of how much evidence a risk documents, so the gradient is produced by the declining applicability of evidence to the most distant missions rather than by differences in documentation. The mission-level analysis therefore recovers, as a strongly significant structural effect, the relationship that the worst-case portfolio summary had collapsed into a null.

Within each Design Reference Mission, the association between documented evidence and the per-mission score was weak and positive, with coefficients of +0.151 and +0.036 for short and long low Earth orbit, +0.176 and -0.044 for the lunar-orbital missions, +0.436 and +0.313 for the lunar orbital and surface missions, and +0.127 and +0.048 for the two Mars missions; the mean coefficient was +0.094 across the low Earth orbit strata and +0.088 across the Mars strata, indicating similar and modest alignment at both ends of the continuum. In plain terms, at neither the near-Earth nor the Mars end of the mission range does documenting more evidence translate into a meaningfully stronger score, and the small positive values mean the two move together only weakly. A complementary degradation analysis, which compares each risk's LoE at the Mars end of the mission range against its value at the low-Earth-orbit end, examined the 22 risks for which a LoE could be scored at both the low-Earth-orbit and Mars ends of the mission spectrum, and found a mean weakening of the LoE of +1.045 across that span, with 18 of the 22 risks degrading; the magnitude of degradation was unrelated to the quantity of direct human spaceflight evidence a risk documented, with a Spearman correlation of +0.047 and a p-value of 0.8371, which reinforces that the gradient reflects mission applicability rather than documentation effort. In plain terms, scores grew weaker for almost every risk as missions reached toward Mars, and how much weaker bore no relation to how much direct astronaut evidence a risk carried, which again points to applicability rather than documentation effort as the driver. Evidence breadth, that is, the number of distinct evidence types a risk drew on, thinned along the same axis: the stratified correlation of the evidence-breadth metric against mission rank was significant for the operations outcome with a p-value of 0.030 and a strong negative coefficient near -0.76, and in the same direction though not significant for long-term health with a p-value of 0.200 and a coefficient near -0.51. The gradient and its supporting structure are shown in Figure 5. Translated into the model's own units,

these coefficients reconstruct the observed gradient directly. The operations intercept of 1.976 is the expected LoE at the average position on the mission continuum, and because the mission term is significant and positive on the 1-4scale where higher numbers denote weaker evidence, the expected score rises step by step from the near-Earth missions to the exploration-class missions, tracing the same path as the observed means that climb from 1.39 at short-duration low Earth orbit to 2.68 at long-duration Mars-planetary missions. The interaction coefficient of +0.003, statistically indistinguishable from zero, means that the evidence-to-score slope estimated within each mission, which the per-mission analysis reports as small values such as +0.151 and +0.036 for short and long low Earth orbit, does not change systematically across the continuum, so that no mission is one in which documenting more evidence measurably earns a stronger score.

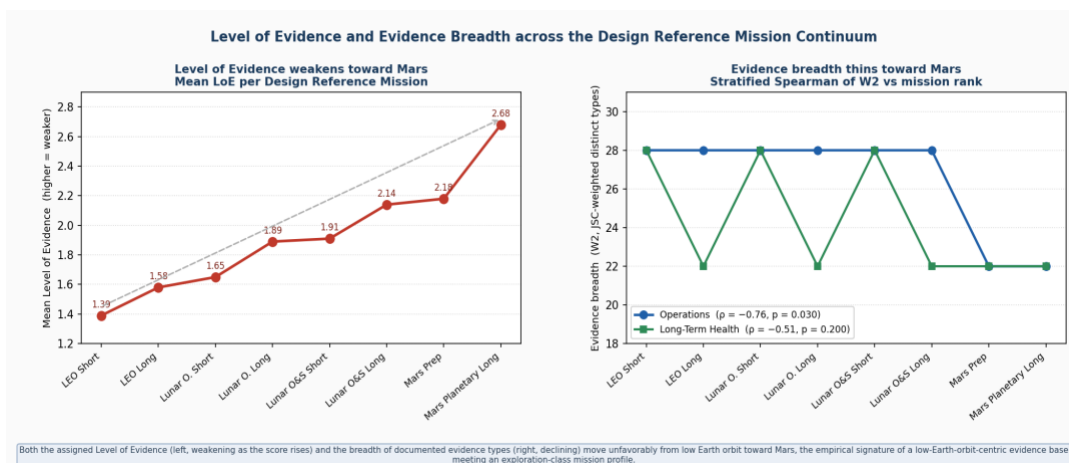


Figure 5. LoE and evidence breadth across the Design Reference Mission continuum, showing the systematic weakening of evidence from low Earth orbit toward Mars.

The evidence-support ratings recorded across the mission observations showed a pronounced ceiling: of the scored observations, 87.5 percent were rated Well-Supported and 12.5 percent Partially-Supported, with no observation rated Unsupported, where a Well-Supported rating means the documented evidence substantively backs the assigned posture, a Partially-Supported rating means it backs the posture only thinly, and an Unsupported rating means the posture rests on little or no documented evidence. The rationale content of these observations was Evidence-Referenced

in 76.6 percent of cases, Narrative-Only in 15.2 percent, and Missing in 8.2 percent. These categories describe the basis the custodian recorded for each posture: an Evidence-Referenced rationale cites a specific evidence type, a Narrative-Only rationale offers prose reasoning without a citation, and a Missing rationale supplies no written basis at all. That roughly one posture in seven rests on narrative alone and nearly one in 12 on no written basis at all indicates that a non-trivial share of the portfolio's scores are not anchored to citable evidence, which is the same alignment gap the portfolio-level tests detected, seen here from the side of documentation rather than correlation. The combination of a near-universal Well-Supported rating with a weakening LoE toward Mars indicates that the agency records confidence in its assessments even where the directly applicable evidence is thin, which is consistent with the role of expert judgment in bridging evidentiary gaps for the most distant missions.

The ceiling reported here is a property of the three-stream evidence-support rating, which classifies each posture as Well-Supported, Partially-Supported, or Unsupported, and its concentration is itself interpretable. The three-stream evidence-support rating is a construct introduced by this study rather than a field of the JSC-66705 risk summary template, and it is defined here explicitly: each mission posture is independently graded as Well-Supported, Partially-Supported, or Unsupported by triangulating the documented rationale, the supporting context surrounding it, and the package-level evidence profile. Its purpose is to capture whether the assigned score is actually backed by documented evidence rather than to restate the score, which is why its concentration near the top of the scale is informative on its own. That 87.5 percent of scored mission observations are rated Well-Supported while the LoE simultaneously weakens toward Mars indicates that the rating expresses confidence in the assessment process even where directly applicable evidence is thin, which is the signature of expert judgment carrying the

assessment for the most distant missions rather than direct mission-specific data. In plain language, the agency rates its Mars assessments as confidently supported even though the directly applicable evidence is thinnest at that end of the mission range, which means the confidence is being carried by expert judgment rather than by mission-specific data. That gap is precisely where additional Mars-relevant evidence would do the most to place the rating on firmer empirical footing.

Longitudinal Trajectories of the Evidence Base

Given that the corpus comprises 93 submissions distributed across the 29 risks over the 2020 to 2025 window, with most risks documented two to four times by way of additional submitted revisions, the design also supports a within-risk longitudinal view of how the evidence base evolves across successive Risk Summary Package revisions. This repeated-measures structure is the defining feature that justifies extracting the full corpus rather than the most recent submission alone, and the trajectory analysis reported here is descriptive and exploratory, characterizing change within each risk across its review cycles rather than testing a confirmatory hypothesis (117). For each risk with two or more submissions, three quantities were tracked from the first to the most recent revision: the raw volume of documented evidence, the diversity of evidence types represented, and the quality-weighted composite that re-weights the evidence counts by the eight-type hierarchy so that growth in higher-grade evidence is distinguished from growth in lower-grade evidence.

Across the 29 parent-level risks, raw evidence volume increased from first to last submission for 20 of 29 risks, or 69 percent; evidence type diversity increased for nine of 29, or 31 percent; and the quality-weighted composite increased for 21 of 29, or 72 percent. The concordance between the volume trajectory and the quality-weighted trajectory was striking: in every risk that gained volume, quality rose as well. 20 risks showed increases in both volume and quality-

weighted evidence, no risk showed rising volume with flat or declining quality, one risk showed rising quality with flat or declining volume, and the remaining eight risks were flat or declining on both measures. The practical implication is that, where the agency added evidence across review cycles, the additions were not concentrated in lower-grade types but tended to raise the quality-weighted score in step with the raw count, indicating that new evidence trended toward higher-grade sources rather than accumulating disproportionately in computational or expert-opinion categories.

The risks with the largest gains in documented evidence were the CO₂, Microhost, Food and Nutrition, EIHSO, and NIR risks, each of which accrued substantial volume and quality-weighted increases across its revisions. At the other end, the risks showing the largest declines or no growth were the Hypoxia, Cardiovascular, Pharmacy, Urinary Retention, and Extravehicular Activity risks, where the documented evidence base contracted or stagnated across the window. For the research, this pattern locates where evidence development has and has not kept pace: the risks gaining the most are those under active investigation, while the stagnant or contracting risks mark the parts of the portfolio where the documented basis is aging relative to the rest, which is the actionable signal a custodian or reviewer would use to target future evidence-gathering. A contraction in documented volume does not necessarily indicate a loss of underlying knowledge, since revisions may consolidate or re-scope the narrative, but a sustained absence of evidence development across review cycles is the pattern most relevant to planning future evidence-gathering activities, and the 8 risks that were flat or declining on both measures are the natural candidates for such attention. These trajectories are reported as descriptive context for the cross-sectional alignment findings; the absence of growth in a documented evidence base is not in itself evidence that the underlying risk is well or poorly understood, a distinction that bears directly on

the interpretation of the null portfolio-level associations reported above (125). The per-risk trajectories and the portfolio-level accrual summary are presented in Appendix E.

Multilayer Network Topology Modeling

Network Structure of the Risk-Evidence Portfolio

The risk-by-risk network projected from the shared-evidence structure, constructed at the sub-risk level so that risks decomposed into several sub-risks contribute their full evidence structure rather than collapsing to a single parent node, contained 124 of 496 possible edges among its 32 nodes, for a density of 0.250 at the similarity threshold applied. A density of one quarter means that, once the similarity threshold is applied, only a minority of risk pairs share enough of their evidence profile to be linked, so the portfolio is neither a loose scatter of unrelated risks nor a single undifferentiated mass but a structured middle ground in which distinct clusters can form. Community detection recovered a modular structure whose strength was statistically significant against a configuration-model null. Finding genuine modularity matters because it means the risks fall into identifiable families that draw on shared evidence bases, which is the structural precondition for the cross-risk evidence-borrowing this review set out to detect. The observed weighted modularity was 0.369 and the unweighted modularity was 0.331, against a null distribution generated from 1000 degree-preserving (126) randomizations with a mean modularity of 0.244 and a 95 percent interval from 0.208 to 0.282, yielding an empirical p-value below 0.0001. Since the observed modularity sits well above the entire null interval, the clustering is a real feature of how evidence is shared across the portfolio rather than a byproduct of how many connections each risk happens to carry. The observed modularity therefore exceeds what the network's connectivity alone would produce, supporting the conclusion that the portfolio contains genuine communities of risks resting on similar evidence foundations rather than an artifact of node

connectivity. For the research this means the recovered communities can be read substantively, as groups of risks that would tend to rise or fall together as shared evidence is added or revised, rather than dismissed as an artifact of how the network was built. The small-world and scale-free properties of the network were examined on the risk-similarity structure through the standard diagnostics. These diagnostics ask whether the portfolio behaves like a small world, in which any risk is reachable from any other through a few similar intermediaries, and whether a few risks act as disproportionately connected hubs, both of which bear on whether evidence insight could plausibly travel across related risks. The network is markedly small-world: its clustering coefficient of 0.479 far exceeds the 0.249 expected of an equivalent random reference graph matched on size and density, while its mean path length of 2.190 remains close to the random baseline of 1.860, so that the Watts-Strogatz (38) small-world coefficient σ , which compares the network's clustering and path length against that random reference, reaches 1.63, above the threshold of 1.5 used to flag small-world structure, and the Telesford ω , a complementary small-world index that runs from minus one to plus one and reads as small-world when it sits near zero (37), is +0.11, sitting near zero on its own scale; both diagnostics place the network firmly in the small-world regime (Figure 6b). Substantively, this means the portfolio is at once locally clustered, with risks forming tight neighborhoods of evidentially similar risks, and globally compact, so that any risk is reachable from any other through a short chain of similar intermediaries, which is the structural property that allows evidence-informed inference to propagate across related risks rather than remaining trapped within isolated groups. The scale-free test was deliberately less conclusive and is reported as such. The observed degree sequence is heavy-tailed, meaning a small number of risks carry far more evidentiary connections than the typical risk while most carry only a few (25), ranging from one to 13 neighbors with a small

number of high-degree hub risks, but a maximum-likelihood power-law fit did not return a finite, stable exponent at this network order, and the likelihood-ratio comparisons could not distinguish a power law from a log-normal alternative (34). The degree distribution is therefore best described as heavy-tailed with identifiable hubs rather than as strictly scale-free; this is the expected outcome for a network of only a few dozen nodes, at which the formal scale-free machinery has limited power to adjudicate between heavy-tailed forms, and the strong distributional claim of scale-freeness is neither needed nor supported by the present data. For the purposes of this study, the practical meaning is that connectivity in the portfolio is concentrated in a few broadly shared evidence types rather than spread evenly, so that revising one of those shared evidence bases would propagate to many risks at once, even though the network remains too small to certify the stronger claim that its connectivity follows a strict power law. The hub structure that motivated the scale-free framing is nonetheless directly visible in the evidence co-occurrence reported below, where a small set of universal evidence types anchors the entire portfolio. The conceptual background for these network structures, including small-world organization and modularity, is developed in the Background, and the corresponding network visualizations are provided in the appendix for reference.

Two resolutions of this network are available, and both are reported because they answer different questions. The community and modularity results above summarize the portfolio at the level of the risk, which is the natural unit for describing how evidence-similar risks group together. The small-world and scale-free diagnostics, by contrast, were computed at the finer resolution of the individual risk posture sub-tables, the 32 scored units into which the portfolio resolves once the risks carrying separately scored variants are expanded, so that the topology is examined at the same granularity at which the evidence and the LoE values are actually recorded. For the research,

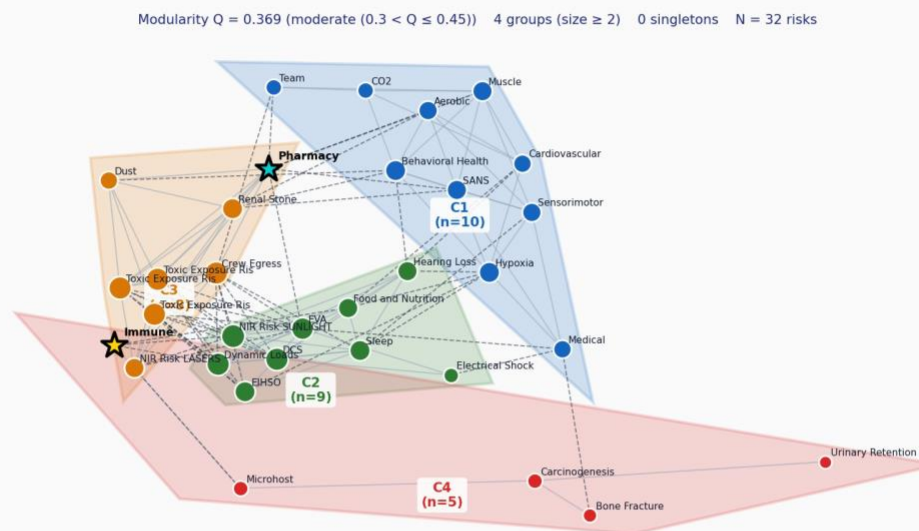
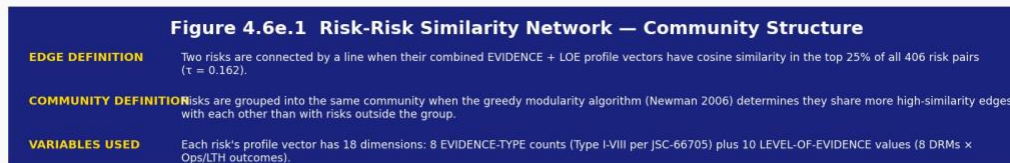
this means the connectivity findings describe evidence relationships among the very units at which LoE values are assigned, so that any inference about shared evidence maps directly onto the entities whose scores are under review. Reading the network at the sub-table resolution is an exploratory choice rather than a corrective one: it neither alters the community structure described above nor depends on it, and it allows the connectivity, clustering, and hub structure to be inspected at the level where the underlying postures are scored. The two views are complementary, the coarser characterizing how risks group and the finer characterizing the shape of the connectivity, and no result requires collapsing them into a single resolution.

Evidence co-occurrence across the packages revealed the universal-hub structure anticipated by the scale-free framing (Figure 7). The strongest co-occurring pair was ground-analog and direct human spaceflight evidence, which appeared together in 56 of 93 packages, or 60 percent, while the most independent pair, animal and computational evidence, co-occurred in only two packages. In practical terms, this means almost every risk is anchored to direct human spaceflight data, with the other evidence types appearing as supplements to that core rather than as independent foundations, so the apparent diversity of the portfolio rests on a narrower base than a simple count of evidence types would suggest. Conditional dependencies were extremely high and consistently directed toward the universal types: whenever terrestrial clinical, ground-analog, animal, cellular, computational, standards, or expert evidence was present, direct human spaceflight evidence was also present 100 percent of the time, and expert opinion was present in 96 to 98 percent of those cases. These dependencies confirm that the portfolio rests on a shared backbone of human spaceflight and expert evidence onto which the other, more specialized evidence types are layered. Expert-opinion evidence also ranked high among the co-occurring types, appearing in roughly two-thirds of the packages, which is worth pausing on given how the evidence was detected. Crew-

flight and analog-mission evidence carry standardized vocabulary, named missions and named analog facilities, that the extraction engine could map cleanly, whereas expert mentions have no comparable standardized keywords and are inherently harder to catch without them. That expert evidence still surfaced as prominent despite lacking those easy keyword anchors suggests it is a genuinely pervasive feature of the portfolio rather than an artifact of what was simplest to map. The evidence-similarity structure and the co-occurrence pattern are shown in Figures 6a and 7. These structural relationships are not only reported statically but are made explorable in an interactive network application, a standalone deliverable of this work described in Appendix F, in which each risk, evidence type, and community can be selected to reveal its connections, centrality, and shared-evidence neighbors in real time. The application turns the portfolio's architecture into a navigable map, allowing a reviewer to trace how evidence is shared across risks and to see at a glance where the structure concentrates and where it thins.

The network is constructed from an 18-dimensional profile of each risk that combines its eight evidence type counts with 10 dimensions encoding the risk's LoE profile. Each risk is represented as a vector, an ordered list of these 18 numbers, the cosine similarity between every pair of vectors is computed, and an edge is drawn for the most similar pairs, retaining the top quarter of all pairs at a similarity threshold of 0.162, which yields the 124 edges among 496 possible pairs that give the density of 0.250. Drawing an edge only where the cosine similarity clears the threshold ensures that a link reflects a genuinely shared evidence profile rather than incidental overlap, so the resulting structure is a conservative map of which risks truly rest on common evidence. Communities are then recovered by greedy modularity maximization (101), an agglomerative procedure that repeatedly merges the grouping that most increases modularity, which partitions the risks so as to maximize the modularity, the excess of within-community edge

weight over the weight expected if the same edges were placed at random, and the partition is validated against a configuration-model null (126) that preserves each node's degree while rewiring edges, with the empirical p-value counting how rarely the random graphs reach the observed modularity, that is, the proportion of degree-preserving null networks whose modularity met or exceeded the observed value (127). The results suggest one recovered community groups the deconditioning and physiological-adaptation risks, including Cardiovascular, Muscle, Aerobic, Sensorimotor, SANS, Behavioral Health, and Team, which share a reliance on long-duration low-Earth-orbit and bed-rest evidence, while a separate community groups operational and acute-exposure risks such as decompression sickness, extravehicular activity, dynamic loads, sleep, and hearing loss. The accompanying network visualization displays these relationships directly: each of the 32 sub-risk nodes is colored by its recovered community and sized in proportion to its degree, edge opacity scales with evidence-profile similarity, and the Immune and Pharmacy nodes are marked as the complex and simple structural poles of the portfolio. One caution applies to how these communities should be read. Because each risk's profile vector combines its eight evidence type counts with ten dimensions that encode the risk's Level-of-Evidence profile, the recovered communities reflect similarity in assigned score as well as similarity in documented evidence; in plain terms, two risks may be grouped together partly because they received similar scores, not only because they rest on similar evidence. The community structure is therefore best read as evidence and score similarity rather than evidence similarity alone, and recomputing the network on evidence-only vectors, with the Level-of-Evidence dimensions removed, is recommended to isolate the structure attributable to documented evidence independently of the score.



(1) WHAT THE FIGURE SHOWS

Each circle represents one of the 32 NASA human-system risks. The spatial layout (spring-force algorithm) places risks near each other when they have similar profiles and farther apart when they do not. The shaded colored regions are COMMUNITIES — groups of risks whose evidence-and-LoE profiles are sufficiently similar that the algorithm clusters them together. The two starred risks (gold = COMPLEX anchor, turquoise = SIMPLE anchor) are highlighted from §4.2.

(2) HOW EDGES AND COMMUNITIES WERE DETERMINED

Step A: for each risk, a profile vector was assembled with 18 dimensions — the 8 evidence-type counts concatenated with the 16 per-DRM-and-outcome LoE values, then z-scored so evidence and LoE contribute equally. Step B: cosine similarity was computed for all 406 risk pairs. Step C: edges were retained where similarity ≥ 75 th percentile ($\tau = 0.162$). Step D: greedy modularity maximization (Newman 2006) was applied to this graph to identify communities.

(3) RESULTS

The detected community structure: C1 (10 risks): Aerobic, Behavioral Health, CO2, Cardiovascular, Hypoxia, Medical, Muscle, SANS, Sensorimotor, Team | C2 (9 risks): DCS, Dynamic Loads, EIBSO, EVA, Electrical Shock, Food and Nutrition, Hearing Loss, NIR Risk SUNLIGHT, Sleep | C3 (8 risks): Crew Egress, Dust, NIR Risk LASERS, Pharmacy, Renal Stone, Toxic Exposure Risk HIGH, Toxic Exposure Risk LOW, Toxic Exposure Risk MEDIUM | C4 (5 risks): Bone Fracture, Carcinogenesis, Immune, Microhost, Urinary Retention. Modularity $Q = 0.369$, which qualifies as moderate ($0.3 < Q \leq 0.45$). The within-community grey edges are visually denser than the between-community dashed edges, confirming that the modularity score corresponds to a real visual distinction. Risks in the same community would respond similarly to a common evidence-acquisition strategy or a common LoE-review protocol. The complex/simple anchors allow the reader to compare where those endpoints sit in the structural map.

(4) DISSERTATION RELEVANCE

Communities reveal the natural sub-structure of NASA's risk portfolio when viewed through the dual lens of evidence support and Level of Evidence — the two outcome dimensions of this dissertation. Risks in the same community would respond similarly to a common evidence-acquisition strategy or a common LoE-review protocol. The complex/simple anchors allow the reader to compare where those endpoints sit in the structural map.

(5) INDIVIDUAL FINDINGS — INTERPRETATION GUIDE

(a) A risk surrounded by many same-community neighbors is a typical exemplar of its group's evidentiary profile. (b) A risk on the edge of its community is a transitional case. (c) A singleton risk requires its own evidence narrative — inferential support cannot be borrowed from any other risk in the portfolio. (d) Two communities far apart in the spatial layout represent maximally different evidence + LoE patterns.



Figure 6a. Evidence-profile similarity network of the portfolio, resolved to 32 sub-risk nodes, with four communities of evidentially similar risks recovered by modularity maximization ($Q = 0.369$) and validated against a degree-preserving configuration-model null ($p < 0.0001$). Node colour denotes community and node size is proportional to degree; the Immune and Pharmacy risks mark the complex and simple structural poles.

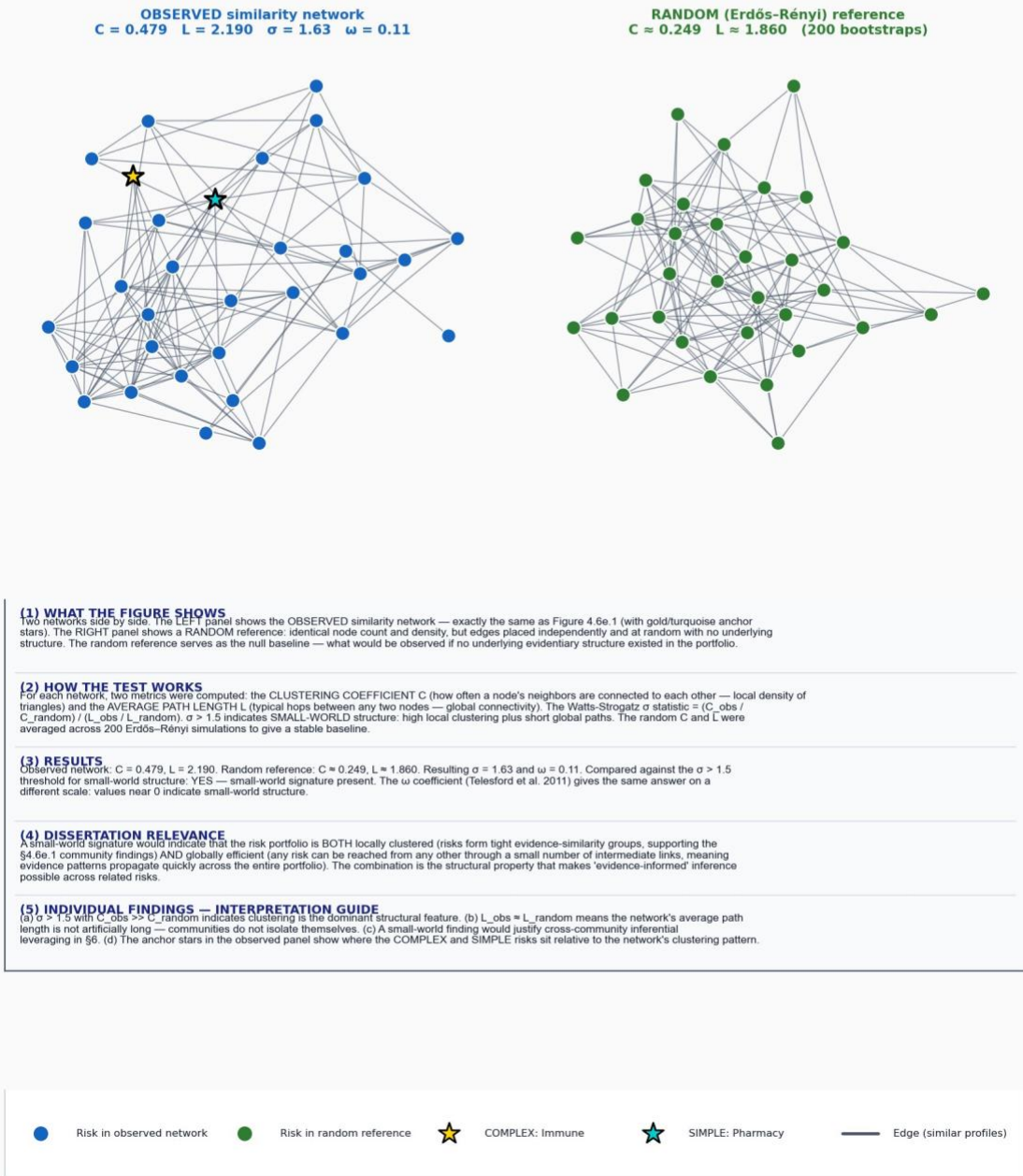
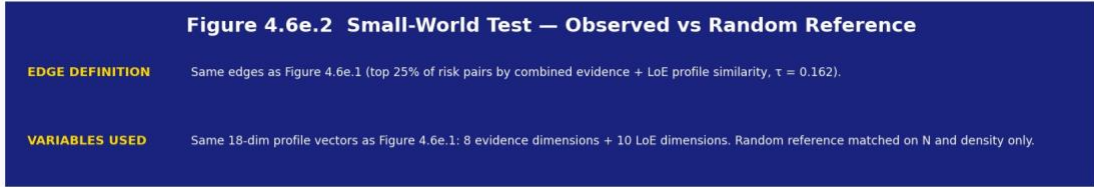


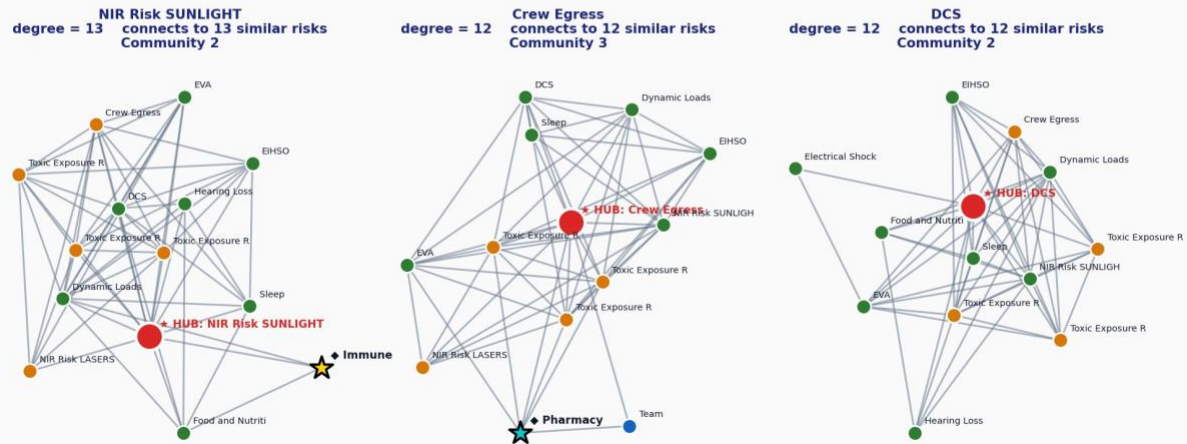
Figure 4.6e.4 Hub Spotlights — Top 3 Most-Connected Risks

EDGE DEFINITION

Hubs are nodes with the highest degree in the Figure 4.6e.1 network — risks whose evidence + LoE profile is similar to the most other risks.

VARIABLES USED

Same profile-similarity network. Each panel shows one hub plus its 1-hop neighborhood (other risks sharing high-similarity edges with that hub).



(1) WHAT THE FIGURE SHOWS

Three panels, each showing one of the three risks with the most similar-profile neighbors in the network. The red circle in the center of each panel is the HUB. The smaller surrounding circles are the risks that share a high-similarity edge with that hub. Where a COMPLEX or SIMPLE anchor risk appears in a hub's neighborhood, it is shown as a gold or turquoise star.

(2) HOW HUBS ARE IDENTIFIED

After building the similarity network in Steps 1-3 above, each node's DEGREE was computed — the number of edges connecting to it. A high-degree risk has many other risks with similar evidence + LoE profiles. Hubs are simply the three risks with the highest degree. Importantly, hubs are CENTRAL within the similarity structure — they may or may not be the most clinically severe risks; that ranking is independent.

(3) RESULTS

The three identified hubs are: NIR Risk SUNLIGHT ($k=13$), Crew Egress ($k=12$), DCS ($k=12$). Each hub's degree indicates how many other risks are evidentially similar to it. The hubs' community memberships show whether they cluster together or anchor different parts of the portfolio.

(4) DISSERTATION RELEVANCE

Hub risks function as EVIDENTIARY EXEMPLARS — their evidence + LoE profile is representative of how a chunk of the portfolio is supported. When NASA updates the evidence base for a hub risk, several similar risks should be flagged for parallel re-review. Hubs are natural candidates for the dissertation's case-study chapters because each one can speak for a structural group of risks.

(5) INDIVIDUAL FINDINGS — INTERPRETATION GUIDE

(a) A hub with many same-color neighbors anchors one community. (b) A hub with mixed-color neighbors bridges communities — these are the most influential risks structurally. (c) A hub whose neighborhood contains the COMPLEX or SIMPLE anchor risk allows direct comparison of that anchor to its evidentiary peers. (d) The absence of an anchor in any hub neighborhood would indicate the anchors are evidence-profile outliers.

● Hub risk ● Neighbor — same community ● Neighbor — other community ★ COMPLEX: Immune ★ SIMPLE: Pharmacy

Figure 6c. Hub spotlights for the three most-connected risks in the similarity network. NIR Risk Sunlight (degree 13), Crew Egress (degree 12), and decompression sickness (degree 12) each anchor a neighborhood of evidentially similar risks; the Immune (complex) and Pharmacy (simple) reference cases are marked where they fall within a hub's neighborhood.

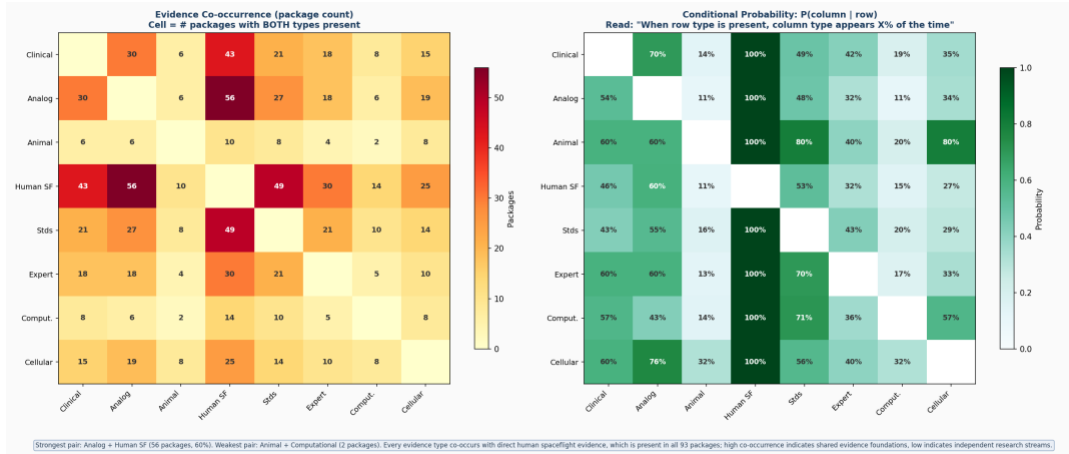


Figure 7. Evidence type co-occurrence and conditional-dependency structure across the 93 packages, showing the universal direct human-spaceflight backbone with which every other evidence type co-occurs.

For the contrasting-complexity comparison specified in Aim 2, the Immune risk emerged as the most complex anchor and the Pharmacy risk as the simplest, the two poles of the composite complexity ranking described below (Figure 9a, 9b). The Immune risk ranks highest because it documents the broadest and most fully developed evidence base in the portfolio, drawing on all eight evidence types, and is centrally positioned in the evidence-similarity network; the Pharmacy risk ranks lowest because it rests on a single dominant evidence type, the most concentrated profile in the portfolio, and occupies a peripheral position. These two risks frame the structural range of the portfolio and are used as the reference cases in the layered network comparison. The same structure can be explored directly in the topology tab of the interactive network application, a priority deliverable of this work described in Appendix F, where the complexity ranking, community membership, and centrality of every risk can be inspected and the two reference cases compared side by side in real time.

Network Centrality and Triad Structure

Degree and betweenness centrality were computed on the risk-evidence network to test Hypothesis H2a, which predicts that structurally complex risks occupy more central positions. In

this network a risk's degree is the number of distinct evidence types it cites, out of the eight in the framework, and its betweenness measures how often it lies on the shortest path linking other risks through shared evidence types. The most central risks by degree were Bone Fracture and Immune, each citing seven of the eight evidence types, followed by Carcinogenesis, Cardiovascular, and Food and Nutrition at six, against a portfolio average degree of 6.59. Betweenness centrality was highest for the Immune risk at 0.030, marking it as the principal bridge across the portfolio's evidence structure. The two reference cases bracket the range as predicted: the complex anchor, Immune, sits at the top of both centrality measures, while the simple anchor, Pharmacy, cites only three evidence types and occupies a peripheral position. The triad census of the network contained 1142 risk-pair triads and 231 evidence-pair triads, quantifying the dense web of three-way overlaps through which evidence is shared. Taken together, this ordering supports H2a: the risks that document the broadest and most fully developed evidence bases are also the ones that sit at the structural center of the portfolio, so that the same risks an expert reviewer would regard as the most thoroughly characterized are confirmed, by an independent network measure, to occupy the structural core of the portfolio's evidence coverage. A complementary reading emerges from the risk-to-risk similarity network, in which degree instead counts how many other risks share a given risk's combined evidence and LoE profile. There the most-connected hubs are the risks with the most representative profiles, NIR Risk Sunlight, Crew Egress, and decompression sickness (Figure 6c), each similar to 12 or 13 others, while the Immune risk, precisely because its eight-type profile is unusually rich, is similar to comparatively few peers and sits toward the periphery of Figure 6a. The two views are consistent rather than contradictory: the complex risks are central in how much evidence they marshal yet distinctive in profile, so that they anchor the portfolio's evidence

coverage without being typical of it. The complete centrality tables, triad counts, and topology figures are maintained in the analysis notebook and interactive dashboard; see Appendix F.

Complexity, Outliers, and Notable Risks

Complexity, as used here, is a structural property of how a risk is documented rather than a statement about its clinical seriousness. It is operationalized as a composite percentile ranking, in which a risk's value on each dimension is converted to its percentile position against the rest of the portfolio before the dimensions are combined, that averages a risk's standing across 14 measured dimensions, which group into seven facets: the diversity of evidence types it documents, the volume of accepted evidence it carries, the completeness of its documentation across the standardized template sections, the strength of its assigned LoE values, the breadth of its Design Reference Mission coverage, the three-stream evidence-support rating of its postures, and its revision activity across the study window. The fourteen dimensions distribute across these seven facets as follows. Evidence type diversity contributes one dimension, the count of distinct evidence types a risk documents. Evidence volume contributes three: the raw count of accepted evidence matches, the quality-weighted evidence composite, and evidence density, the amount of evidence per documentation page. Documentation completeness contributes four: the proportion of standardized template sections populated, the number of cited references, the total page count, and the depth of the causal-pathway (DAG) narrative. The strength of the assigned Level-of-Evidence values contributes three, taken separately for the in-mission operations track, the long term health track, and the overall worst-case score, each inverted so that stronger evidence ranks higher. Design Reference Mission coverage contributes one dimension, the number of mission profiles for which the risk is scored. The three-stream evidence-support rating contributes one dimension, and review activity contributes the last, the number of approved submissions the risk accumulated

across the study window. Each dimension is converted to a percentile rank across the portfolio before the fourteen are averaged into the composite, so that every dimension contributes on a common scale. A risk ranks as complex when it scores high across these dimensions together, so that it is triangulated from many independent directions and tends to sit centrally in the evidence-similarity network. This composite is therefore a measure of documentary dimensionality rather than of evidence diversity alone: a risk can cite many evidence types yet still rank as simple if it is thin on volume, completeness, mission coverage, or revision activity, so the ranking captures how many independent directions support a risk, not merely how varied its evidence types are. Furthermore, the composite weights breadth and documentation heavily, it must be read as a property of the documentation and not of the hazard: a structurally simple, single-stream risk is not a less serious or less dangerous one, and a thin evidence base may instead mark a risk that warrants more research attention and Board scrutiny, not less. For the overall package, a high complexity ranking signals a risk whose posture is corroborated from many independent directions, so that its assigned scores rest on a broad and mutually reinforcing base, whereas a low ranking flags a posture that, however confidently stated, leans on a narrow foundation a reviewer may wish to probe further.

The risks varied widely in the breadth and concentration of their evidence. The Immune risk was the most evidentially comprehensive, documenting all eight evidence types with more than 200 matches, and the Carbon Dioxide, Cardiovascular, and Immune risks each documented the complete set of eight types. At the opposite extreme, the Pharmacy risk drew on a single dominant evidence type and recorded the maximum concentration index of 1.0, standing as the clearest single-type outlier in the portfolio. The same two risks anchor the composite complexity ranking (Figure 8): the Immune risk was the most complex by the composite measure and the Pharmacy

risk the least, so that the most comprehensive risk is also the most complex and the most concentrated is also the simplest. These outliers illustrate that the portfolio spans a wide structural range, from richly triangulated risks documenting every evidence type to narrowly supported risks resting on a single source, and that this range is not captured by the LoE alone. The complexity ordering and the complex-versus-simple comparison are shown in Figures 8 and 9.

It is worth distinguishing these structural descriptions from the simpler question of how much evidence a risk has accumulated. The composite complexity ranking and the breadth and balance description coincide: the Immune risk is both the most comprehensive, documenting all eight evidence types, and the most complex by the composite measure, while the Pharmacy risk is both the most concentrated, resting on a single dominant type, and the simplest. Raw documentation volume and diversity tell a related but separate story, in which the Cardiovascular risk, carrying the largest citation count and representing all eight evidence types, is the most heavily documented and most diverse risk, and the Electrical Shock risk, resting on four types and the fewest citations, is the least. A risk can therefore be the most heavily cited without being the most complex, which is why Cardiovascular leads on volume yet Immune, with comparably broad but more evenly developed documentation, anchors the complex pole.

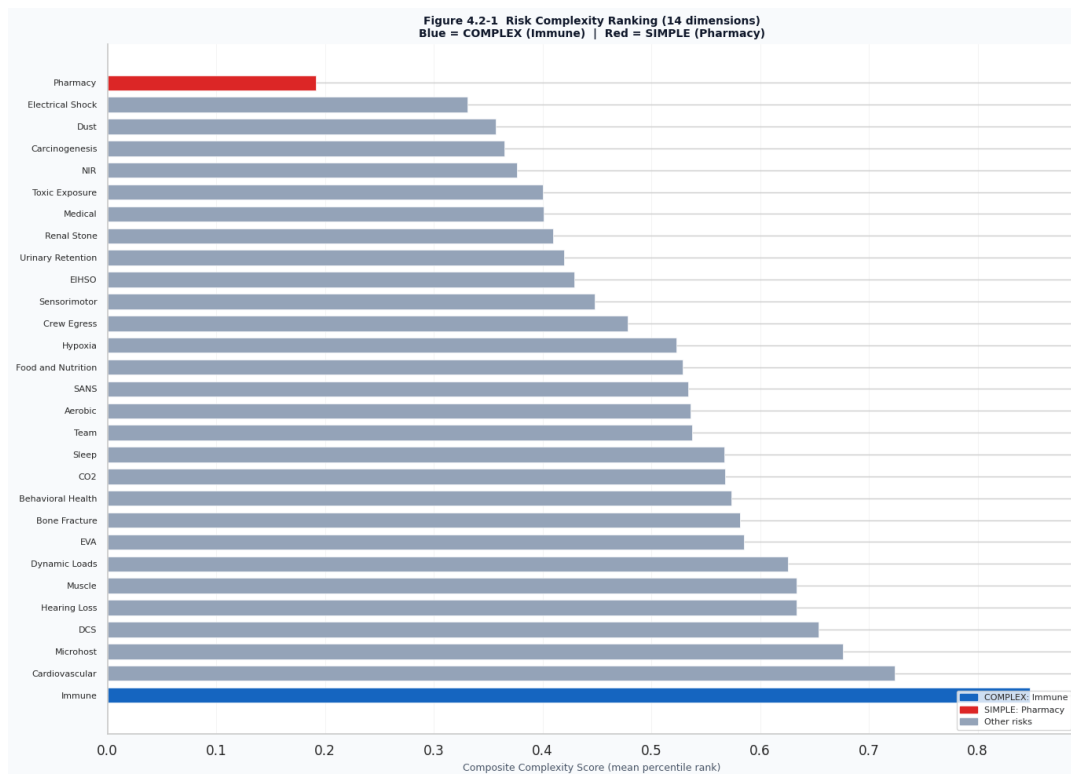


Figure 8. Composite complexity ordering of the 29 risks, derived from a composite percentile across evidence diversity, volume, documentation, LoE strength, mission coverage, and the evidence-support rating.

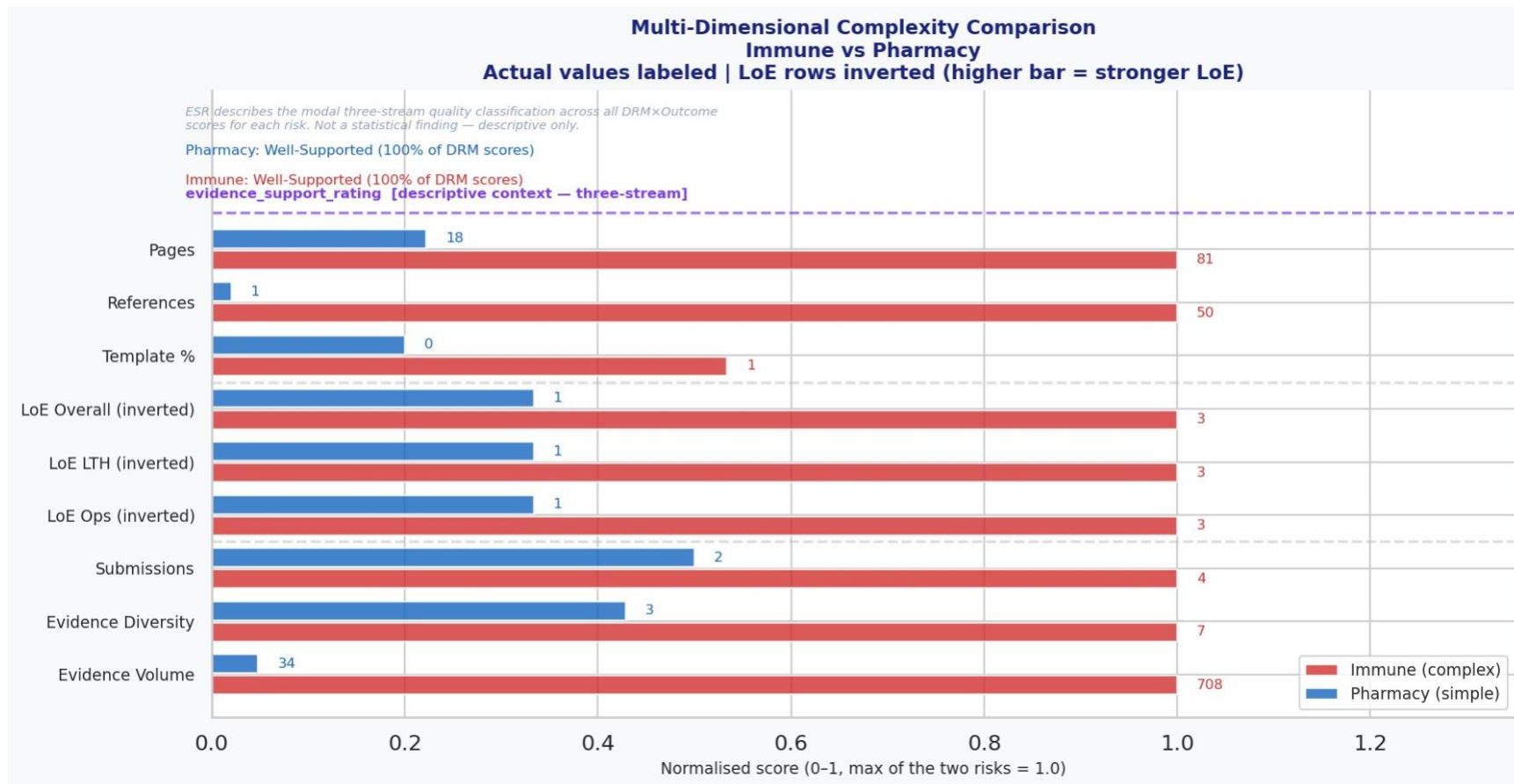


Figure 9a. Multi-dimensional comparison of the most complex risk, Immune, and the simplest risk, Pharmacy, across the documentation dimensions underlying the composite complexity ranking, with actual values labeled and LoE rows oriented so that a longer bar denotes stronger evidence.

Evidence Type Profile Comparison
(all submissions summed, normalised to each risk's maximum)

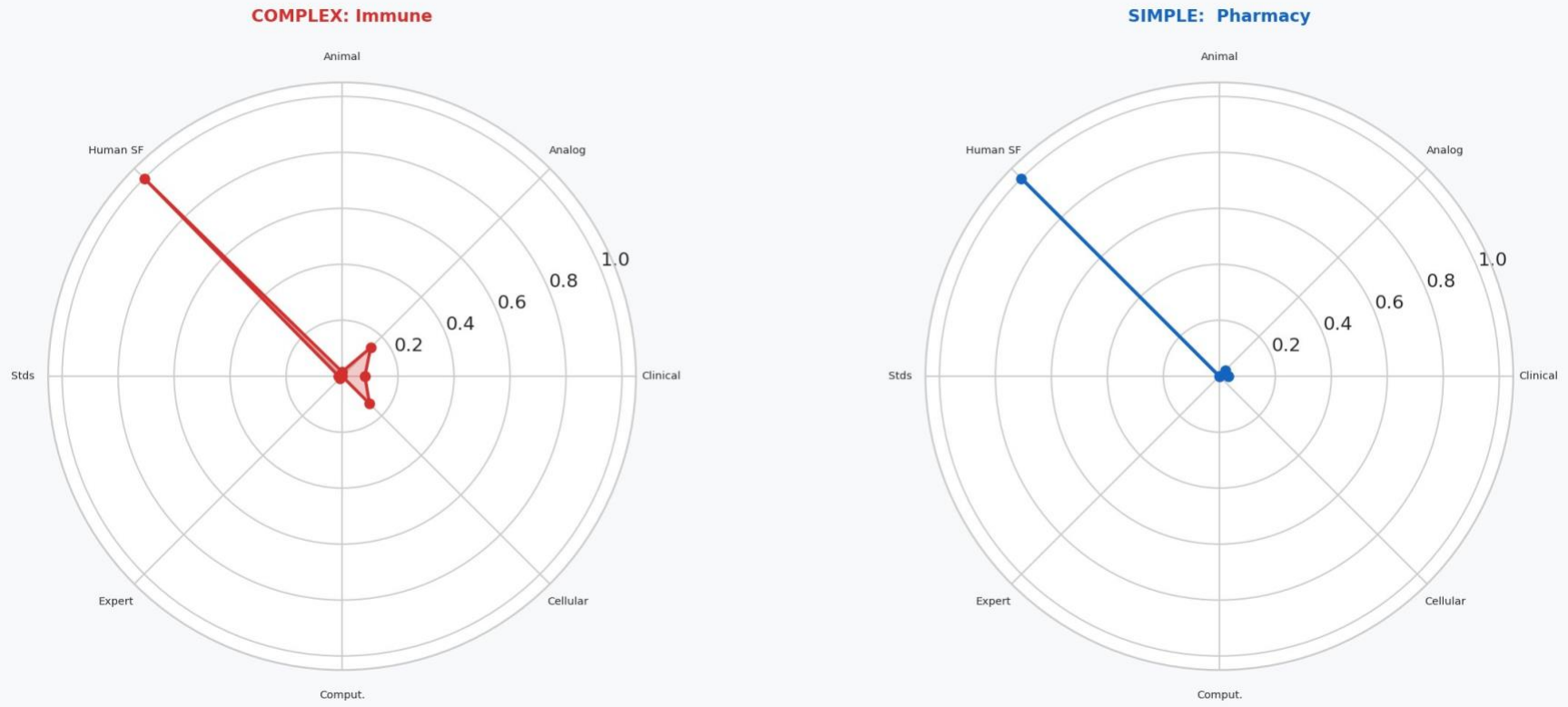


Figure 9b. Evidence type profile comparison of the most complex risk, Immune, and the simplest risk, Pharmacy, with each risk's eight-type profile normalized to its own maximum. Immune reaches modestly into several evidence types beyond direct human spaceflight, whereas Pharmacy resolves to a single dominant spoke, illustrating the difference in documentary dimensionality that underlies the complexity ranking.

Two risks, Carcinogenesis and Urinary Retention, carry no valid LoE in any mission cell within the study window and are held out of the scoring analyses for the documentary reason set out in the Methods; the specifics of their evidence bases reinforce that the missing scores mark unfinalized adjudications rather than thin evidence. Their absence from the scored set does not mean these risks have never carried a LoE; it reflects that no revised, Board-approved record assigning one fell within the 2020 to 2025 approved-submission window analyzed here, a status consistent with an adjudication that was in progress but had not been revised and re-approved during the study period. Space-radiation carcinogenesis is among the most quantitatively instrumented of all human system risks: a dedicated, peer-reviewed model, the NASA Space Cancer Risk model, projects an astronaut's risk of exposure-induced death using physics-based radiation transport, radiation-quality and dose-rate factors, population-specific cancer incidence rates, and explicit uncertainty propagation, and it underpins the career dose limit of three percent risk of exposure-induced death evaluated at the upper 95-percent confidence bound (128, 129); the model and its data foundation have been independently reviewed by the National Academies (130). The absence of a current LoE for cancer therefore reflects how the risk is managed, through a standalone quantitative dose-limit framework, rather than any deficiency of evidence. Urinary Retention is likewise well documented and anchored by direct human spaceflight data: a Lifetime Surveillance of Astronaut Health query identified 16 cases of urinary retention across 908 person-flights, including cases that required catheterization, a Type I observation measured on crew in flight (131), around which the documented record adds terrestrial clinical, computational, cellular, expert, and standards and requirements evidence. In the documented record each risk carries one of the larger and more diverse evidence profiles in the portfolio, so that for both the missing score indicates an adjudication not yet completed rather than an evidentiary gap.

Documentation Completeness, Missingness, and Bias

Documentation completeness was assessed across the standardized template sections of the 93 packages. Mean template completeness was 35.1 percent, and 81 of the 93 packages, or 87 percent, populated fewer than half of the tracked sections, so the packages are thorough on a small core of sections and sparse elsewhere. Two complementary missingness measures locate where the gaps fall. At the section level, of the 31 template sections tracked, only four were consistently present, populated in more than 80 percent of packages, while 15 were systematic gaps, appearing in fewer than 30 percent; the consistently completed sections concentrate on the risk's evidence and posture, whereas backup-guidance, contributing-factor, and requirements-flow sections are routinely left empty. At the rationale level, of the 512 mission-by-outcome posture cells, 8.2 percent carried no written rationale at all, so that 91.8 percent of postures rest on at least some narrative basis. A documentation-bias check using the Herfindahl-Hirschman concentration index (111) found that 80 of the 93 packages, or 86 percent, exceed an index of 0.5, the conventional threshold for high concentration, indicating that each package's evidence is dominated by a single type; that dominant type is direct human spaceflight evidence in 97 percent of cases, consistent with the universal-backbone pattern reported above. Together these measures show that the portfolio's documentation is not missing at random but structured: complete where it records the headline evidence and posture, thin in the supporting and backup sections, and concentrated on a single evidence stream. The full completeness, missingness, and bias outputs are maintained in the analysis notebook and dashboard; see Appendix F.

Thematic Synthesis and Recommendations

Thematic Structure of the Documentation

The deductive thematic coding of the package narratives revealed both pervasive and selective themes (Table 5). Three themes were universal, appearing in 100 percent of packages: explicit acknowledgment of evidence gaps, the status of countermeasures, and operational impact, the last two carrying the highest total frequencies in the corpus. Temporal progression of the risk across missions also appeared in 100 percent of packages. Knowledge maturity was discussed in 73 percent of packages, and explicit acknowledgment of specific evidence gaps in 54 percent. The rationale was evidence-referenced in 86 percent of packages and narrative-only in 15 percent, consistent with the package-level rationale-content finding from the mission observations.

The cross-risk dependency theme is the result of greatest structural significance. Language referencing dependencies between one risk and another appeared in only 38 of the 93 packages, or 41 percent, which is fewer than half and confirms the corresponding hypothesis. The portfolio is therefore documented in substantial isolation: although the network analysis demonstrates that risks are structurally interconnected through shared evidence and similar profiles, the narratives themselves acknowledge those interconnections less than half the time. This juxtaposition of structural interconnection with documentary isolation is one of the study's central qualitative findings and motivates a recommendation for more explicit cross-referencing of related risks. The thematic prevalences are shown in Figure 10.

These prevalences are computed at the level of the package: each of the 93 packages was coded for the presence or absence of each deductively defined theme, and the reported percentage is the share of packages in which the theme appears at least once. The cross-risk dependency theme illustrates the coding: a package was credited with the theme when its narrative explicitly

referenced another named risk or a shared mechanism, for example language linking bone demineralization to the renal-stone risk through mobilized calcium, and it was not credited when the narrative treated the risk in isolation. That only 38 of the 93 packages, or 41 percent, met this criterion, against a backdrop in which the network analysis shows the same risks to be evidentially interconnected, is the documentary expression of risks that are structurally linked but described as though they were independent.

Table 5. Prevalence of deductive themes across the 93 package narratives.

Theme	Present in	% of packages
Evidence gap (general)	93/93	100%
Countermeasure status	93/93	100%
Operational impact	93/93	100%
Temporal progression	93/93	100%
Evidence-referenced rationale	80/93	86%
Knowledge maturity	68/93	73%
Evidence gap (explicit/specific)	50/93	54%
Cross-risk dependency	38/93	41%
Narrative-only rationale	14/93	15%

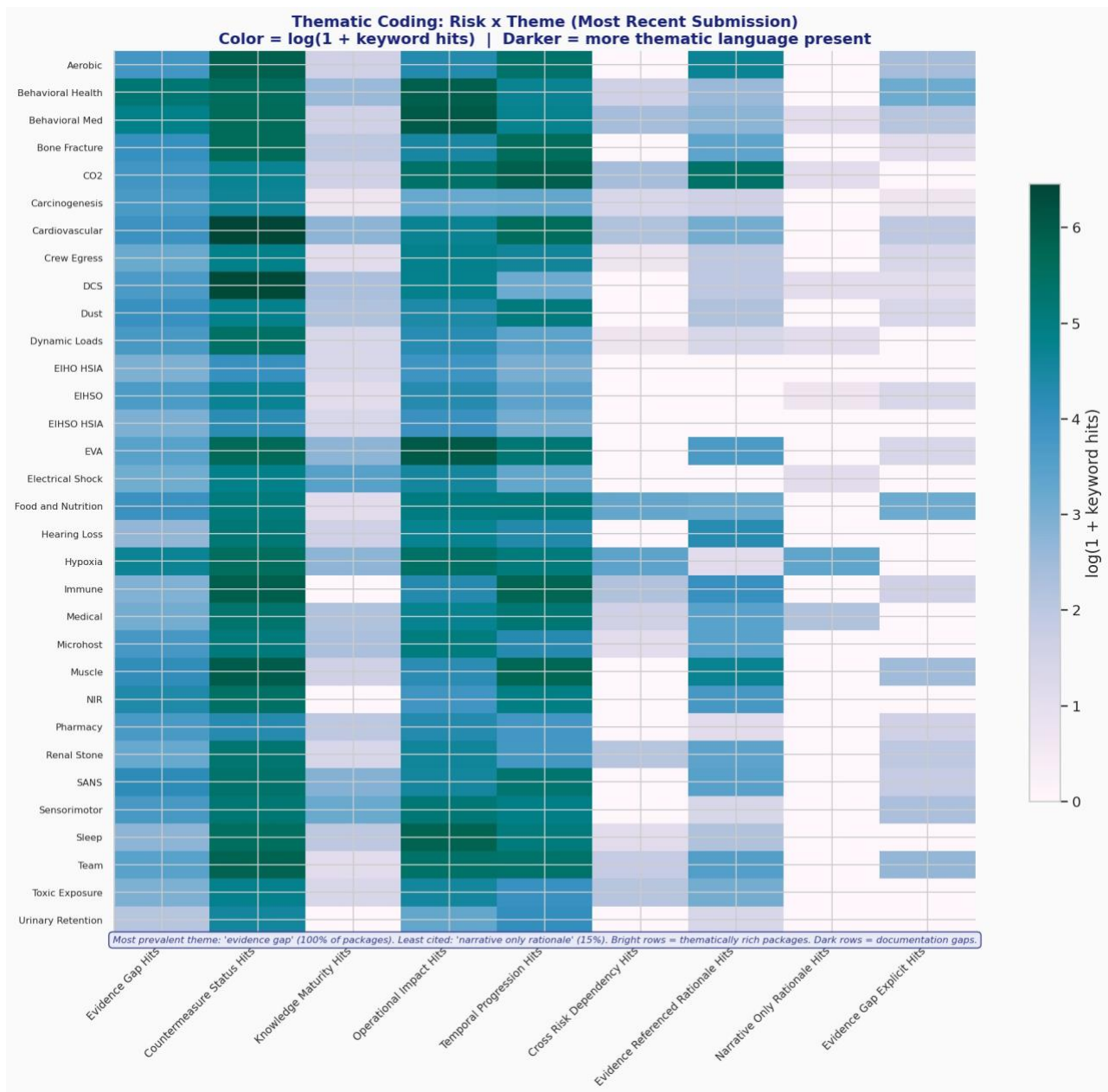


Figure 10. Prevalence and frequency of the deductive themes across the corpus, highlighting the universal evidence-gap, countermeasure, and operational-impact themes and the minority cross-risk-dependency theme.

Validation and Robustness Across the Aims

Validation against the human-annotated gold standard revealed that the initial substring-based evidence type classifier, which matched keywords without regard to the surrounding context, agreed with human judgment at a level well below the threshold required of a defensible

measurement instrument and produced systematic false positives where generic terms triggered a type without supporting evidence, “requirements” mapped to standards, “model” to computational even when the model rested on spaceflight or animal data, hazard labels such as “isolation” to analog studies, and hedging or process language such as “assumed” or “HSRB Risk Presentation” to expert consensus. A refined classifier (v3) was therefore implemented with word-boundary matching, a standards gazetteer restricting Type VIII to genuine standard identifiers, explicit exclusion of adjudication-process metadata and negated and reference-list contexts, a precedence rule favoring direct empirical evidence over modeling and opinion, disambiguation of models built on spaceflight or animal data, and a context window widened from 200 to 400 characters. Together these changes raised agreement with the human gold standard to 96.2 percent, a Cohen kappa of 0.923. The refinement was built around the structure the packages are expected to contain; fuller standardization of the source packages would permit a simpler and more reliable classifier in future work.

Furthermore, the misclassification is non-differential noise, meaning it is unrelated to a risk’s LoE score and so falls evenly across well- and poorly-scored risks, in the evidence measure rather than in the outcome, its expected effect is attenuation of any evidence-to-score association toward the null; it cannot generate a spurious null, a direction of bias established for non-differential measurement error (124). The non-significant primary result is therefore conservative with respect to this error, and re-analysis under the refined classifier confirmed the null, with the corrected per-type prevalences, network structure, and complexity rankings reported in the classification-robustness analysis (Section 6.6a). The thematic analysis, which uses an independent theme lexicon, and the LoE coverage statistics, which derive from the structured posture tables rather than the text classifier, are unaffected by this refinement.

Agreement between the engine and the human annotator is reported as per-type precision (57), for each evidence type, the proportion of engine assignments the reviewer confirmed, together with an overall Cohen kappa interpreted on the Landis and Koch bands, on which a kappa of 0.61 or above is the conventional minimum for a measurement instrument and 0.81 or above is near-perfect (70, 71, 72), and read additionally against the Krippendorff convention, under which a value of 0.80 or above supports firm conclusions and a value between 0.667 and 0.80 supports only tentative ones (64). Across the 287-item stratified validation sample, which drew 40 items from each of the larger evidence types and the full available pool from the two sparsest, Animal at 22 and Computational at 25, the engine reproduced the human annotator's evidence type assignments with an overall Cohen kappa of 0.923 (95% CI 0.879 to 0.968) and 96.2 percent precision. Both figures sit above the near-perfect band of the Landis-Koch scale and clear the Krippendorff threshold for firm conclusions, so the evidence counts that enter every downstream test can be treated as a faithful measurement of the documented record rather than an artifact of automated extraction; the per-type precision breakdown is retained in the validation table in Appendix F. Because the portfolio contains only 29 scored risks, the confirmatory tests are intrinsically small-n and modestly powered; the longitudinal design, however, 93 approved submissions across the window, with most risks documented two to four times, together with the prevalence of evidence drawn from human studies and published reports across the corpus, supports a more robust preliminary assessment than the risk count alone would imply, while keeping the findings explicitly preliminary.

Linking the Results to the Research Question and Aims

Read against the research question, the results give a two-level answer. At the portfolio level, the types, sources, and volume of documented evidence do not by themselves predict the LoE the

risk custodian assigns, which indicates that the scoring integrates mission applicability and deliberative expert judgment beyond what citation counts capture. At the mission level, the LoE and the breadth of the evidence base weaken systematically from low Earth orbit toward Mars, which locates the portfolio's evidentiary thinness precisely where the operational margin is smallest. The network analysis answers the structural portion of the question by showing universal evidence hubs and statistically meaningful communities of evidentially similar risks, while the thematic analysis shows that this structural interconnection is acknowledged in fewer than half of the narratives. Together these results address all three aims: Aim 1 through the portfolio and mission-stratified evidence analyses, Aim 2 through the network topology and the complex-versus-simple comparison, and Aim 3 through the thematic synthesis and the documentation-completeness findings. A consolidated crosswalk mapping each pre-specified hypothesis to its analytic approach, result, and preliminary, exploratory finding is provided in Appendix I (Table I7).

Discussion

This study conducted the first systematic computational examination of the relationship between the documented evidence and the assigned LoE across NASA's active human system risk portfolio, and it produced a finding that operates at two levels. The portfolio-level analysis found that simple characteristics of the documented evidence, its volume, its quality-weighted composite, the presence of direct human spaceflight data, and its diversity, do not predict a risk's worst-case LoE. Rather than indicating that evidence is irrelevant to scoring, this result is best interpreted as evidence that the risk custodian's scoring is not a mechanical function of citation counts but an integrative judgment that weighs the applicability of evidence to specific mission contexts and the deliberative assessment of expert teams. The universality of direct human spaceflight evidence and expert opinion across every risk reinforces this reading: because these foundational types are present everywhere, they cannot by themselves discriminate among scores, and the discriminating work is done by the applicability and interpretation of evidence rather than by its mere presence.

The contribution of the study rests as much on the pattern of these decisions as on any single one. Failing to reject the evidence-volume nulls is not a null result in the dismissive sense; in a domain that has long assumed that accumulating evidence earns a stronger score, demonstrating that volume, quality, breadth, and type do not predict the assigned LoE is a substantive and, for the field, counterintuitive finding. It is also a disciplined one, because it is reported with its effect sizes and confidence intervals and is bounded by pre-specified equivalence and Bayes-factor analyses rather than asserted from a single p-value. These two analyses ask different and complementary questions and are reported even where they diverge. The equivalence test asks whether the association is small enough to be treated as effectively zero; at this sample size it was inconclusive, since the confidence interval was too wide to place the correlation inside the pre-

specified region of practical equivalence, so it could neither confirm nor rule out a meaningful association. The Bayes factor asks instead how much more probable the observed data are under the null than under a plausible alternative, and here it favored the null by roughly five to one, a Bayes factor near 0.19 for the alternative that counts as moderate evidence for the absence of an association rather than its mere non-detection. The rejection of the mission-effect null supplies the positive complement: the assigned score is governed by mission applicability, with the documented evidence base strongest for near-Earth missions and weakest for exploration-class missions, and with expert judgment bridging the gap where direct evidence is sparse. It is worth being precise about what drives this gradient: in the mixed-effects models the mission profile itself was the significant predictor of the score, whereas documented evidence breadth was not, so the weakening toward Mars reflects the diminishing applicability of existing evidence to exploration-class conditions rather than a measured thinning of evidence the Board mechanically tracks. The further observation that evidence-to-score alignment is itself weaker for Mars than for low Earth orbit is directional and exploratory, clearer for the long-term-health outcome than for operations, and is best read as a signal to watch rather than a firmly estimated effect. Read together, the retained nulls and the rejected mission null converge on the same conclusion from two directions, and that convergence, reinforced by the qualitative finding that rationales cite applicability and judgment rather than evidence quantity, is what allows the central claim to be advanced as the interpretation best supported by the data rather than as an artifact of limited power.

The mission-stratified analysis revealed the structural signal that the portfolio summary conceals. The LoE weakens monotonically and significantly from low Earth orbit toward Mars, and the breadth of the evidence base thins along the same axis, while the relationship is unaffected by how much evidence a risk documents. This is the empirical signature of a low-Earth-orbit-

centric evidence base confronting an exploration-class mission profile: the missions for which the operational margin is smallest are precisely those for which the directly applicable evidence is weakest. The near-universal Well-Supported rating recorded across mission observations, coexisting with this weakening, indicates that the agency expresses confidence in its assessments even where direct evidence is sparse, which is consistent with, and dependent on, the bridging role of expert judgment for the most distant missions. The contrast between the two tiers is itself a methodological lesson, in that a portfolio-level null can coexist with a strong and consequential mission-level gradient, and an examination that considered only the portfolio summary would have missed the finding of greatest operational importance.

The structural analyses extend this account. The network exhibits statistically significant community structure, with groups of risks resting on similar evidence foundations, and a universal backbone of human spaceflight and expert evidence onto which more specialized types are layered. Yet the thematic analysis shows that cross-risk dependency is acknowledged in fewer than half of the narratives. The portfolio is thus structurally interconnected but documented in isolation, a juxtaposition that points to a concrete opportunity: risks that share evidence foundations and mechanisms could be reviewed and documented in a coordinated way, with explicit cross-referencing, which would both improve the efficiency of the review process and surface dependencies that isolated documentation obscures.

These findings translate into recommendations across three horizons. In the near term, the agency could institute explicit cross-referencing of mechanistically and evidentially related risks within the standardized template, and could use the alignment-tension cases identified by the examination, those risks whose documented evidence sits in apparent tension with their disposition, as a prioritized list for review. In the medium term, the systematic mapping of the low-

Earth-orbit-to-Mars evidence gradient provides a data-informed basis for directing risk-reduction research toward the mission contexts where the evidence base is thinnest relative to the missions being planned. In the longer term, the reproducible pipeline developed here could be integrated into the documentation workflow itself, so that evidence extraction, classification, and alignment checking are performed continuously as packages are revised, converting a one-time examination into an ongoing monitoring capability.

Several directions for future work follow directly from the bounded scope of this study. The quantitative analysis of the causal-graph structure within each package, as distinct from the textual narrative analyzed here, would add a complementary structural dimension and is a natural extension of the framework (24, 132). A formal appraisal of the methodological quality of individual cited sources, the Quality-of-Evidence dimension acknowledged but not assessed here, would refine the evidence characterization. The human-annotated gold standard was completed for this study by a single domain expert, establishing the classifier’s agreement with expert judgment. Adding a second independent annotator, an annotation team, or an automated large-language-model judge to report a formal inter-rater reliability estimate is the most immediate methodological priority for future work. The addition of negation detection to the natural language engine would address a known source of misclassification. Each of these extensions builds on, rather than revises, the architecture established in this work. The study delivers three concrete artifacts to the field: a reproducible extraction and analysis pipeline portable to other evidence-grading domains, a quantitative baseline of the documented evidence base against which future revisions can be measured, and a framework for identifying alignment-tension cases for operational review (133). These artifacts are made operational through an interactive dashboard that turns the pipeline’s outputs into a usable tool: from a portfolio-level overview the user can

drill into any individual risk to inspect its evidence profile across the eight evidence types and its LoE across mission contexts, trace the risk's position within the evidence-similarity network, and surface the alignment-tension cases flagged for review, so that an analysis that is intricate to compute is delivered through a single, navigable interface that is straightforward to use.

The inferential status of the statistical findings should be separated from the standing of the artifacts the study delivers, because the two do not rise or fall together. The portfolio-level correlations are pilot results, powered only for large effects and offered as hypothesis-generating, and their strength is appropriately bounded by the equivalence and Bayes-factor analyses. The three deliverables, by contrast, do not depend on that inferential strength and retain their value whatever a larger future sample eventually shows. The reproducible pipeline is a working instrument: it ingests the standardized packages, extracts and classifies evidence against the published eight-type hierarchy at validated accuracy, and runs end to end from raw submissions to analysis, so that any future revision of the corpus can be re-examined on demand rather than rebuilt from scratch. The quantitative baseline is a fixed, dated measurement of the documented evidence base across all 29 active risks and the 2020-to-2025 window, against which future submissions can be compared to detect whether the evidence base is strengthening, where, and for which missions; its worth is as a reference point, and a reference point does not require a significant correlation to be useful. The alignment-tension framework is an operational screening tool: it flags the specific risks whose documented evidence sits in apparent tension with their disposition, giving the Board a prioritized, reproducible list for review regardless of whether evidence volume predicts scores in aggregate. Each of these is a concrete, reusable contribution to how a graded evidence base can be examined, and each would stand even if every portfolio-level correlation were eventually overturned, because they are tools and measurements rather than claims. The

exploratory framing governs how confidently the numerical associations may be read; it does not diminish the methods, the baseline, or the framework that the study contributes to the field.

A supplementary exploratory analysis examined whether the documented evidence could reconstruct a LoE for the two risks that carry none, namely Carcinogenesis and Urinary Retention, an analysis framed to ask whether a missing LoE represents a true evidentiary gap or an unfinalized adjudication, given that both risks carry complete posture information and a substantial evidence record despite lacking a posted score. A model trained on the scored risks performed at or below a majority-class baseline, the accuracy obtainable by always predicting the single most common score, reaching 0.40 against that baseline of 0.50, while the risk custodian's qualitative evidence-support rating did not by itself determine a score, indicating that a missing LoE cannot be reconstructed mechanically from the evidence record. This corroborates the central finding at the level of predictive reconstruction: the LoE encodes mission-applicability and deliberative judgment that the citation record does not contain. What the evidence can do is bound the plausible posture: both unscored risks are evidence-rich and Well-Supported, the validated model places Carcinogenesis at a model-implied LoE of 2 and Urinary Retention at 3, and both fall within the band of 2 to 3, median 3, occupied by comparable well-supported risks, so that the bound is indicative rather than assertable and frames their missing scores as unfinalized adjudications rather than evidentiary gaps. This reconstruction is exploratory and rests on only the 27 scored risks, a sample far too small for a predictive model to carry independent weight, so it is reported as a bounding and corroborative exercise rather than as a stand-alone result. As the active-risk taxonomy expands with future missions and more risks accrue posted scores, the same procedure could be repeated on a larger sample, where it would gain the resolution to move from an indicative bound toward a more dependable estimate.

A natural extension of this work is to examine the fidelity of the data-collection process one level upstream of the documented evidence. Future study could characterize, for each evidence stream feeding a risk, the provenance and instrument fidelity of the source measurements, the consistency of capture across crews and missions, and the latency and completeness with which results propagate into the documented package. This matters because the central finding could reflect either genuine deliberative judgment or variation in how faithfully the upstream data were captured; distinguishing the two would require a data-collection-fidelity examination and would further strengthen a causal interpretation of the evidence-to-score relationship.

A further exploratory direction, pursued in the accompanying analytical notebook rather than here, asks whether a small set of frequently invoked risks, among them the medical, pharmacy, food and nutrition, and closed-environment risks, function as structural “core” or foundation risks within the evidence similarity and narrative dependency networks. In this framing, a core or foundation risk is one whose evidence profile and causal narrative are repeatedly drawn upon by other risks, a structural hub on which many risks depend, rather than one that is merely severe or extensively studied in isolation. Formalizing that question through directed causal modeling could reveal how such foundational risks propagate to others across the portfolio and represents a promising avenue for future work (134).

Limitations

The findings must be read against several limitations, the most consequential of which concerns the data-generating process itself. The documented evidence and the assigned LoE are not produced independently. Within a single submission package the risk custodian team compiles the evidence record and the assigned score together, as one coupled deliverable, rather than fixing a score for each mission and operational or long-term health track in isolation and then assembling

evidence to defend it. The evidence and the score are therefore developed and presented jointly, not independently, so an observed alignment between them, or its absence, cannot be read as a test of a relationship between two separately established quantities. This coupling is a form of information bias that no statistical adjustment can remove, because it is intrinsic to how the records are created, and it means that an observed alignment, or its absence, between documented evidence and score cannot be interpreted as a test of an independent relationship. This is the single most important limitation of the study and bounds the causal interpretation of every alignment finding; the results characterize the documented record as it exists, not an independent evidentiary basis for the scores. The standardized package template does not currently separate the evidence record from the scoring rationale, so this pilot could analyze the documentation only in the coupled form in which it is produced; decoupling the two, for instance by capturing the evidence base independently of the score within the template, lies outside the scope of this study, which characterizes the records as they exist, and is offered as a design recommendation for any future iteration of the framework that aims to test the evidence-to-score relationship as a genuinely independent one. A related limitation follows from authorship: each risk is maintained by a different custodian team, and even within the shared template the teams recorded narratives and evidence with differing conventions and levels of detail, so that some of the variation observed across packages reflects documentation practice rather than substantive differences in the underlying evidence.

A second set of limitations arises from the natural language methodology. The lexicon-driven engine was tuned to favor precision over recall, favoring a low rate of false matches over capturing every possible mention, so that accepted matches are well supported by their context but some genuine references phrased in unanticipated ways are missed, which means the evidence counts

are conservative lower bounds rather than exhaustive enumerations. This precision-oriented configuration was a deliberate response to the absence of standardized evidence formatting across packages, since terminology and structure vary between custodians and a recall-oriented engine would have admitted many spurious matches; broadening recall is a natural direction for future work once a more standardized corpus is available. The engine does not perform negation detection, the automated recognition of when a sentence asserts that something is absent rather than present, so a statement that a given type of evidence is absent could in principle be counted as a mention of that evidence type, although the precision-oriented matching and the subject-role disambiguation limit the practical effect, so that in practice such miscounts are rare, since a negated mention must still survive the context and subject-role checks to be accepted, though they remain possible where a phrase is read out of context. The optical character recognition that recovers text from figures and tables is imperfect, and some graphically presented evidence may be missed or misread. This recognition step was necessary precisely because the packages lack a standardized, machine-readable format for the content of figures and tables, and it recovered most of that material well; its imperfection is therefore a limitation that follows from the non-standardized source material rather than from the method itself, and it marks a direction for future analysis in which a more structured input format or a purpose built table and figure parser would extend the recall demonstrated here. The human-annotated gold standard that quantifies the classifier’s accuracy against expert judgment was completed by a single domain expert on a stratified sample of 287 matches, with per-type precision and the overall Cohen kappa reported in the validation section; the principal residual limitation is the absence of a second independent annotator or an automated large-language-model judge, an approach in which a language model is prompted to

adjudicate annotations in place of a second human rater (74); adding either a second annotator or such a model would permit a full inter-annotator agreement estimate and is left to future work.

A related limitation follows from the structure of the source documents rather than from the engine alone. The natural language engine is designed to pull clearly expressed evidence types from each package, and it performs best where a package follows the standardized template and states its evidence in the expected sections using the expected terminology. The packages, however, are authored by different risk custodians and are not uniform in how their content is structured or worded, so the consistency on which a lexicon-driven engine depends is not guaranteed. Where a package departs from the anticipated structure, names its evidence in idiosyncratic language, or records a relevant statement such as a rationale in a location or phrasing the lexicon does not anticipate, a genuine piece of evidence can be missed even though the design incorporates a human-annotated gold standard to benchmark the automated classification against expert judgment. The rationale content is a concrete example: a package may include a substantive rationale, but its presence, placement, and wording are not standardized across custodians, and the cue terms that signal an evidence type are not used consistently, so the engine recovers what is expressed in the form it was built to expect and can under-count evidence that is present but phrased in an unanticipated way. The provenance distribution reported in the Methods, in which nearly half of the accepted matches were drawn from non-canonical locations, is the empirical signature of this heterogeneity and reinforces that the evidence counts are conservative lower bounds tied to the documentary form of each package.

A third set of limitations concerns the sample and the analytic design. The study analyzes a complete census of 29 active risks, which fixes the portfolio-level sample size and confers approximately 80 percent power only for large effects, so that the non-significant portfolio findings

cannot distinguish a true null from an undetectable small or moderate association, a limitation that the equivalence testing made explicit and could not resolve. The worst-case aggregation of the LoE to a single value per risk, chosen for its safety relevance, necessarily discards the within-risk variation across missions that the mission-stratified tier then recovers, and it eliminates Strong scores at the portfolio level by construction. The risks are documented in prose and structured tables rather than in machine-readable causal graphs, so the network analysis rests on the documented evidence relationships rather than on the formal causal structure of each risk. The thematic coding, while systematic and reproducible, applies a keyword-based detection of themes that shares the precision-over-recall character of the evidence extraction.

Finally, the study is conducted within a single institution's documentation system under a specific governance framework, and the specific quantitative findings are particular to NASA's portfolio and practices. The methodological framework is designed to be transferable, but the substantive results should not be generalized to other evidence-grading systems without re-analysis. The strength of the framework lies in this very separation: the pipeline, the classification scheme, and the alignment-tension diagnostics are domain-general and could be repointed at any portfolio in which structured scores are paired with documentary evidence, whereas the specific quantitative findings remain particular to NASA's portfolio and should not be carried over without being re-derived. This distinction is likely to grow in importance as human spaceflight expands: commercial providers and international partners are taking on a rising share of crewed activity, and NASA's Moon-to-Mars program envisions a sustained lunar presence, with an Artemis Base Camp and a planned lunar outpost intended to precede crewed Mars missions (135). As more organizations generate their own human-system risk evidence under differing governance frameworks, a reusable and transparent method for examining how evidence maps to risk scores

becomes more valuable, even as the substantive benchmarks established here remain specific to the system that produced them. The network and thematic components are exploratory and hypothesis-generating, and the patterns they reveal are intended to motivate review and further study rather than to support confirmatory claims. These limitations together define the study as a pilot-scale, proof-of-concept analysis whose value lies in establishing a reproducible methodological framework to test architecture, demonstrated here on a human-spaceflight risk architecture but applicable in principle to other public-health evidence architectures, and an empirical baseline, and whose specific findings are preliminary pending the extensions identified in the Discussion.

A further boundary concerns data granularity. The analysis examines the evidence as documented in the Risk Summary Packages; where a risk's support resides in data not reproduced in the package, the analysis can characterize what is recorded but not the finer substructure beneath it. Hardware-governed risks illustrate this: a package may record that an engineering minimum requirement applies, an entry in the standards and requirements evidence type, without enumerating the underlying verification dataset or test history that justified it, so the requirement is observed as a single coded citation rather than as the body of engineering evidence behind it (16). The same limit applies when a package points to a fuller source by reference title or link to support a claim whose underlying evidence it does not reproduce. In those cases, the engine captures the citation at the title level only, so the source is recorded as a documented pointer to its evidence type rather than appraised from its full text. Resolving this substructure would require different data sources and a different study design and is therefore outside the present scope.

Finally, the portfolio of active human system risks analyzed here is the count applicable to the study window rather than a fixed quantity; the taxonomy is periodically revised, and risks may

be added, split, merged, or retired as understanding matures. The present data already exhibit this pressure, since sub-risks under a single parent can receive different LoE values. The set of recognized human system risks has itself shifted over time as missions and ground analogs have revealed new hazards and as existing entries have been split, merged, or retired, so that the portfolio is best understood as a living set rather than a fixed list. This is precisely why a reproducible examination framework is valuable: as future missions surface risks not yet in the taxonomy, the same pipeline can characterize their evidence base and test how that evidence would be scored, extending the examination to risks the agency has yet to formalize. Venous thromboembolism is an illustrative candidate for future addition: the incidental discovery of an asymptomatic internal jugular vein thrombosis in an otherwise healthy crew member aboard the International Space Station (136) prompted a coordinated response and a dedicated medical technical brief (137), and the condition, presently adjacent to the cardiovascular adaptations risk and mechanistically linked to spaceflight fluid shifts, may warrant tracking in its own right. That the risk set can evolve is itself an argument for a reproducible examination framework able to re-characterize the portfolio whenever its membership changes.

Conclusion

Project CASCADE found that NASA's LoE scoring is most consistent with tracking the mission applicability of evidence and the deliberative judgment of expert teams rather than the sheer quantity or diversity of documented evidence, a reading the portfolio data support but, with only 29 risks and power to detect only a Spearman correlation of roughly 0.48 or larger, cannot establish to the exclusion of a weaker association; the analysis further indicates that the evidence base is structurally thinnest for the most demanding exploration-class missions, weakening systematically from low Earth orbit toward Mars even as the agency records confidence in its

assessments. The risk portfolio is structurally interconnected through a universal backbone of human spaceflight and expert evidence and through statistically meaningful communities of evidentially similar risks, yet it is documented in substantial isolation, with cross-risk dependencies acknowledged in fewer than half of the records. Beyond these substantive findings, the study delivers a reproducible computational pipeline for examining graded evidence, a quantitative baseline of the documented evidence base against which future revisions can be assessed, and a framework for flagging alignment-tension cases for operational review. The approach is portable to other domains that grade expert-reviewed evidence under conditions of limited data, and its principal contribution is to demonstrate that the evidence architecture underlying high-stakes risk governance can be made transparent, comparable, and reproducible. As human spaceflight extends toward the Moon and Mars, and as terrestrial public-health settings, including disaster response and other situations that impose harsh and often recurring conditions on affected populations, face the analogous task of grading sparse evidence to guide decisions, the capacity to examine an evidence base systematically, rather than to read it one record at a time, is a contribution to the protection of the people, like astronauts, whose safety those evidence judgments ultimately serve.

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Appendix A. Risk Roster, Mission Taxonomy, Evidence Hierarchy, and Abbreviations

A1. The 29 Active Human System Risks

Table A1. The 29 active human system risks (32 posture sub-tables).

#	Risk	#	Risk
1	Aerobic	16	Hypoxia
2	Behavioral Health	17	Immune
3	Bone Fracture	18	Medical
4	Carbon Dioxide (CO ₂)	19	Microhost
5	Carcinogenesis	20	Muscle
6	Cardiovascular	21	Non-Ionizing Radiation (NIR)
7	Crew Egress	22	Pharmacy
8	Decompression Sickness (DCS)	23	Renal Stone
9	Dust	24	Spaceflight Associated Neuro-ocular Syndrome (SANS)
10	Dynamic Loads	25	Sensorimotor
11	EVA-Induced Hand/Shoulder Overuse (EIHSO)	26	Sleep
12	Extravehicular Activity (EVA)	27	Team
13	Electrical Shock	28	Toxic Exposure
14	Food and Nutrition	29	Urinary Retention
15	Hearing Loss		

A2. Design Reference Mission Continuum

Table A2. The eight Design Reference Missions, ordered along the mission continuum.

Order	Design Reference Mission	Profile
1	LEO Short	Low Earth orbit, ≤30 days
2	LEO Long	Low Earth orbit, >30 days (ISS)
3	Lunar Orbital Short	Lunar orbit, ≤14 days
4	Lunar Orbital Long	Lunar orbit, >14 days
5	Lunar Orbital and Surface Short	Lunar orbit + surface, ≤14 days
6	Lunar Orbital and Surface Long	Lunar orbit + surface, >14 days
7	Mars Preparation	Mars-preparation missions
8	Mars Planetary Long	Mars surface, long duration

A3. The Eight-Type Evidence Hierarchy

Table A3. The eight evidence types, ordered from strongest to weakest, with the analytic weight applied.

Type	Category	Weight
I	Direct human spaceflight data	8
II	Human ground-analog data	7
III	Terrestrial clinical / epidemiological	6
IV	Animal data	5
V	Cellular / tissue data	4
VI	Computational / modeling	3
VII	Expert opinion	1
VIII	Standards and requirements	2

The Type VIII weight of two reflects the codified but indirect character of standards evidence, placing it above expert opinion in the analytic composite. The eight-to-one ordering is the author's analytic device; the agency supplies only the ordinal ranking of the types.

A4. Abbreviations

Table A4. Abbreviations and acronyms used throughout the dissertation.

Abbreviation	Definition
CASCADE	Computational Analysis of Spaceflight Conditions, Astronaut Data, & Epidemiological Evidence
CFR	Code of Federal Regulations
CHS	Crew Health and Safety
CI	Confidence Interval
CO ₂	Carbon Dioxide
CR	Change Request
CRM	Continuous Risk Management
DAG	Directed Acyclic Graph

Abbreviation	Definition
DCS	Decompression Sickness
DRM	Design Reference Mission
EIHSO	Earth-Independent Human-Systems Operations
ESR	Evidence Support Rating
EVA	Extravehicular Activity
FWER	Family-Wise Error Rate
HDT	Head-Down Tilt
HERA	Human Exploration Research Analog
HHI	Herfindahl-Hirschman Index
HHPD	Human Health and Performance Directorate
HMTA	Health and Medical Technical Authority
HRP	Human Research Program
HSIA	Human Systems Integration Architecture
HSRB	Human System Risk Board
HZE	High atomic number (Z) and high-energy particles
ISS	International Space Station
KW	Kruskal-Wallis
LEO	Low Earth Orbit
LoE	LoE
LSAH	Lifetime Surveillance of Astronaut Health
LSDA	Life Sciences Data Archive
LTH	Long-Term Health
LxC	Likelihood-by-Consequence
NEEMO	NASA Extreme Environment Mission Operations
NIR	Non-Ionizing Radiation
NLP	Natural Language Processing
NLSP	NASA Life Sciences Portal
NSCR	NASA Space Cancer Risk model
OCHMO	Office of the Chief Health and Medical Officer
OCR	Optical Character Recognition
Ops	In-Mission Operations
OR	Odds Ratio
PBPK	Physiologically Based Pharmacokinetic
PCA	Principal Component Analysis
PEL	Permissible Exposure Limit
PRA	Probabilistic Risk Assessment
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QoE	Quality of Evidence
REID	Risk of Exposure-Induced Death
RIDM	Risk-Informed Decision Making
ROPE	Region of Practical Equivalence
RSP	Risk Summary Package

Abbreviation	Definition
SANS	Spaceflight-Associated Neuro-ocular Syndrome
SMAC	Spacecraft Maximum Allowable Concentrations
SME	Subject Matter Expert
SRD	System Requirements Document
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology
TOST	Two One-Sided Tests
TtO	Transition to Operations

Appendix B. Analytic Workflow and Standardized Risk Summary Package

Template Sections

B1. Analytic Workflow Across the Three Aims

The conceptual workflow of the study is summarized in Figure B1, which traces the research question through the construction of the evidence corpus, the three analytic aims, and the integrated outputs that bear on human-risk buydown. The figure orients the reader to how the evidence-alignment, network, and thematic components of the analysis fit together and feed a single synthesis.

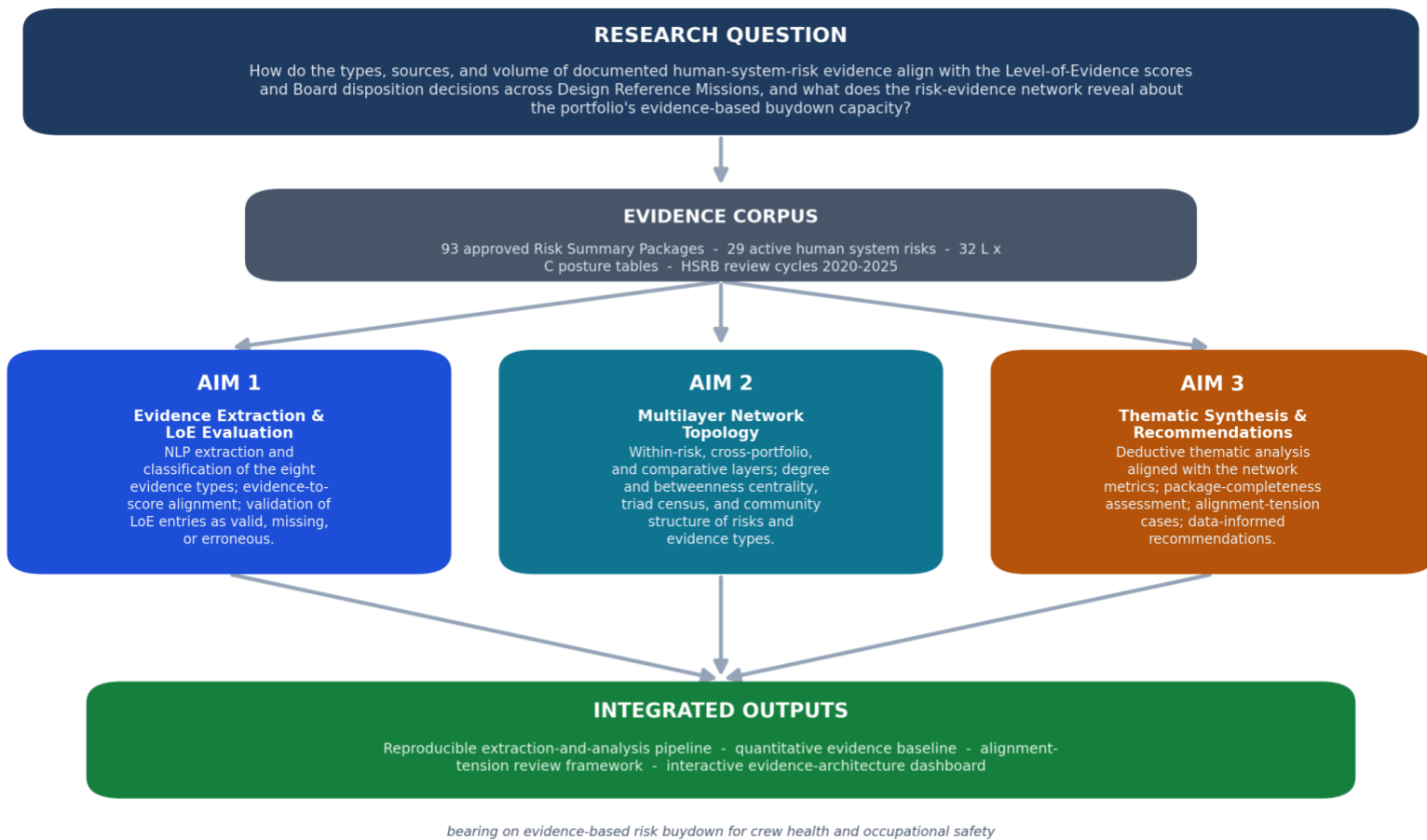


Figure B1. Conceptual analytic workflow linking the research question, the evidence corpus, the three specific aims, and the integrated outputs.

B2. Standardized Risk Summary Package Template Sections

The standardized Risk Summary Package template specifies the sections each package is expected to contain. The principal sections are the directed acyclic graph of the risk's causal architecture, the accompanying DAG Narrative, the Likelihood by Consequence matrix, the per-mission risk-posture tables for In-Mission Operations and Long-Term Health, the Overall Assessment of the Evidence, the State of Knowledge, the supporting metrics, the standards and requirements flow, the risk mitigation framework, and the high-value risk mitigation targets. Conformance to these sections varied across the corpus, with a mean template completeness of 35.1 percent, and the presence of each section was recorded during ingestion as part of the documentation-completeness assessment.

B3. The Risk Summary Package Development Process

Each Risk Summary Package is the product of a defined, multidisciplinary process rather than a single author's account. Ownership of a given human system risk rests with a risk custodian team, which the Human System Risk Board defines as a researcher, an operational researcher or physician, and an epidemiologist, each contributing distinct expertise; this team synthesizes the scientific and operational evidence in the context of spaceflight, identifies and evaluates the metrics for the risk, and communicates the resulting posture to the agency (Antonsen et al., NASA/TM-20220006812; NASA Human System Risk Board, JSC-66705 Rev B). The team first constructs the risk's directed acyclic graph and its accompanying narrative, which together externalize the causal reasoning linking the spaceflight hazards to the contributing factors and ultimately to the mission-level outcomes; against this causal model the team assembles the supporting evidence, classifies it by directness and applicability, and records

it in the standardized template sections enumerated in Section B2. The team then positions the risk on the Likelihood by Consequence matrix for each Design Reference Mission and for each of the two outcome classes, operational and long-term health, assigns the LoE from the standardized hierarchy, and documents the state of knowledge, the supporting metrics, the governing standards and requirements, and the mitigation framework. The completed package is configuration-managed and vetted by the broader NASA community before the Board sets the official posture, so that the same construction sequence is applied consistently across custodian teams and across successive revisions.

B4. Directed Acyclic Graphs as Causal-Inference Tools

A directed acyclic graph is a causal diagram in which nodes represent variables or contributing factors and directed edges, drawn as arrows, represent claims of causation from one node to another; the graph is acyclic because no path may lead from a node back to itself, which enforces a coherent forward flow of cause to effect. Directed acyclic graphs are a standard instrument of modern epidemiology, where they are used to make causal assumptions explicit and to determine, from those assumptions, which variables must be adjusted for to obtain unconfounded estimates and which must not, so that confounders, mediators, and colliders can be distinguished before any analysis is run (Greenland, Pearl, and Robins, 1999); the same formalism is implemented in widely used tools such as DAGitty (Textor et al., 2016). The Human System Risk Board adopted this epidemiological method directly, formalizing each risk's causal architecture as a directed acyclic graph that functions as a configuration-managed knowledge graph and shared mental model for the Board's stakeholders (Antonsen et al., NASA/TM-20220006812; NASA/TP-20220015708). For the present study the significance of

this structure is twofold. First, it is a genuine source of methodological rigor: the standardized template, the multidisciplinary custodian team, the explicit causal diagramming, and the configuration-managed review impose consistency on how evidence is assembled and reasoned about across the portfolio and its successive revisions. Second, it nonetheless leaves room for structured bias, because the causal assumptions encoded in a graph are themselves expert judgments, the evidence admitted into a package is selected by the custodian team, and the LoE is carried by expert opinion wherever directly applicable data are sparse. The analyses reported in this dissertation are best read against this dual character, as an empirical examination of a documentation process that is deliberately structured yet still shaped, at its margins, by judgment.

Appendix C. Statistical and Network Formulas

Spearman rank correlation:

$$\rho_s = 1 - (6 \sum d_i^2) / (n(n^2 - 1))$$

Point-biserial correlation:

$$r_{pb} = [(M_1 - M_0) / s_n] \cdot \sqrt{(p/q)}$$

Fisher / Woolf odds ratio:

$$OR = (a d) / (b c); \text{Var}(\ln OR) = 1/a + 1/b + 1/c + 1/d$$

Firth penalized log-likelihood:

$$\ell^*(\beta) = \ell(\beta) + \frac{1}{2} \log |I(\beta)|$$

Kruskal-Wallis H:

$$H = [12 / (N(N+1))] \sum (R_j^2 / n_j) - 3(N+1)$$

Bonferroni corrected alpha:

$$\alpha^* = 0.05 / 3 = 0.0167$$

Herfindahl-Hirschman index:

$$HHI = \sum s_i^2$$

Degree centrality:

$$C_D(i) = k_i / (N - 1)$$

Betweenness centrality:

$$C_B(i) = \sum \sigma_{st}(i) / \sigma_{st}$$

Clustering coefficient:

$$C_i = 2E_i / (k_i (k_i - 1))$$

Average geodesic distance:

$$L = [1/(N(N-1))] \sum d_{ij}$$

Modularity:

$$Q = (1/2m) \sum_{ij} [A_{ij} - (k_i k_j)/(2m)] \delta(c_i, c_j)$$

Cosine similarity:

$$\cos(\theta) = (A \cdot B) / (\|A\| \|B\|)$$

Mixed-effects model:

$$LoE_{ij} = \beta_0 + \beta_1 Ev_{ij} + \beta_2 DRM_j + \beta_3 (Ev \times DRM)_{ij} + \beta_4 Yr_{ij} + \beta_5 Cx_i + u_i + \varepsilon_{ij}$$

Appendix D. Lexicon, Catalogs, and Extraction Provenance

Table D1. Worked examples of the eight JSC-66705 evidence types: detection cues, an illustrative package sentence, and the classification decision for each.

Evidence type (weight)	Definition	NLP detection cues	Illustrative RSP sentence	Classification note
Type I: Spaceflight, Human (weight 8, highest)	Direct physiological or medical data from humans in spaceflight (ISS, Shuttle, Apollo, crew medical records, LSAH).	in-flight, on orbit, ISS, astronaut(s), crewmember(s), LSAH, person-flights, post-flight, mission names (Apollo, Mir, Artemis).	“An LSAH query identified 16 cases of urinary retention across 908 person-flights, including cases requiring catheterization.”	Typed I because the observation is of actual crew in flight (person-flights, LSAH). An empirical statistic measured on astronauts, not a model output or a judgment, anchors the assignment.
Type II: Ground-Analog, Human (weight 7)	Human data from spaceflight analogs: head-down bed rest, Antarctic winter-over, undersea (NEEMO), isolation and confinement.	bed rest, head-down tilt, HDT, Antarctic, winter-over, NEEMO, analog, isolation and confinement, parabolic flight.	“In 70-day head-down bed-rest studies, subjects showed cardiovascular deconditioning comparable to short-duration flight.”	Typed II rather than I because subjects are terrestrial analog participants, not crew in flight; and rather than III because the design deliberately simulates spaceflight stressors.
Type III: Terrestrial Clinical & Epidemiological (weight 6)	Human clinical or epidemiological data from the general (non-analog) population.	cohort, case-control, prevalence, incidence, clinical trial, meta-analysis, patients, registry, epidemiological.	“A meta-analysis of terrestrial cohorts reported a solid-cancer latency of ≥ 10 years following radiation exposure.”	Typed III because the population is terrestrial patients or cohorts, neither crew nor analog subjects.
Type IV: Animal (weight 5)	In-vivo studies in non-human animals.	murine, mouse/mice, rat, rodent, canine, in vivo, animal model, species names, irradiated mice.	“Irradiated murine models demonstrated dose-dependent tumorigenesis at HZE exposures.”	Typed IV on the species and in-vivo cues; distinguished from V by the presence of a whole organism.
Type V: Cellular & Tissue (weight 4)	In-vitro cellular, tissue, or molecular studies.	in vitro, cell line, cultured, tissue, DNA damage, double-strand breaks, assay, fibroblast, molecular.	“In-vitro assays of irradiated human fibroblasts showed elevated double-strand-break frequency.”	Typed V on in-vitro and cellular cues; distinguished from IV by the absence of an organismal model.
Type VI: Computational & Modeling (weight 3)	Models, simulations, and risk projections (e.g., NSCR, PRA, dosimetry).	model, simulation, projected, estimated risk, NSCR, %REID, Monte Carlo, PRA, dosimetry, predicted.	“The NSCR model projects $\%REID \leq 1\%$ for a short-duration LEO mission.”	Typed VI because the value is a model projection (%REID, NSCR), not an observation. This is the key contrast with Type I: a modeled %REID is VI; an observed crew outcome is I.
Type VII: Expert Opinion (weight 1, lowest)	Professional judgment, consensus, or board/panel opinion not tied to a specific dataset.	expert opinion, consensus, the panel judged, it is believed, in the judgment of, SME, clinical experience suggests, Board consensus; often no citation.	“In the judgment of the discipline experts, onset timing and available treatment options keep the operational consequence low.”	Typed VII because the claim rests on professional judgment rather than a dataset. The contrast with Type I is the crux of the scheme: Type I is grounded in measured crew data (“16 cases / 908 person-flights”); Type VII is grounded in expert assessment (“in the judgment of ...”). The engine keys on judgment- and consensus-framing and on

				the absence of an empirical referent.
Type VIII: Standards & Requirements (weight 2)	Normative documents, NASA standards, requirements, permissible limits, that set the acceptable bound rather than supply observational evidence.	NASA-STD-3001, permissible exposure limit, PEL, requirement, shall, standard, fitness-for-duty, document/standard numbers (JSC-, NASA-STD-).	“Career exposure is bounded by the NASA-STD-3001 permissible exposure limit for a 3% REID.”	Typed VIII, and categorized OUTSIDE the strict I→VII evidentiary ranking, because a standard is normative, not observational: it states the accepted limit, not what was measured. The engine routes standard- and requirement-citations (document numbers, “shall,” “permissible limit”) here even when they appear beside empirical evidence, so that normative bounds are never miscounted as data. Its fixed weight reflects a regulatory rather than evidentiary role.

Table D2. Composition of the classification lexicon and reference catalogs.

Resource	Count
Canonical evidence keywords	117
Keyword aliases (surface forms)	514
Crewed missions cataloged	261
Crewed-mission families	13
Crewed-mission aliases	66
Ground-analog environments	25
Ground-analog groups	15
Ground-analog aliases	64

Table D3. Provenance of the 9,098 accepted evidence matches across the document structure.

Source location	Share
Canonical template sections	14.9%
Tables	25.5%
Optically recovered image text	13.3%
Non-canonical narrative locations	46.3%

Table D4. Structured LoE entry validation across the corpus.

Status	Count	Note
Valid	419	88.6% of 473 expected (Ops 89.7%, LTH 87.3%)
Missing	40	no score where one was expected
Not applicable	39	risk by mission combination not scored
Erroneous	14	7 range entries (e.g. 2-3) + 7 numeric-to-label mismatches

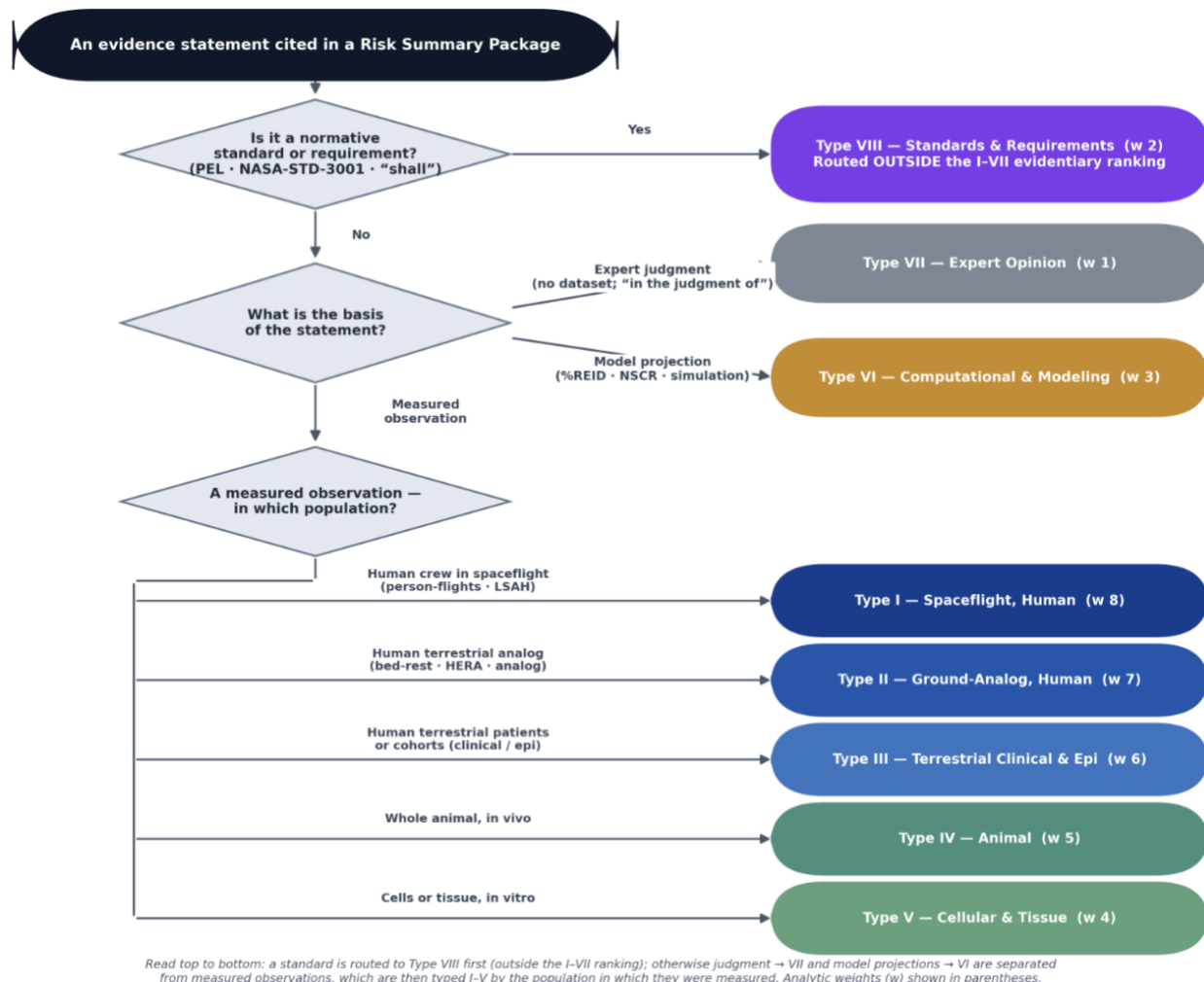


Figure D1. Decision logic of the evidence type classifier: how a cited evidence statement is routed to one of the eight JSC-66705 evidence types.

D5. Thematic codebook (Aim 3). The deductive thematic analysis coded each package narrative against a fixed list of themes set in advance rather than allowed to emerge from the text, so that every package was read against the same defined categories. The themes were drawn from the structural features of the corpus, and the coding was applied programmatically using fixed, keyword-anchored rules, so identical language was credited the same way in every package. Table D5 states, in plain language, what each theme captures, how it was identified, and why it was included, together with the share of the 93 packages in which it appeared.

The purpose of the coding is to connect how evidence is described and reasoned about in the narratives to the quantitative evidence and network findings, and in particular to test whether mechanistically related risks are documented in isolation or with explicit reference to one another. Two of the themes both concern evidence gaps but at different levels of specificity: the general evidence-gap theme captures any acknowledgment that the evidence is incomplete or uncertain, whereas the explicit (specific) evidence-gap theme captures cases in which the narrative names a particular missing study, dataset, or measurement. The first measures how often uncertainty is admitted at all; the second measures how often that uncertainty is tied to a concrete, addressable need.

Table D5. Deductive thematic codebook: what each narrative theme captures, how it was identified, and its prevalence across the 93 packages.

Theme	What it captures (and why it matters)	How it was identified	Present in
Evidence gap (general)	Any acknowledgment that the evidence base is incomplete or uncertain; shows how routinely packages recognize the limits of what is known.	Language flagging missing, limited, or insufficient evidence, or open questions (e.g., “limited data,” “not well characterized,” “further study needed”).	93/93 (100%)
Countermeasure status	Mention of actions taken or planned to reduce the risk; links the narrative to the agency’s risk-reduction (buydown) process.	References to countermeasures, mitigations, protective measures, treatments, or monitoring.	93/93 (100%)
Operational impact	How the risk affects crew health or mission performance during a mission; ties each risk to its real operational consequences.	Language describing effects on the crew, performance, operations, or mission success.	93/93 (100%)
Temporal progression	Whether the narrative describes how the risk changes over longer or more distant missions; connects the narrative to the mission gradient from low Earth orbit to Mars.	Language about mission duration, exposure time, or change across mission profiles.	93/93 (100%)
Evidence-referenced rationale	Reasoning that points to specific evidence or data; indicates how often scoring rationale is anchored in documented evidence.	References to studies, datasets, measurements, or cited sources supporting a claim.	80/93 (86%)
Knowledge maturity	Discussion of how well the underlying mechanism or causal pathway is understood; gauges how developed the science behind a risk is.	Language about mechanism, pathway, understanding, or maturity of knowledge.	68/93 (73%)
Evidence gap (explicit/specific)	Naming of a particular missing study, dataset, or measurement; distinguishes concrete, actionable gaps from general expressions of uncertainty.	Statements identifying a specific unknown or a named data need.	50/93 (54%)
Cross-risk dependency	Language in which one risk’s narrative explicitly refers to another named risk or a shared mechanism; tests whether related risks are documented in isolation, and is the basis for the cross-risk-isolation finding.	Explicit mention of another risk by name, or of a shared causal pathway.	38/93 (41%)
Narrative-only rationale	Reasoning given in prose without pointing to specific evidence; the counterpart to evidence-referenced rationale, flagging where claims rest on narrative alone.	Explanatory text asserting a basis for the posture without a citation or dataset.	14/93 (15%)

Appendix E. Supplementary Figure Series

The following supplementary figures, generated by the analysis pipeline, document the methodology and exploratory analyses in greater detail than the main text. They are provided for completeness and are referenced collectively in the Methods and Results.

Data Flow — Two Input Paths to Two Master Files to Analysis

How df_master_n93_longitudinal (Tier 1) and loe_drm_detail_final (Tier 2) are built and consumed

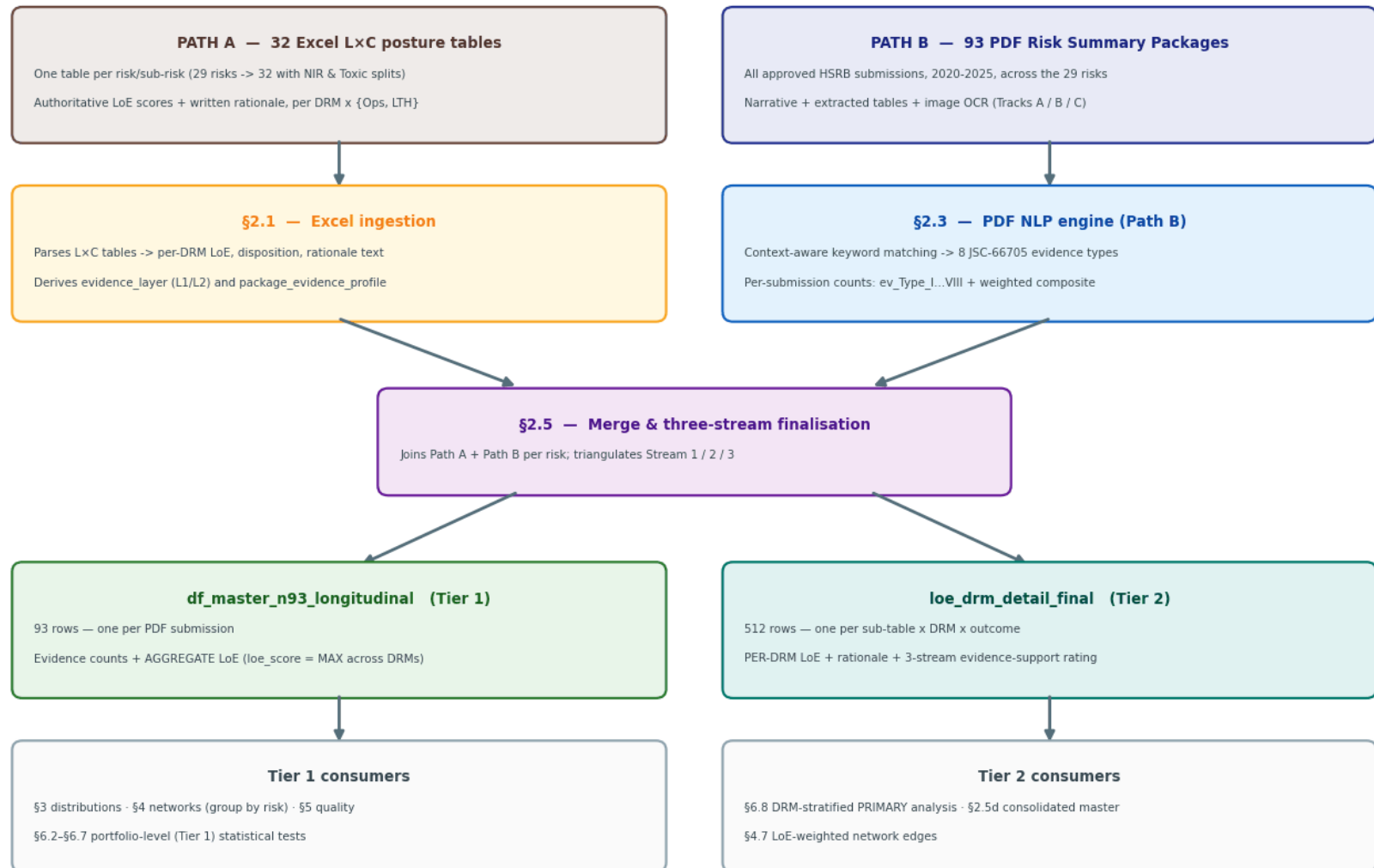


Figure E1. Data-flow hierarchy from the two ingestion paths to the consolidated master files.

Table 1 — Study Population and Variable Characteristics
Project CASCADE | N=29 risks | N=93 RSP submissions | 2020-2025 | JSC-66705 framework

STUDY CHARACTERISTICS		
Study Design	Cross-sectional computational audit with network analysis	
Study Population	NASA HSRB active human system risks	N = 29 parent-level (complete census); N =
Analysis Window	2020-2025 HSRB review cycles	5 years
Total Submissions	All HSRB-approved risk summary packages	N = 93
DRM×Outcome Scores	OPS+LTH LoE: 32 sub-tables x 8 DRMs x 2 outcomes	512 poss.; 419 valid (227 Ops/192 LTH)
Data Sources	PDF risk packages + Excel LxC posture tables	2 source types
Unit of Analysis	Risk (Tier 1, N=29 parent-level) and sub-risk×DRM (Tier 2, N=32 sub-risks x 8 DR	Two-tier design
EXPOSURE VARIABLES — EVIDENCE TYPES (STREAM 3, §2.3 NLP)		
Human Spaceflight	Type I — crewed mission data (Mercury through Artemis)	JSC-66705 weight: 8
Ground Analog Human	Type II — HERA, NEEMO, bed rest, parabolic flight	JSC-66705 weight: 7
Terrestrial Clinical	Type III — RCTs, cohort studies, meta-analysis	JSC-66705 weight: 6
Animal Data	Type IV — in vivo rodent, primate, whole-organism	JSC-66705 weight: 5
Cellular / Tissue	Type V — in vitro, omics, biomarkers	JSC-66705 weight: 4
Computational	Type VI — Monte Carlo, PBPK, OLTARIS models	JSC-66705 weight: 3
Standards	Type VIII — NASA-STD-3001, SMAC, SRD references	JSC-66705 weight: 2
Expert Opinion	Type VII — SME judgment, Delphi, consensus	JSC-66705 weight: 1
EXPOSURE VARIABLES — THREE-STREAM FRAMEWORK (§2.1 + §2.3)		
rationale_content_type	Stream 1: Evidence-Referenced / Narrative-Only / Missing	Per-DRM×Outcome
supporting_context_type	Stream 2: Substantive / Thin / Missing	Per-DRM×Outcome
package_evidence_profile	Stream 3: Well-Supported / Partially-Supported / Unsupported	Per-risk RSP
evidence_support_rating	Combined: Well-Supported / Partially-Supported / Unsupported	Per-DRM×Outcome
evidence_layer	L1 = rationale present; L2 = rationale missing	Per-DRM×Outcome
ev_weighted_composite	Quality-adjusted evidence score (JSC-66705 weights §2.5b)	Per-submission
evidence_diversity	Count of JSC-66705 evidence types with >0 hits	Integer 0-8
OUTCOME VARIABLES		
loe_score (Tier 1)	MAX LoE across all 8 DRMs — most conservative posture	Ordinal 2-4 (no LoE=1)
2 = Moderate	Limited human SF data supported by analogs	Tier 1 unit
3 = Weak	Primarily analog or ground-based evidence	Tier 1 unit
4 = Speculative	Expert opinion or insufficient data	Tier 1 unit
ops_loe (Tier 2)	Operational LoE per risk×DRM — primary Tier 2 outcome	Ordinal 1-4
1 = Strong	Compelling body of human spaceflight data	Tier 2 only
2-4	Moderate through Speculative as above	All 4 levels present
disposition	HSRB risk management decision	Categorical 6 classes
Accepted	Risk accepted at current LoE level	
Requires Mitigation	Active countermeasure development required	
STRATIFICATION VARIABLES		
DRM	Design Reference Mission (8 official categories)	LEO to Mars spectrum
LEO Short	Low Earth Orbit ≤30 days	
LEO Long	Low Earth Orbit >30 days (ISS)	
Lunar Orbital Short	Lunar orbit ≤14 days	
Lunar Orbital Long	Lunar orbit >14 days	
Lunar Orbital and Surface Short	Lunar orbit + surface ≤14 days	
Lunar Orbital and Surface Long	Lunar orbit + surface >14 days	
Mars Prep	Mars preparation missions	
Mars Planetary Long	Mars surface long duration	Most speculative
outcome	OPS (operational) or LTH (long-term health)	2 outcomes per DRM
STATISTICAL METHODS		
Tier 1 primary	Spearman (H1a/H1b), Fisher (H1c), KW+stratified-Spearman (H1d Part A+B)	N=29, Bonferroni k=3
Bonferroni α	0.05/3 = 0.0167	Pre-specified §6.1
Tier 2 primary	DRM-stratified Spearman, LMM, ESR vs LoE (H1e/H1g) — H1f for DRM-stratified, H1g	N≤256/outcome (419 valid), §6.8
Power (N=29)	80% power requires rho ≥0.48 or Pr≥0.52	Large effects only
Null interpretation	TOST equivalence + BF approximation (§6.7)	

Figure E2. Epidemiological study-characteristics table for the risk cohort.

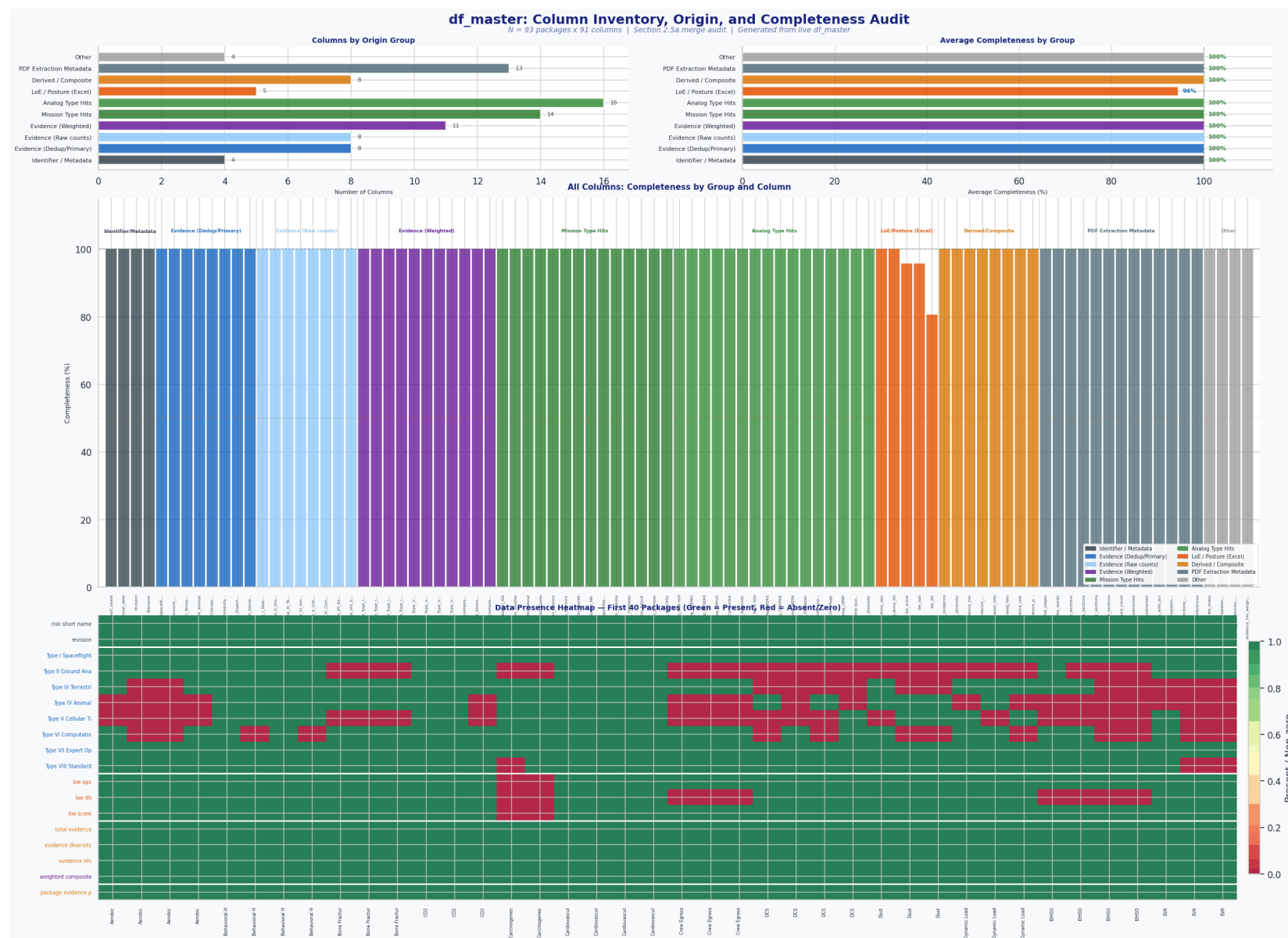


Figure E3. Master-dataset structure and completeness examination.

Figure 2.5b-1 JSC-66705 Evidence Type Weight Hierarchy & Impact

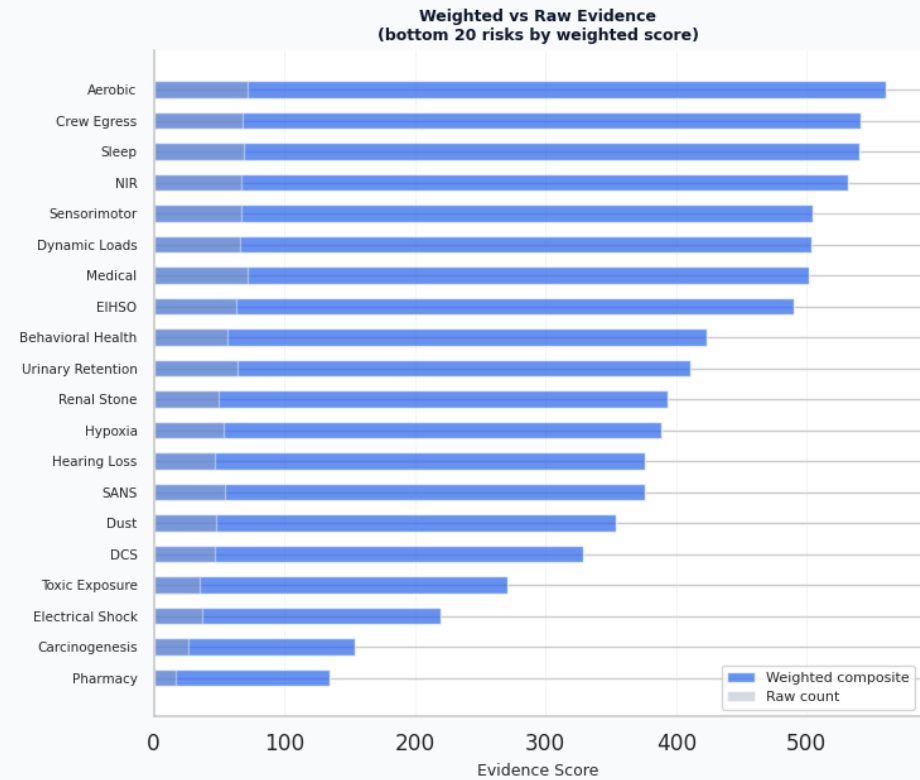


Figure E4. Evidence type quality-weighting scheme.

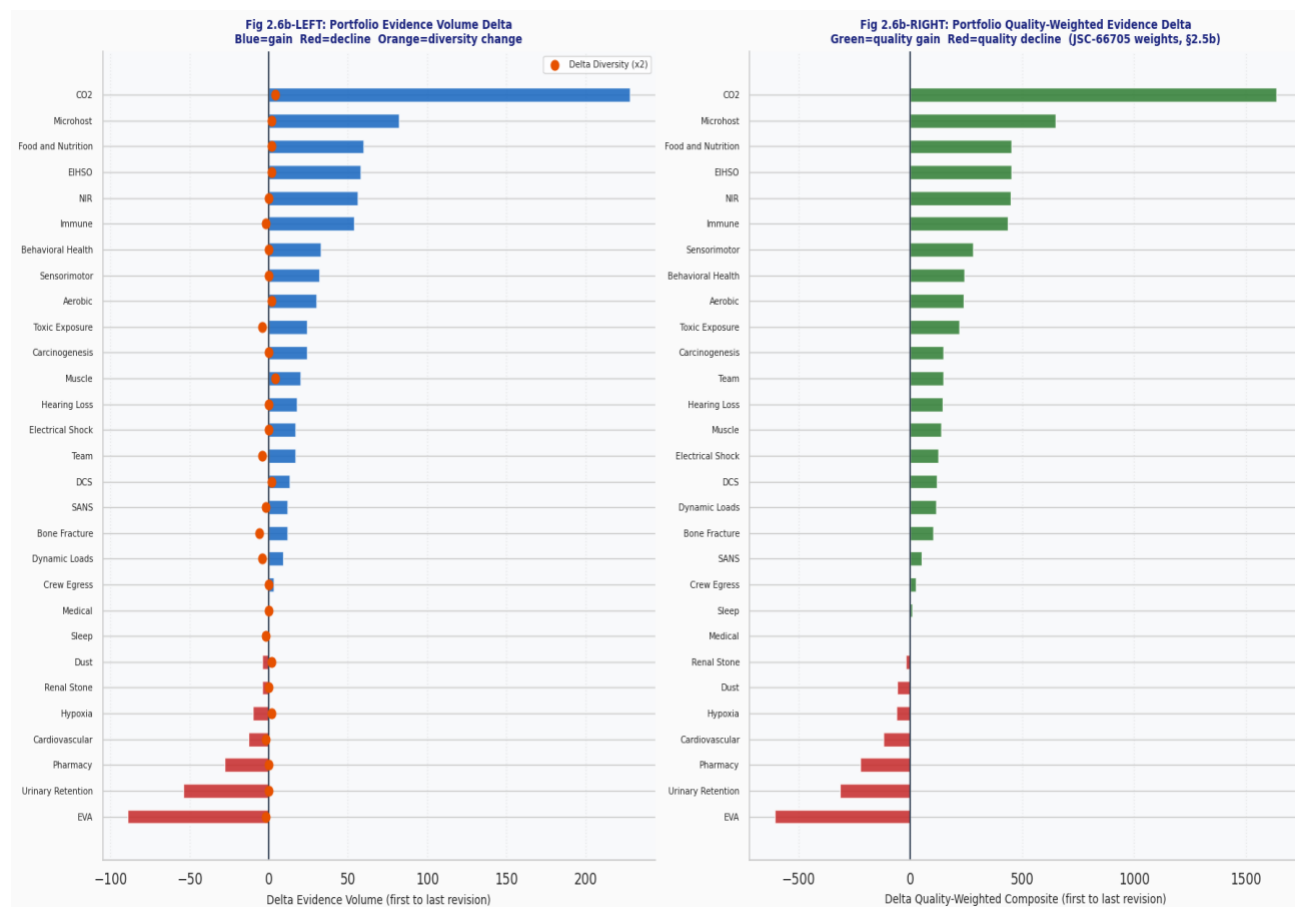


Figure E5. Longitudinal evidence trajectory across successive approved submissions.

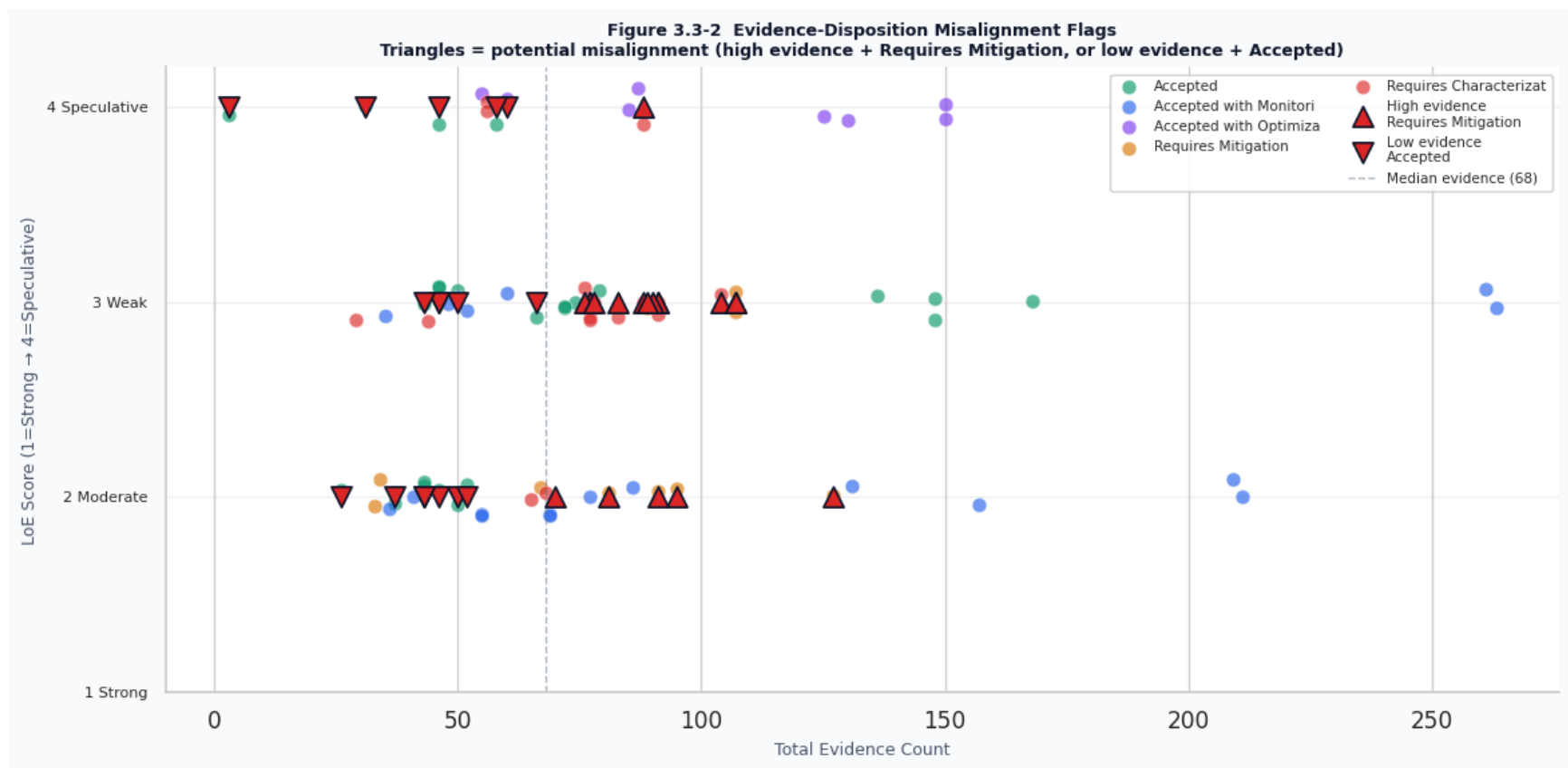
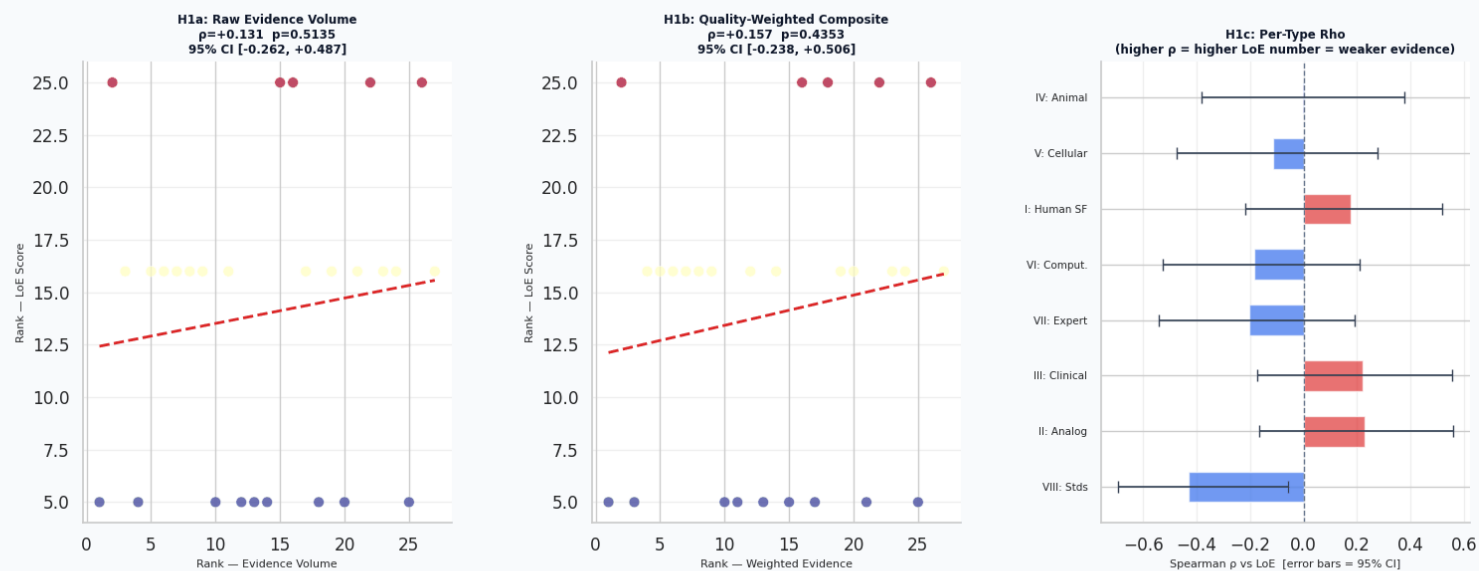


Figure E6. Disposition-evidence alignment across the portfolio.

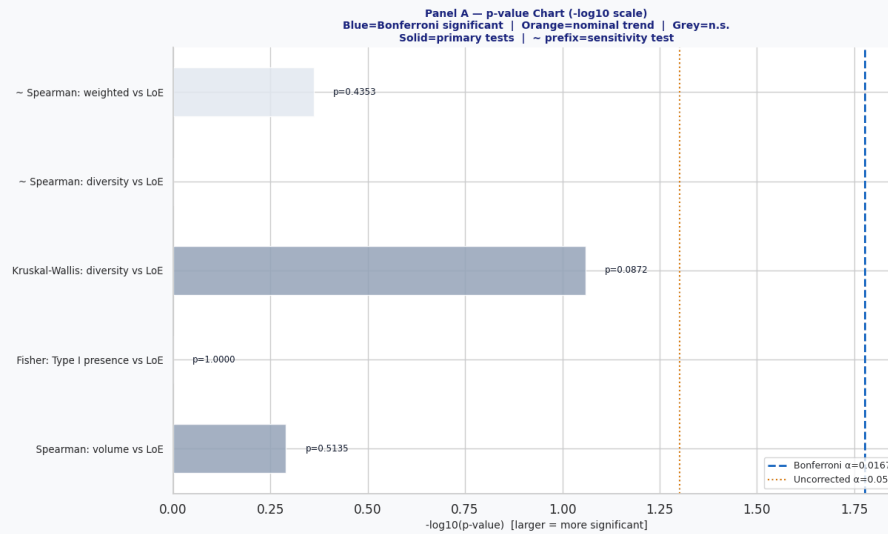
Figure 6.2-1 Spearman Rank Correlation — Tier 1 Portfolio Level
95% CI via Fisher Z transformation | Unit = risk (N=29)



H1a: Raw evidence volume shows a positive correlation with LoE (not significant, $\rho = +0.131$, 95% CI [-0.262, +0.487], $p = 0.5135$). H1b: Quality-weighted composite is not significant ($\rho = +0.157$, 95% CI [-0.238, +0.506], $p = 0.4353$). Wide CIs at N=29 are expected — see §6.1 power table. Bonferroni $\alpha = 0.0167$. Per-type H1c results (error bars) are exploratory. See §6.7 for TOST equivalence bounds and Bayes Factor interpretation.

Figure E7. Spearman correlation of documented evidence against LoE.

Section 6.5 — Bonferroni Correction: Multi-Test Summary
Project CASCADE N=29 Human System Risks Family of 3 primary tests $\alpha=0.0167$



Results Summary				
Test	p	rho	95% CI	Sig
Spearman: volume vs	0.5135	+0.131	[-0.26, +0.49]	X
Fisher: Type I prese	1.0000	-	-	X
Kruskal-Wallis: dive	0.0872	-	-	X
Spearman: diversity	-	-	-	X
Spearman: weighted v	0.4353	+0.157	[-0.24, +0.51]	X

$\alpha=0.0167$ | $r=sig$ → trend | $r=ns$ | CI via Fisher-Z

INTERPRETATION — Section 6.5 Bonferroni Correction

PANEL A: -log10(p). Bars past blue line = Bonferroni-significant.
PANEL B: Full table with 95% CI on rho where available (Fisher-Z).

RESULT: All 3 primary tests non-significant at $\alpha=0.0167$. H1a: $p=+0.131$ [95% CI -0.262, +0.487] — wide CI at N=29 does not exclude a moderate true association. Evidence counts do not predict LoE posture at the portfolio level — consistent with the Board integrating DRM-applicability and quality beyond raw citation volume. See §6.7 for TOST equivalence bounds.

CONSERVATISM: Bonferroni is conservative when tests share data. Bayes Factor analysis in §6.7 provides a complementary perspective.

Figure E8. Bonferroni correction and multiple-test summary for the primary family.

§6.8 Analyses A-C: DRM-Stratified Primary Analysis
Does evidence-LoE alignment degrade from LEO toward Mars?

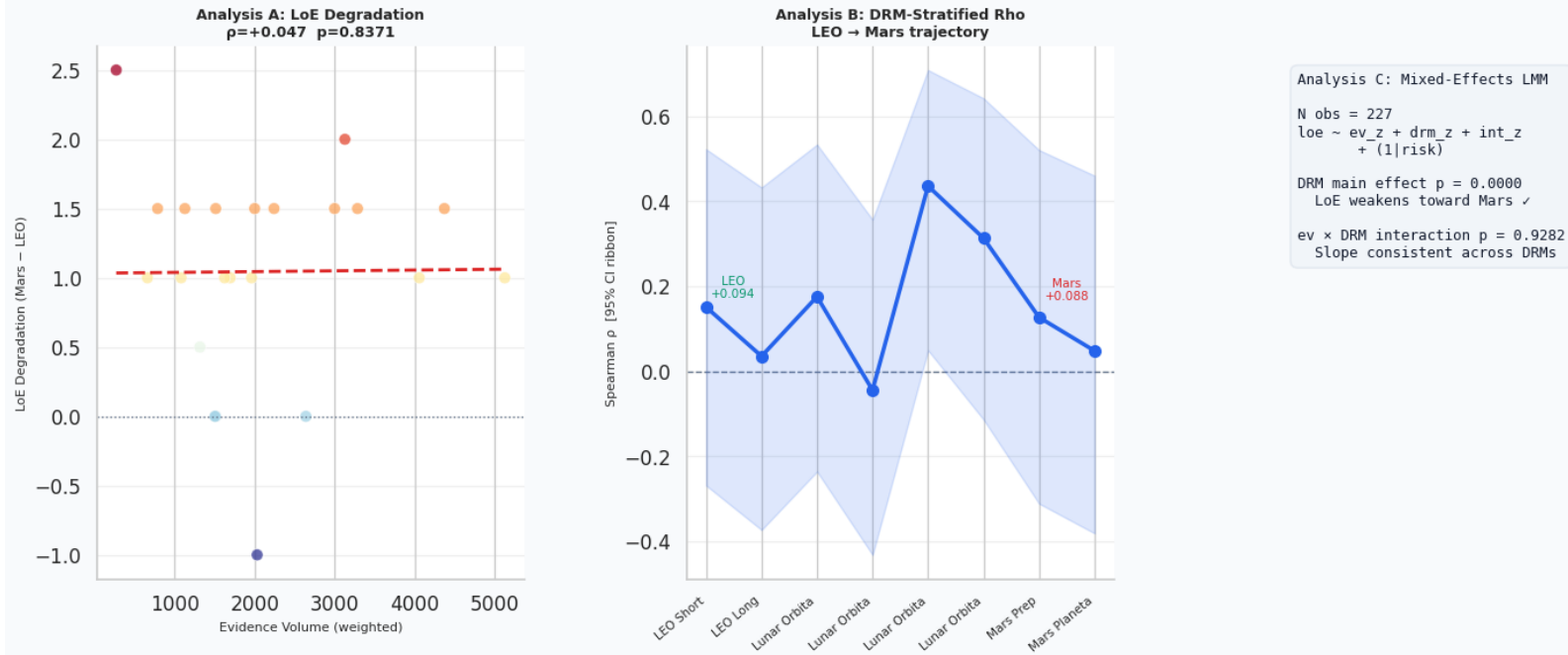


Figure E9. Mission-stratified Tier-2 primary analysis across Design Reference Missions.

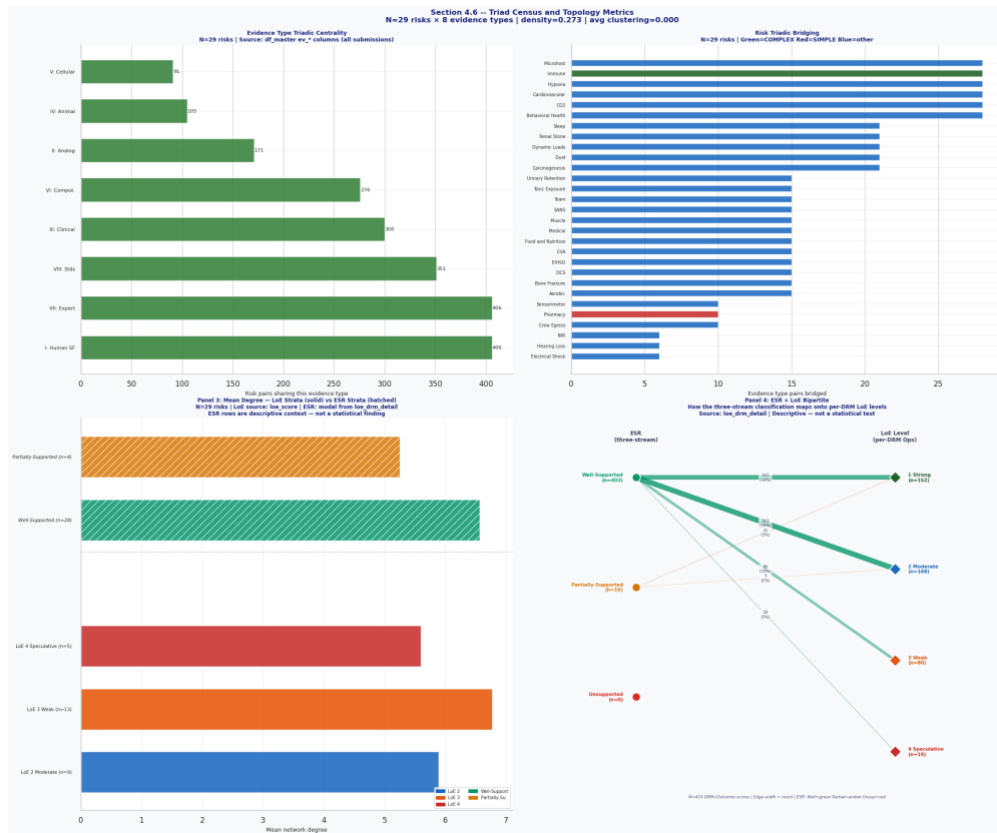


Figure E10. Triad census and topology metrics for the risk-evidence network.

214

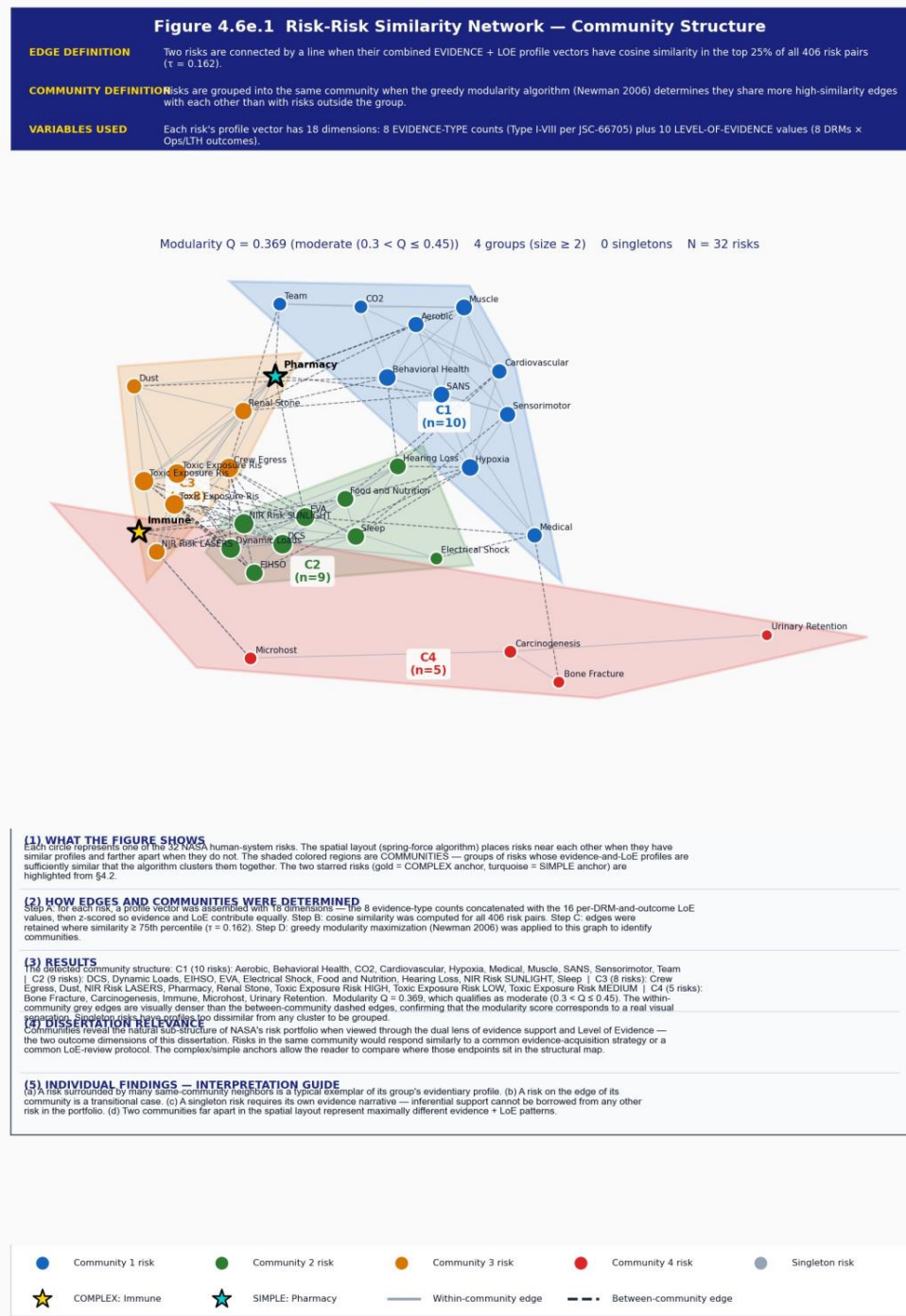


Figure E12. Community structure of the risk-risk evidence-similarity network (32 sub-risk nodes resolved into four communities).



Figure E13. Portfolio heatmap of risks by evidence type.

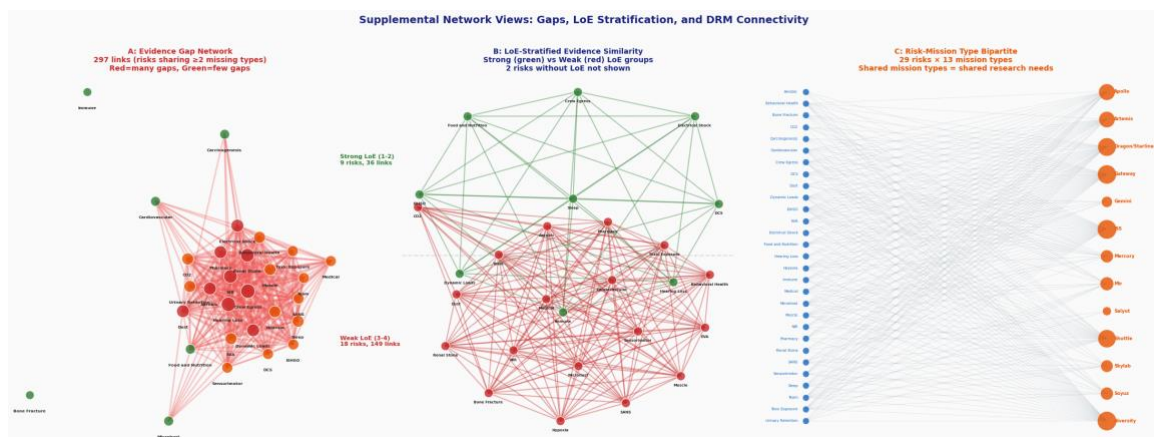


Figure E14. Focused network analysis of quantified risk-evidence relationships.

Figure 6.6c-1 Weight Robustness — N=29 risks
Source: ev_* columns from df_master (\$2.5b weights) | LoE: loe_score (MAX across DRMs)
Primary scheme = \$2.5b pre-specified | Bonferroni $\alpha=0.0167$

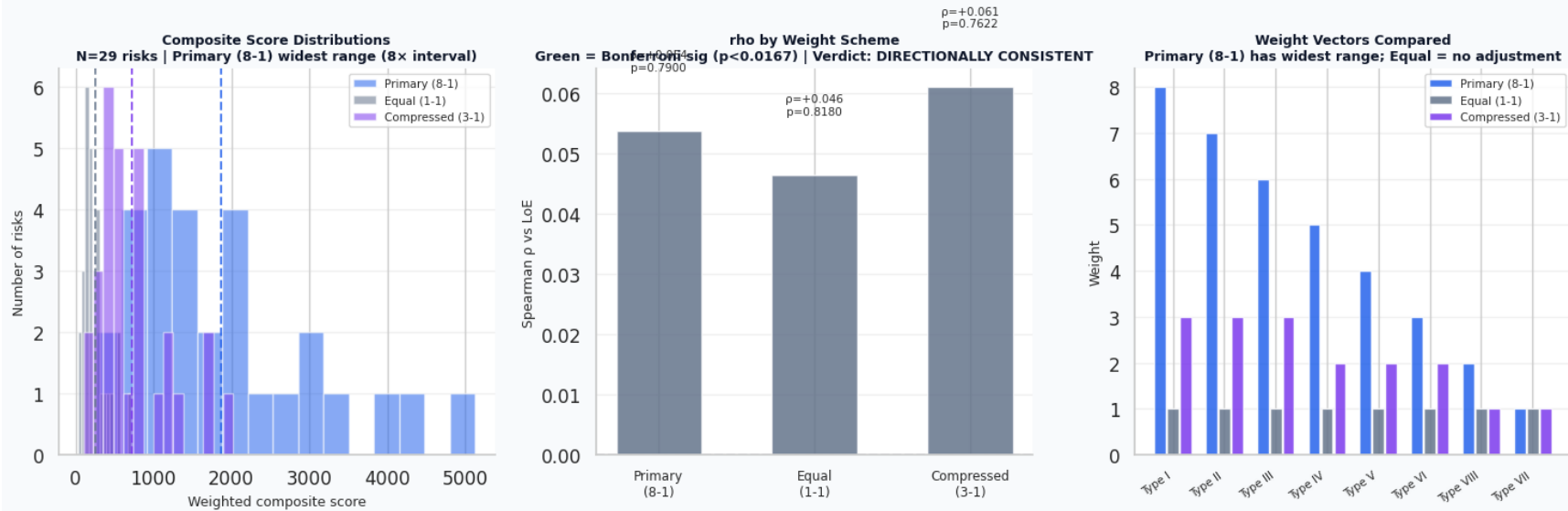


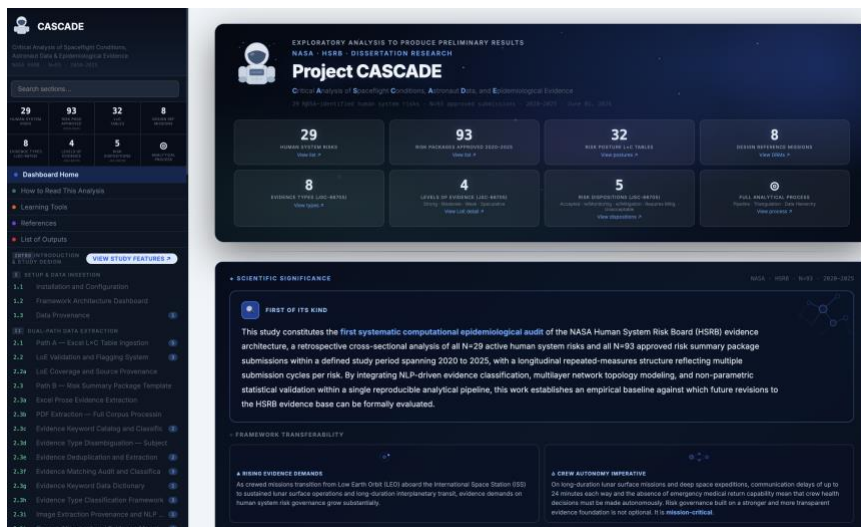
Figure E15. Weight-scheme robustness of the evidence-to-LoE association: the Spearman correlation between documented evidence and the Level of Evidence stays near zero and non-significant under the primary eight-to-one weighting, an equal weighting, and a compressed three-to-one weighting, so the central null finding does not depend on the particular weights chosen.

Appendix F. Reproducibility, Data, and Code Availability

The analysis underlying this dissertation was implemented as a single, fully scripted computational pipeline so that every figure, table, and statistic can be regenerated from the source corpus without manual intervention. The pipeline performs the dual-path ingestion, the NLP and classification against the eight-type hierarchy, the construction of the two analytic tiers, the statistical tests, the network projection and community analysis, and the thematic coding, and it writes the derived datasets and figures that the main text reports. This appendix records where the materials may be accessed so that the work can be inspected and reproduced.

F1. Interactive Dashboard and Notebook

The interactive dashboard, the analytic notebook, and the interactive application described here were developed as completed deliverables of Project CASCADE, built to make the full analysis transparent, navigable, and independently reproducible. The interactive dashboard, presenting the portfolio, the mission-stratified results, the network, and the thematic findings, is publicly available at <https://kmmori01.github.io/ProjectCASCADE-Dashboard/>. The complete analytic notebook, including the documented code, the inline methodological narrative, and the full statistical output, is published at <https://kmmori01.github.io/ProjectCASCADE-Dashboard/notebook.html>.



Project CASCADE

Critical Analysis of Spaceflight Conditions, Astronaut Data, and Epidemiological Evidence
NASA Human System Risk Board — N=93 risk summary packages · 29 risks · 2020-2025

Released: June 22, 2025 · Code: [Open Source](#)

Project CASCADE (2020-2025): N=93 All Submissions, Longitudinal Analysis

Critical Analysis of Spaceflight Conditions, Astronaut Data, and Epidemiological Evidence

Multilayer Network Topology, Natural Language Processing, and Level of Evidence Concordance Across NASA Human System Risk Board Risk Breakdown Architecture
Version 1.0 | N=29, most recent submission per risk | Framework: Evidence Type Extraction, L1 Validation, Multilayer Network Modeling

LIVING ANALYTICAL NOTEBOOK

This notebook is designed as a **living tool** — re-run with updated HSRB risk summary packages to track how the evidence landscape evolves across future review cycles. The extraction pipeline, statistical framework, and network models are fully reproducible. The [§8.7-8.8 Deliverables Catalog](#) (`CASCADE_Section_Catalog_Auto_v1.xlsx`) auto-generates from the live notebook on every run, providing a current section-by-section audit trail for version tracking and reproducibility review.

Note: A comprehensive acronym and abbreviation reference list is provided at the end of this notebook ([Section 8.1 Acronym Reference](#)). Consult this list for any unfamiliar abbreviations.

Table of Contents

Introduction & Study Design

INTRODUCTION & STUDY DESIGN — SECTION GUIDE (PRE-ANALYSIS)	
Research Questions	Bradford Hill Criteria — Exploratory
Specific Aims	NLP Evidence Classification Pipeline
Hypotheses — Exploratory	Network Analysis Pipeline
Executive Summary	Epidemiological 'Table T' Variables
Project Objective	29-Risk Census & DRMs
Public Health Impact	Project Flow Visualization

F1. Project CASCADE Dashboard (top), Project CASCADE Notebook (bottom)

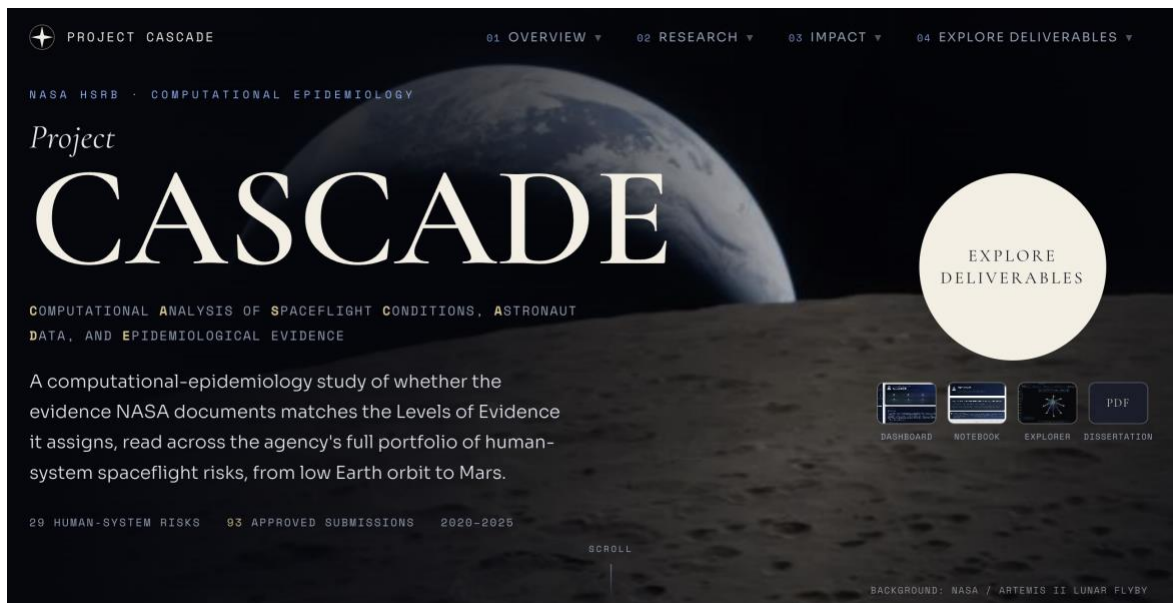
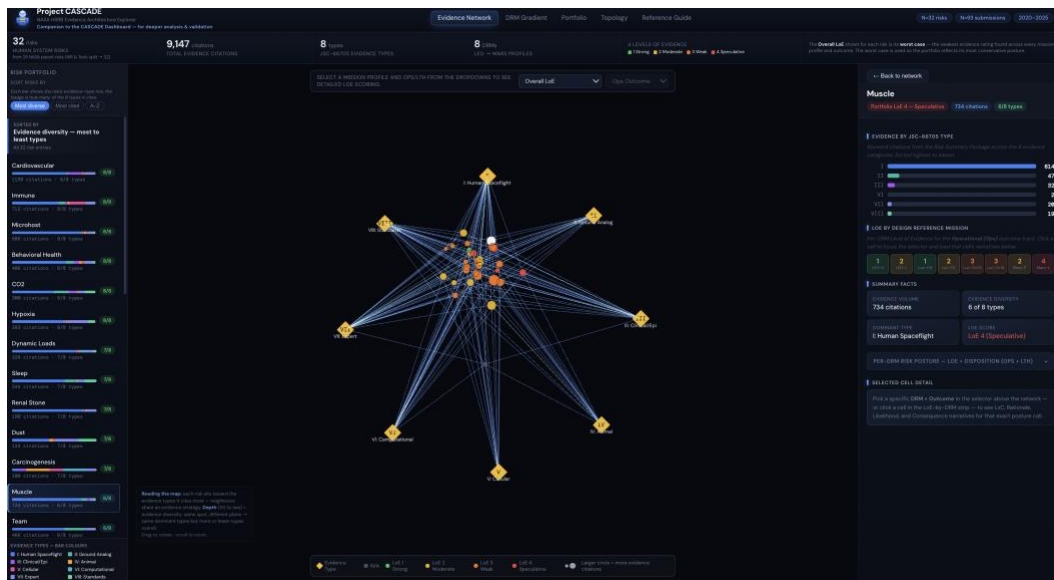
reported in this document, and they extend the work beyond a static report into a set of tools that others can interrogate, validate, and build upon.

F2. Interactive Application

Complementing these, the interactive application, available at <https://kmmori01.github.io/ProjectCASCADE-Dashboard/explorerapp.html>, is a key resource for mapping the relationship between the documented evidence and the assigned LoE, providing a queryable interface to the risk-level evidence profiles, the mission-stratified LoE gradients, and the network and thematic results so that any risk can be examined against its evidence base directly. Together, these deliverables provide a navigable, reproducible record of the analysis behind every result

A dedicated project website, which will house the dashboard, notebook, and application as a single portfolio that digitizes this body of work, is available at <https://kmmori01.github.io/ProjectCASCADE-Dashboard/website.html>.

The presentation layer of the project website was informed by publicly available front-end design resources rather than authored from a blank page. Structural and responsive layout patterns were adapted from challenge briefs and reference solutions on Frontend Mentor, and the visual direction, including the spacing, the typographic hierarchy, and the dark, space-themed treatment carried across the deliverables, drew on design concepts curated on Dribbble. These resources informed only the framework and the visual presentation of the site; the underlying content, data, and interactive logic are the programmatic outputs of the analytic pipeline described above.



F2. Project CASCADE Interactive App (top), Project CASCADE Website (Bottom)

F3. Data and Code Availability Statement

The primary source materials for this study are the approved Risk Summary Packages and the standardized Likelihood by Consequence posture tables maintained by the NASA Human System Risk Board; these are agency documents whose distribution is governed by NASA. The derived, de-identified analytic datasets generated by the pipeline, the classification lexicon, and the analysis code are available through the dashboard and notebook linked above, and further materials may be made available on reasonable request, subject to NASA distribution policy. No individually identifiable astronaut data were used in this analysis; the units of analysis are documented human system risks and their approved evidence packages.

The three deliverables share a common architecture designed for portability and independent verification. Each is a single self-contained HTML file generated programmatically from the analytic notebook rather than authored by hand, so that the published artifacts are reproducible outputs of the same pipeline that produced the dissertation results. The interactive application serializes the analytic tables into JavaScript Object Notation embedded directly in the page, so that it queries the evidence profiles, the mission-stratified gradients, and the network and thematic results entirely in the browser with no server or external database, and the dashboard likewise embeds its tables and figures inline. Static figures are rendered to images and encoded in base64, with photographs and screenshots stored in the WebP format to keep file sizes small, while the interactive charts are written as self-contained scripts. The files are version-controlled and hosted as static pages, which makes each deliverable a fixed, citable artifact that a reviewer can download and open without installing any software.

All figures, tables, and intermediate outputs generated by the analysis are retained in full within the project notebook and the interactive dashboard described above. To keep this dissertation focused, only the figures and tables central to the three aims are reproduced here; the complete output series, the underlying datasets, and the source code are available on request and are catalogued in the notebook output inventory.

Appendix G. Mission-Stratified and Risk-Level Data Summary

The following landscape table summarizes the mission-stratified results across the Design Reference Mission continuum for the operations outcome, combining the mean LoE, the within-mission evidence-to-score correlation, and the number of scored observations.

Table G1. Mission-stratified summary across the Design Reference Mission continuum (operations outcome).

Design Reference Mission	Mean LoE	Within-DRM ρ (evidence vs LoE)	Direction
LEO Short	1.39	+0.185	strongest evidence
LEO Long	1.58	+0.262	
Lunar Orbital Short	1.65	+0.158	
Lunar Orbital Long	1.89	+0.156	
Lunar Orbital and Surface Short	1.91	+0.363	
Lunar Orbital and Surface Long	2.14	+0.356	
Mars Preparation	2.18	+0.184	
Mars Planetary Long	2.68	+0.140	weakest evidence

Mixed-effects model (operations): Design Reference Mission main effect $p < 0.001$ (LoE weakens toward Mars); evidence-by-mission interaction $\beta = -0.025$, $p = 0.512$ (non-significant). Mean within-mission correlation +0.094 across the low Earth orbit strata

and +0.088 across the Mars strata. Across the 22 risks scored at both ends of the continuum, the mean weakening of the LoE was +1.045, with 18 of 22 risks degrading.

G2. Risk-Level Evidence and LoE Summary

The following landscape table summarizes, for each of the 29 active risks, the documented evidence volume, the diversity of evidence types represented, the maximum (worst-case) LoE used as the primary risk-level outcome, the Herfindahl-Hirschman concentration of the evidence profile, the total documentation page count, and the number of cited references. These curated risk-level values underlie the portfolio-level analyses reported in the Results.

Table G2. Risk-level evidence and LoE summary across the 29 active human system risks.

Risk	Evidence	Diversity	MAX LoE	HHI	Pages	Refs
Aerobic	124	6	4.0	0.60	33	20
Behavioral Health	102	7	3.0	0.41	57	21
Bone Fracture	137	6	3.0	0.65	93	107
CO2	121	8	3.0	0.31	88	116
Carcinogenesis	62	7	n/a	0.20	55	78
Cardiovascular	288	8	3.0	0.46	48	34
Crew Egress	90	5	2.0	0.76	50	3
DCS	87	4	2.0	0.42	30	31
Dust	37	5	3.0	0.38	31	15
Dynamic Loads	167	6	2.0	0.59	34	9
EIHSO	15	3	2.0	0.45	58	13
EVA	36	3	3.0	0.57	37	11
Electrical Shock	18	4	2.0	0.40	31	16
Food and Nutrition	55	6	2.0	0.48	60	26

Risk	Evidence	Diversity	MAX LoE	HHI	Pages	Refs
Hearing Loss	101	4	2.0	0.75	37	13
Hypoxia	102	7	3.0	0.46	46	26
Immune	217	8	2.0	0.38	77	89
Medical	120	6	3.0	0.52	75	20
Microhost	206	8	3.0	0.69	50	25
Muscle	189	6	4.0	0.58	59	45
NIR	53	4	3.0	0.68	35	8
Pharmacy	2	1	4.0	1.00	18	1
Renal Stone	89	4	3.0	0.80	47	21
SANS	75	4	4.0	0.49	44	16
Sensorimotor	110	5	4.0	0.81	51	12
Sleep	133	7	2.0	0.45	69	136
Team	106	4	3.0	0.48	43	16
Toxic Exposure	34	5	3.0	0.54	41	14
Urinary Retention	62	6	n/a	0.46	50	187

Appendix H. Study Characteristics (Epidemiological Summary)

This appendix presents the study’s design, population, variables, and analytic plan in a single epidemiological Table H1, in the standard “Table 1” format. It is the orienting reference for the variable definitions and coding used throughout the dissertation; the documented-evidence landscape (Table 1) and evidence type prevalence (Table 2) reported in the Results supplement it with the corpus’s observed composition.

Table H1. Study population, variables, and analytic design (STROBE-aligned).

Variable	Value	Notes
Study characteristics		
Study design	Cross-sectional computational analysis with network modeling	
Study population	NASA HSRB active human system risks	N = 29 parent-level (complete census); N = 32 sub-risks (Toxic Exposure → HIGH/MED/LOW; NIR → LASERS/SUNLIGHT)
Analysis window	2020–2025 HSRB review cycles	5 years
Total submissions	All HSRB-approved risk summary packages	N = 93
DRM × outcome scores	OPS + LTH LoE: 32 sub-tables × 8 DRMs × 2 outcomes	512 possible; 419 valid (227 Ops / 192 LTH)
Data sources	PDF risk packages + Excel L×C posture tables	2 source types
Unit of analysis	Risk (Tier 1, N = 29 parent-level) and sub-risk × DRM (Tier 2)	Two-tier design; Tier 2 = 512 cells, 419 valid
Exposure variables: evidence types (Stream 3, NLP)		
Human spaceflight	Type I: crewed mission data (Mercury through Artemis)	JSC-66705 weight: 8
Ground analog human	Type II: HERA, NEEMO, bed rest, parabolic flight	JSC-66705 weight: 7
Terrestrial clinical	Type III: RCTs, cohort studies, meta-analysis	JSC-66705 weight: 6
Animal data	Type IV: in vivo rodent, primate, whole-organism	JSC-66705 weight: 5
Cellular / tissue	Type V: in vitro, omics, biomarkers	JSC-66705 weight: 4
Computational	Type VI: Monte Carlo, PBPK, OLTARIS models	JSC-66705 weight: 3
Standards	Type VIII: NASA-STD-3001, SMAC, SRD references	JSC-66705 weight: 2
Expert opinion	Type VII: SME judgment, Delphi, consensus	JSC-66705 weight: 1
Exposure variables: three-stream framework		
rationale_content_type	Stream 1: Evidence-Referenced / Narrative-Only / Missing	Per DRM × outcome
supporting_context_type	Stream 2: Substantive / Thin / Missing	Per DRM × outcome
package_evidence_profile	Stream 3: Well-Supported / Partially-Supported / Unsupported	Per-risk RSP
evidence_support_rating	Combined: Well- / Partially- / Unsupported	Per DRM × outcome

Variable	Value	Notes
evidence_layer	L1 = rationale present; L2 = rationale missing	Per DRM × outcome
ev_weighted_composite	Quality-adjusted evidence score (JSC-66705 weights)	Per submission
evidence_diversity	Count of evidence types with > 0 hits	Integer 0–8
Outcome variables		
loe_score (Tier 1)	MAX LoE across all 8 DRMs (worst-case posture)	Ordinal 2–4 (no LoE = 1 at portfolio level)
2 = Moderate	Limited human spaceflight data supported by analogs	Tier 1 unit
3 = Weak	Primarily analog or ground-based evidence	Tier 1 unit
4 = Speculative	Expert opinion or insufficient data	Tier 1 unit
ops_loe (Tier 2)	Operational LoE per risk × DRM (primary Tier 2 outcome)	Ordinal 1–4
1 = Strong	Compelling body of human spaceflight data	Tier 2 only
2–4	Moderate through Speculative, as above	All four levels present
disposition	HSRB risk management decision	Categorical, 6 classes
Stratification variables		
DRM	Design Reference Mission (8 official categories)	LEO-to-Mars spectrum
LEO Short / LEO Long	Low Earth Orbit ≤ 30 days / > 30 days (ISS)	
Lunar Orbital Short / Long	Lunar orbit ≤ 14 / > 14 days	
Lunar Orbital and Surface Short / Long	Lunar orbit + surface ≤ 14 / > 14 days	
Mars Prep / Mars Planetary Long	Mars preparation / Mars surface long duration	Mars Planetary Long most speculative
outcome	OPS (operational) or LTH (long-term health)	2 outcomes per DRM
Statistical methods		
Tier 1 primary	Spearman (H1a/H1b), Fisher (H1c), Kruskal–Wallis + stratified Spearman (H1d A+B)	N = 29; Bonferroni k = 3
Bonferroni α	0.05 / 3 = 0.0167	Pre-specified
Tier 2 primary	DRM-stratified Spearman, linear mixed models, ESR vs LoE (H1e)	N ≤ 256 / outcome (419 valid)
Power (N = 29)	80% power requires $ \rho \geq 0.48$ or $r^2 \geq 0.52$	Large effects only
Null interpretation	TOST equivalence + Bayes-factor approximation	

Appendix I. Core Datasets and Data Manifest

This appendix documents the analytic datasets that underlie the results. Each file is named with its dimensions and contents so that any reported quantity can be traced to its source table. The complete files, including the two large master tables, which are too wide and too long to reproduce legibly in print, are available for inspection and download in the “Core Datasets” section of the interactive dashboard and in the project repository (see Appendix F). The smaller summary tables are reproduced here in full.

I.1 Dataset manifest

Table II. Core analytic datasets, dimensions, and contents.

Dataset (file)	Rows × Cols	Contents	Key variables
df_master_n93_longitudinal.csv	93 × 91	Submission-level master: one row per approved RSP submission; document metrics, raw and accepted evidence type counts, mission/analog hits, derived analytic variables, and LoE outcomes.	total_evidence, evidence_diversity, evidence_hhi, ev_weighted_composite, loe_score
cascade_master_consolidated.csv	513 × 32	Posture-level master: one row per risk × DRM × outcome; LoE score and status, L×C position, disposition, and rationale/likelihood/consequence narrative text.	drm, outcome, loe_score, lxc_disposition_category, evidence_support_rating
evidence_type_presence_audit.csv	32 × 9	Per-risk presence (1/0) of each of the eight JSC-66705 evidence types.	risk_short_name, Type_I...Type_VIII
evidence_type_presence_audit_summary.csv	32 × 3	Per-risk count and list of evidence types present.	risk, n_types, types
evidence_support_rating_by_drm.csv	8 × 3	Counts of Well- vs Partially-Supported postures by DRM.	drm, Well-Supported, Partially-Supported
h1d_partB_w2_per_drm_outcome.csv	16 × 8	Weighted evidence-diversity (w2) ranking by DRM × outcome (H1d Part B).	drm, outcome, w2, n_distinct_types
bradfordhill_detection_n93.csv	30 × 13	Exploratory Bradford–Hill cue detection per risk (nine criteria + totals).	risk, bh_total, bh_pct
conditional_evidence_audit.csv	0 records	Log of conditional/ambiguous analog references flagged for review; no records required override in this corpus.	(header only)
disambiguation_flags.csv	0 records	Log of subject-role disambiguation reclassifications; no records required reclassification in this corpus.	(header only)

I.2 Data dictionary: df_master_n93_longitudinal.csv (93 × 91)

The submission-level master has 91 columns, grouped below by function; the complete column-by-column file is available in the dashboard’s Core Datasets section.

Column group	Representative columns	What it captures
Identifiers / metadata	file_index, risk_short_name, approval_date, revision, filename	One row per approved submission; risk identity and submission vintage.

Column group	Representative columns	What it captures
Document metrics	total_pages, total_words, n_sections, tables_extracted, images_extracted, images_with_ocr, template_completeness, n_references	Size, structure, and template completeness of each package.
Raw evidence counts	raw_ev_Type_I ... raw_ev_Type_VIII	Pre-acceptance keyword-match counts for the eight evidence types.
Mission / analog hits	mission_ISS ... mission_Chinese; analog_Bed rest ... analog_Neutral buoyancy	Counts of named crewed-mission and ground-analog references.
Accepted evidence	ev_Type_I ... ev_Type_VIII, total_evidence, evidence_diversity, evidence_hhi, dominant_evidence_type	Context-accepted evidence by type and derived volume, diversity, and concentration.
Weighted evidence	ev_weighted_Type_I ... VIII, ev_weighted_composite, ev_weighted_diversity, ev_weighted_composite_norm, evidence_hhi_weighted	JSC-66705 quality-weighted composites used in sensitivity analyses.
Profile / outcomes / network	package_evidence_profile, loe_score, loe_ops, loe_lth, n_drms_ops, n_drms_lth, network_community_id	Stream-3 support profile, LoE outcomes, and assigned network community.

I.3 Data dictionary: cascade_master_consolidated.csv (513 × 32)

The posture-level master has one row per risk × DRM × outcome (513 rows). The rationale, likelihood, consequence, and assumption fields contain long free text and are best read in the dashboard.

Column group	Columns	What it captures
Identity / stratifiers	parent_risk, risk_short_name, drm, drm_raw, mission_type_duration, outcome	Risk, mission profile, and outcome (OPS/LTH) for each posture cell.
Scores / status	loe_score, loe_status, loe_ops, loe_lth	Assigned LoE and its validity status per cell.
Narrative text	rationale_text, rationale_content_type, likelihood_text, consequence_text, assumption_text	The package's prose rationale and posture narrative (long free text).
L×C posture	lxc_position, lxc_likelihood, lxc_consequence, lxc_disposition_category	LXC position and disposition category.
Evidence	evidence_support_rating, total_evidence, evidence_diversity, package_evidence_profile, ev_weighted_composite, ev_Type_I ... VIII	Per-cell evidence support and the eight evidence type counts.

I.4 Reproduced summary tables

Table I2. Evidence-support rating by DRM (posture counts).

drm	Partially-Supported	Well-Supported
LEO Long	9	55
LEO Short	11	53
Lunar Orbital Long	6	58
Lunar Orbital Short	9	55

drm	Partially-Supported	Well-Supported
Lunar Orbital and Surface Long	4	60
Lunar Orbital and Surface Short	9	55
Mars Planetary Long	5	59
Mars Prep	8	56

Table I3. Per-risk evidence type presence (1 = present). Columns I–VIII are the eight JSC-66705 evidence types.

Risk	I	II	III	IV	V	VI	VII	VIII
Aerobic	1	1	1	0	0	1	1	1
Behavioral Health	1	1	1	1	1	0	1	1
Behavioral Med	1	1	1	1	1	1	1	1
Bone Fracture	1	0	1	1	0	1	1	1
CO2	1	1	1	1	1	1	1	1
Carcinogenesis	1	0	1	1	1	1	1	1
Cardiovascular	1	1	1	1	1	1	1	1
Crew Egress	1	0	1	0	0	1	1	1
DCS	1	0	0	1	1	1	1	1
Dust	1	0	1	1	1	1	1	1
Dynamic Loads	1	0	1	1	1	1	1	1
EIHO HSIA	1	0	0	0	0	0	1	1
EIHSO	1	1	1	0	0	1	1	1
EIHSO HSIA	1	0	0	0	0	0	1	1
EVA	1	1	0	0	1	1	1	1
Electrical Shock	1	1	0	0	0	0	1	1
Food and Nutrition	1	1	1	0	1	0	1	1
Hearing Loss	1	0	1	0	0	0	1	1
Hypoxia	1	1	1	1	1	1	1	1
Immune	1	1	1	1	1	1	1	1
Medical	1	1	1	0	0	1	1	1
Microhost	1	1	1	1	1	1	1	1
Muscle	1	1	1	0	0	1	1	1
NIR	1	0	1	0	0	0	1	1
Pharmacy	1	0	1	1	0	1	1	0
Renal Stone	1	1	1	1	0	1	1	1
SANS	1	1	1	1	0	1	1	0
Sensorimotor	1	1	0	0	1	0	1	1

Risk	I	II	III	IV	V	VI	VII	VIII
Sleep	1	1	1	1	0	1	1	1
Team	1	1	1	0	0	1	1	1
Toxic Exposure	1	1	1	0	0	1	1	1
Urinary Retention	1	0	1	0	1	1	1	1

Table I4. Per-risk count and list of evidence types present.

risk	n_types	types
Aerobic	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Behavioral Health	7	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_V_Cellular_Tissue
Behavioral Med	8	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Bone Fracture	6	Type_III_Terrestrial_Clinical_Epi, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
CO2	8	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Carcinogenesis	7	Type_III_Terrestrial_Clinical_Epi, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Cardiovascular	8	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Crew Egress	5	Type_III_Terrestrial_Clinical_Epi, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
DCS	6	Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Dust	7	Type_III_Terrestrial_Clinical_Epi, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Dynamic Loads	7	Type_III_Terrestrial_Clinical_Epi, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
EIHO HSIA	3	Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion
EIHSO	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
EIHSO HSIA	3	Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion
EVA	6	Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Electrical Shock	4	Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion

risk	n_types	types
Food and Nutrition	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_V_Cellular_Tissue
Hearing Loss	4	Type_III_Terrestrial_Clinical_Epi, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion
Hypoxia	8	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Immune	8	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Medical	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Microhost	8	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue
Muscle	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
NIR	4	Type_III_Terrestrial_Clinical_Epi, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion
Pharmacy	5	Type_III_Terrestrial_Clinical_Epi, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Renal Stone	7	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
SANS	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Sensorimotor	5	Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_V_Cellular_Tissue
Sleep	7	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_IV_Animal, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Team	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Toxic Exposure	6	Type_III_Terrestrial_Clinical_Epi, Type_II_Ground_Analog_Human, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling
Urinary Retention	6	Type_III_Terrestrial_Clinical_Epi, Type_I_Spaceflight_Human, Type_VIII_Standards_Requirements, Type_VII_Expert_Opinion, Type_VI_Computational_Modeling, Type_V_Cellular_Tissue

Table I5. Weighted evidence-diversity (w_2) ranking by DRM $\times$ outcome (H1d Part B).

drm	outcome	rank	w2	n_distinct_types	n_rows	n_cited_rows	types_present
LEO Short	LTH	0	28	5	32	3	Type I Type II Type III Type V Type VI
LEO Long	LTH	1	22	4	32	3	Type I Type II Type V Type VI
Lunar Orbital Short	LTH	2	28	5	32	3	Type I Type II Type III Type V Type VI

drm	outcome	rank	w2	n_distinct_types	n_rows	n_cited_rows	types_present
Lunar Orbital Long	LTH	3	22	4	32	3	Type I Type II Type V Type VI
Lunar Orbital and Surface Short	LTH	4	28	5	32	3	Type I Type II Type III Type V Type VI
Lunar Orbital and Surface Long	LTH	5	22	4	32	3	Type I Type II Type V Type VI
Mars Prep	LTH	6	22	4	32	4	Type I Type II Type V Type VI
Mars Planetary Long	LTH	7	22	4	32	3	Type I Type II Type V Type VI
LEO Short	OPS	0	28	5	32	3	Type I Type II Type III Type V Type VI
LEO Long	OPS	1	28	5	32	3	Type I Type II Type III Type V Type VI
Lunar Orbital Short	OPS	2	28	5	32	3	Type I Type II Type III Type V Type VI
Lunar Orbital Long	OPS	3	28	5	32	3	Type I Type II Type III Type V Type VI
Lunar Orbital and Surface Short	OPS	4	28	5	32	3	Type I Type II Type III Type V Type VI
Lunar Orbital and Surface Long	OPS	5	28	5	32	3	Type I Type II Type III Type V Type VI
Mars Prep	OPS	6	22	4	32	3	Type I Type II Type V Type VI
Mars Planetary Long	OPS	7	22	4	32	2	Type I Type II Type V Type VI

Table 16. Exploratory Bradford–Hill cue detection per risk. Str=strength, Con=consistency, Spec=specificity, Temp=temporality, Grad=biological gradient, Plaus=plausibility, Coh=coherence, Exp=experiment, Anlg=analogy; Pct=percent of cues detected.

Risk	Total	Len	Str	Con	Spec	Temp	Grad	Plaus	Coh	Exp	Anlg	Pct
CO2	8	179096	1	1	1	1	0	1	1	1	1	88.8888888888889
Behavioral Health	8	134043	1	1	1	1	1	1	0	1	1	88.8888888888889
EVA	7	199068	1	1	1	1	0	1	0	1	1	77.7777777777777
Sleep	7	186574	0	1	1	1	1	1	0	1	1	77.7777777777777
Carcinogenesis	7	93650	1	1	1	1	1	1	0	1	0	77.7777777777777
Cardiovascular	7	204436	1	1	1	1	0	1	0	1	1	77.7777777777777
Immune	7	214978	0	1	1	1	0	1	1	1	1	77.7777777777777
Team	7	127953	1	1	1	1	1	1	0	1	0	77.7777777777777
Muscle	6	139897	1	0	1	1	1	0	0	1	1	66.6666666666667
Medical	6	163404	1	1	1	0	1	1	0	1	0	66.6666666666667
Hypoxia	6	155751	0	1	1	1	0	1	0	1	1	66.6666666666667
SANS	6	129792	0	0	1	1	1	1	0	1	1	66.6666666666667
Pharmacy	6	137966	0	1	1	1	1	0	0	1	1	66.6666666666667
Renal Stone	6	78844	1	1	1	1	0	1	0	1	0	66.6666666666667
EIHSO	6	156733	1	1	1	1	0	0	0	1	1	66.6666666666667
Dust	6	128599	1	1	1	0	0	1	0	1	1	66.6666666666667
Bone Fracture	6	170931	0	1	1	1	0	1	0	1	1	66.6666666666667
DCS	5	93475	0	0	1	1	0	1	0	1	1	55.5555555555556
Crew Egress	5	93165	0	0	1	1	1	0	0	1	1	55.5555555555556
Microhost	5	89583	0	0	1	0	1	1	0	1	1	55.5555555555556
NIR	5	71803	0	0	1	1	0	1	0	1	1	55.5555555555556

Risk	Total	Len	Str	Con	Spec	Temp	Grad	Plaus	Coh	Exp	Anlg	Pet
Aerobic	5	99369	0	0	1	1	0	1	0	1	1	55.55555555555556
Hearing Loss	4	83390	0	0	1	1	0	0	0	1	1	44.44444444444444
Electrical Shock	4	83560	0	1	1	0	0	1	0	1	0	44.44444444444444
Dynamic Loads	4	82433	0	0	1	1	0	1	0	1	0	44.44444444444444
Sensorimotor	3	120261	0	1	1	0	0	0	0	1	0	33.333333333333336
Toxic Exposure	3	77615	0	0	1	1	0	0	0	1	0	33.333333333333336
Food and Nutrition	2	79358	0	0	0	0	1	0	0	1	0	22.22222222222222
Urinary Retention	2	111053	0	0	0	1	0	0	0	1	0	22.22222222222222

The full master files (df_master and cascade_master_consolidated) and every table above can be opened, sorted, and downloaded in the “Core Datasets” section of the interactive dashboard, consisted of)

Table I7 maps each of the eleven pre-specified hypotheses to the analytic approach used, the corresponding result, and the preliminary finding. All findings are exploratory and preliminary, derived from a single-agency pilot census (N = 29 risks; N = 93 packages, 2020–2025); they characterize the documented record as it exists and are not, on their own, a basis for operational or scoring changes. Tier 1 portfolio-level tests applied a Bonferroni-corrected significance threshold of $\alpha = 0.0167$.

Table I7. Results of pre-specified hypotheses to analytic approach, result, and preliminary finding.

Hypothesis	Analytic approach	Result	Preliminary finding
H1a — Evidence volume is positively associated with LoE strength (Tier 1).	Spearman rank correlation	$\rho = +0.05$; 95% CI $[-0.34, +0.42]$; $p = 0.82$	Null retained; no preliminary evidence of association.
H1b — The quality-weighted evidence composite is associated with LoE (Tier 1).	Spearman rank correlation	$\rho = +0.05$; $p = 0.79$	Null retained; no preliminary evidence of association.
H1c — Presence of Type I (direct human spaceflight) evidence is associated with stronger LoE (Tier 1).	Fisher exact test; Firth-penalized logistic regression	Degenerate table (Type I universal); Firth OR = 0.72; $p = 1.00$	Not evaluable as a differentiator because the exposure is universal; no preliminary association.
H1d — Evidence diversity (breadth of types) is associated with LoE (Tier 1).	Kruskal–Wallis H-test	$H = 0.45$; $df = 2$; $p = 0.80$	Null retained; no preliminary evidence of association.
H1e — The modal evidence-support rating is associated with LoE (Tier 1).	Spearman rank correlation	$\rho = -0.36$; 95% CI $[-0.66, +0.04]$; $p = 0.08$	Suggestive negative trend, not significant; exploratory only.
H2a — Higher composite-complexity risks show greater degree and betweenness centrality (hubs).	Network centrality; triad census	Complex risks (e.g., Immune, Bone Fracture) bridge the most evidence-type pairs; simple risks (e.g., Pharmacy) the fewest	Preliminary, descriptive support; exploratory.
H2b — Risks with similar evidence profiles cluster into identifiable communities.	Cosine-similarity network; modularity vs configuration-model null	$Q = 0.37$ (weighted) vs null mean 0.24 $[0.21, 0.28]$; $p < 0.001$	Preliminary support; exploratory.
H2c — Cross-risk narrative references reveal dependencies beyond profile similarity.	Deductive thematic coding; co-occurrence	Cross-risk dependency language present in 41% (38/93) of packages	Partial, exploratory; dependencies appear but in a minority of packages.
H3a — Greater package completeness is associated with stronger LoE.	Spearman rank correlation (exploratory)	$\rho = +0.02$; $p = 0.91$	Null retained; no preliminary evidence of association.
H3b — Cross-risk dependency language appears in fewer than half of packages.	Deductive thematic coding	Present in 41% (38/93) of packages	Preliminary support; exploratory.
H3c — Greater evidence-type concentration is associated with weaker LoE.	Spearman rank correlation (exploratory)	$\rho = -0.18$; $p = 0.40$	Null retained; no preliminary evidence of association.

Note. All results are preliminary and exploratory. Tier 1 portfolio analyses used worst-case (maximum) LoE across Design Reference Missions for the complete census; H3a and H3c were evaluated as exploratory correlations on worst-case LoE. ρ = Spearman correlation coefficient; CI = confidence interval; OR = odds ratio; LoE = Level of Evidence.